

This is a part of SFQED-Loops script collection
developed for calculating loop and tree-level processes in
Strong-Field Quantum Electrodynamics.
The scripts are available on <https://github.com/ArsenyMironov/SFQED-Loops>

If you use this script in your research, please, consider
citing our papers:

- A. A. Mironov, S. Meuren, and A. M. Fedotov, PRD 102, 053005 (2020); arXiv:2003.06909
<https://doi.org/10.1103/PhysRevD.102.053005>
- A. A. Mironov, A. M. Fedotov, 105 (3), 033005 (2022); arXiv:2109.00634
<https://doi.org/10.1103/PhysRevD.105.033005>
- Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)

If you have any questions, please, don't hesitate to contact:
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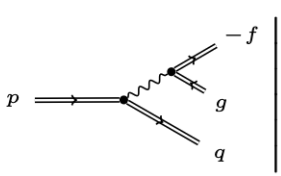
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This notebook is a supplement for our work
[Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)]

The notebook was prepared by A. A. Mironov and Y. V. Selivanov

The notebook presents a tree-level calculation of the trident process rate (the direct term contribution) initiated by a relativistic electron propagating in a strong constant crossed electromagnetic field, W_{dir}

The corresponding diagram considered in this notebook reads

$$\sum_{\text{spin}} \int d\Pi(q, g, f) \left| \begin{array}{c} \text{diagram} \\ \text{"direct" diagram} \end{array} \right|^2$$


The notebook depends on two additional files
 definitions.nb
 TridentDiracSimplify.nb
 which should be contained in the same directory

```
In[2]:= NotebookEvaluate[NotebookOpen[NotebookDirectory[] <> "definitions.nb"]]
```

FeynCalc 10.1.0 (stable version). For help, use the
 online documentation, visit the forum and have a look at the supplied
 examples. The PDF-version of the manual can be downloaded [here](#).

If you use FeynCalc in your research, please
 evaluate FeynCalcHowToCite[] to learn how to cite this software.

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 of our work is crucial to ensure the future development of this package!*

Trident process ($e > 0$)

$$S_{fi}^{\text{trident, full}} = S_{fi}^{\text{dir}} + S_{fi}^{\text{ex}},$$

where S_{fi}^{dir} is the direct term and S_{fi}^{ex}
 is the exchange term. We will consider only S_{fi}^{dir}

$$\text{Note that } \int d\Pi \left| S_{fi}^{\text{dir}} \right|^2 = \int d\Pi \left| S_{fi}^{\text{ex}} \right|^2,$$

where $\int d\Pi$ is the integral over all the outgoing particle momenta

The direct term matrix element is defined by*

$$S_{fi}^{\text{dir}} = (ie)^2 \int d^4x_1 d^4x_2 \bar{\psi}_q(x_1) \gamma^\mu \psi_p(x_1) D_{\mu\nu}^c(x_1 - x_2) \bar{\psi}_f(x_2) \gamma^\nu \psi_g^+(x_2);$$

x_1^μ, x_2^μ – position of the left and right vertices of the diagram;

p^μ, q^μ – initial and final electron momenta;

$S^c, D_{\mu\nu}^c$ – electron and photon casual propagators;

Superscript + stands for Hermitian conjugation

The Ritus E_p – function

$$E_p(x_1) = \left[1 - \frac{e(\gamma k)(\gamma a)}{2(kp)} (kx_2) \right]$$

$$\text{Exp}\left[-i(\mathbf{p} \cdot \mathbf{x}_1) + i \frac{e(a \cdot \mathbf{p})}{2(k \cdot \mathbf{p})} (k \cdot \mathbf{x}_1)^2 + i \frac{e^2 a^2}{6(k \cdot \mathbf{p})} (k \cdot \mathbf{x}_1)^3\right];$$

$$\bar{E}_q(x_2) = \gamma^0 E_q(x_2) \gamma^0;$$

Volkov states

$$\psi_{p\lambda}(x_1) = E_p(x_1) u_{p\lambda}$$

$$\bar{\psi}_{p\lambda}(x_1) = \bar{u}_{p\lambda} \bar{E}_p(x_1)$$

$$\psi_{p\lambda}^+(x_1) = E_{-p}(x_1) v_{p\lambda}$$

$$\bar{u}_{p\lambda} = u_{p\lambda} \gamma^0$$

Photon propagator

$$D_{\mu\nu}^c(x) = \int \frac{d^4 l}{(2\pi)^4} e^{ilx} \frac{-i}{l^2 + i0} g_{\mu\nu}$$

* Note some differences in notations used in this

Notebook and in section 6.1 of [arXiv 2605.12479]. Here, $e > 0$ denotes the absolute value of the electron charge, and momenta f and g are interchanged. While this slightly changes some intermediate steps, this does not affect the final result.

Main

Preliminary definitions

We assume that

F is the crossed field tensor,

k is the 'wave' 4 - vector, normalised such that $k^0 = m$,

a is the transverse (space - like) constant 4 - vector,

χp denotes the χ parameter of a particle with 4 - momentum p ,

All objects such as vectors and tensors introduced via commands from `definitions.nb` are assumed to be D - dimensional. However, since there is no regularization involved in this calculation, we will set $D \rightarrow 4$ along the Notebook

Command `NewMomentum` is defined in `definitions.nb`. It introduces a set of 4 - vectors and scalars induced by a 4 - momenta and tensor F , see `definitions.nb` for more details

```

In[3]:= NewMomentum["p"] (*Initial electron*)
NewMomentum["q"] (*Scattered electron*)
NewMomentum["g"] (*New positron, created by a photon*)
NewMomentum["f"] (*New electron, created by a photon*)
NewMomentum["l"] (*Intermediate photon*)
NewMomentum["l1"] (*Also intermediate photon,
l1 is used when mod-sqaring the amplitude*)
NewMomentum["l2"] (*Same as above*)
NewCoordinate["x1"] (*Coordinate at which emission happens*)
NewCoordinate["x2"] (*Coordinate at which pair creation happens*)
ScalarProduct[p, p] = m^2;
ScalarProduct[q, q] = m^2;
ScalarProduct[f, f] = m^2;
ScalarProduct[g, g] = m^2;

{p^α, p^2, k · p, Fp^α, FFp^α, FDp^α, a · p, 0, 0, 0, -a^2 (k · p), 0, 0, - $\frac{m^6 \chi p^2}{e^2}$ , - $\frac{m^6 \chi p^2}{e^2}$ ,  $\frac{m^6 \chi p^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{q^α, q^2, k · q, Fq^α, FFq^α, FDq^α, a · q, 0, 0, 0, -a^2 (k · q), 0, 0, - $\frac{m^6 \chi q^2}{e^2}$ , - $\frac{m^6 \chi q^2}{e^2}$ ,  $\frac{m^6 \chi q^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{g^α, g^2, g · k, Fg^α, FFg^α, FDg^α, a · g, 0, 0, 0, -a^2 (g · k), 0, 0, - $\frac{m^6 \chi g^2}{e^2}$ , - $\frac{m^6 \chi g^2}{e^2}$ ,  $\frac{m^6 \chi g^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{f^α, f^2, f · k, Ff^α, FFf^α, FDf^α, a · f, 0, 0, 0, -a^2 (f · k), 0, 0, - $\frac{m^6 \chi f^2}{e^2}$ , - $\frac{m^6 \chi f^2}{e^2}$ ,  $\frac{m^6 \chi f^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{l^α, l^2, k · l, Fl^α, FFl^α, FDll^α, a · l, 0, 0, 0, -a^2 (k · l), 0, 0, - $\frac{m^6 \chi l^2}{e^2}$ , - $\frac{m^6 \chi l^2}{e^2}$ ,  $\frac{m^6 \chi l^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{l1^α, l1^2, k · l1, Fl1^α, FFl1^α, FDll1^α, a · l1, 0, 0, 0, -a^2 (k · l1), 0, 0, - $\frac{m^6 \chi l1^2}{e^2}$ , - $\frac{m^6 \chi l1^2}{e^2}$ ,  $\frac{m^6 \chi l1^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{l2^α, l2^2, k · l2, Fl2^α, FFl2^α, FDll2^α, a · l2, 0, 0, 0, -a^2 (k · l2), 0, 0, - $\frac{m^6 \chi l2^2}{e^2}$ , - $\frac{m^6 \chi l2^2}{e^2}$ ,  $\frac{m^6 \chi l2^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{x1^α, x1^2, k · x1, a · x1, Fx1^α, FFx1^α, FDx1^α, k · x1, 0, 0, 0,
-a^2 (k · x1), 0, 0, - $\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}$ , - $\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}$ ,  $\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{x2^α, x2^2, k · x2, a · x2, Fx2^α, FFx2^α, FDx2^α, k · x2, 0, 0, 0,
-a^2 (k · x2), 0, 0, - $\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}$ , - $\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}$ ,  $\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

```

Commands Ep and EpC introduce Ep –

functions in the form of an array {prefactor, phase},
see definitions.nb for details

In[16]:= **Ep**x1 = **Ep**[x1, p]

EqBarx1 = **EpC**[x1, q]

EfBarx2 = **EpC**[x2, f]

Emingx2 = **Ep**[x2, -g]

$$\text{Out[16]} = \left\{ 1 - \frac{e(k \cdot x1)(\gamma \cdot k)(\gamma \cdot a)}{2(k \cdot p)}, \frac{a^2 e^2 (k \cdot x1)^3}{6(k \cdot p)} + \frac{e(a \cdot p)(k \cdot x1)^2}{2(k \cdot p)} - p \cdot x1 \right\}$$

$$\text{Out[17]} = \left\{ 1 - \frac{e(k \cdot x1)(\gamma \cdot a)(\gamma \cdot k)}{2(k \cdot q)}, -\frac{a^2 e^2 (k \cdot x1)^3}{6(k \cdot q)} - \frac{e(a \cdot q)(k \cdot x1)^2}{2(k \cdot q)} + q \cdot x1 \right\}$$

$$\text{Out[18]} = \left\{ 1 - \frac{e(k \cdot x2)(\gamma \cdot k)(\gamma \cdot k)}{2(f \cdot k)}, -\frac{a^2 e^2 (k \cdot x2)^3}{6(f \cdot k)} - \frac{e(a \cdot f)(k \cdot x2)^2}{2(f \cdot k)} + f \cdot x2 \right\}$$

$$\text{Out[19]} = \left\{ \frac{e(k \cdot x2)(\gamma \cdot k)(\gamma \cdot a)}{2(g \cdot k)} + 1, -\frac{a^2 e^2 (k \cdot x2)^3}{6(g \cdot k)} + \frac{e(a \cdot g)(k \cdot x2)^2}{2(g \cdot k)} + g \cdot x2 \right\}$$

The dressed vertex (with an outgoing photon line)

$$\Gamma^\mu(l, q, p) = \int d^4x \bar{E}_q(x) \gamma^\mu E_p(x) e^{i l x};$$

x^μ – position of the vertex;

p^μ, q^μ – initial and final electron momenta;

l^μ, q^μ – momentum of the outgoing photon

$-\infty < s < \infty$ – measure for the momentum exchanged

with the field (may be viewed as the 'photon number')

$$\Gamma^\mu(l, q, p) = \int ds (2\pi)^4 \delta(p + sk - q - l) \tilde{\Gamma}^\mu(s, q, p)$$

$$\tilde{\Gamma}^\mu(s, q, p) = \gamma^\mu A_0(s, q, p) -$$

$$\frac{e}{2} \left(\frac{(\gamma a)(\gamma k) \gamma^\mu}{kq} + \frac{\gamma^\mu (\gamma k)(\gamma a)}{kp} \right) A_1(s, q, p) - \frac{e^2 a^2 k^\mu (\gamma k)}{2 kq kp} A_2(s, q, p)$$

$$A_0(s, q, p) = \int \frac{d\phi}{2\pi} e^{is\phi} e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} = \frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \text{Ai}(y)$$

$$A_1(s, q, p) = -i \frac{dA_0}{ds} = \int \frac{d\phi}{2\pi} e^{is\phi} \phi e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} =$$

$$\frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \left\{ \frac{\alpha}{8\beta} \text{Ai}(y) - i \frac{1}{(4\beta)^{1/3}} \text{Ai}'(y) \right\}$$

$$A_2(s, q, p) = (-i)^2 \frac{d^2 A_0}{ds^2} = \int \frac{d\phi}{2\pi} e^{is\phi} \phi^2 e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} =$$

$$\frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \left\{ \left[2\left(\frac{\alpha}{8\beta}\right)^2 - \frac{s}{4\beta}\right] \text{Ai}(y) - i \frac{\alpha}{(4\beta)^{4/3}} \text{Ai}'(y) \right\}$$

where

$$\alpha(q, p) = -e \left(\frac{ap}{kp} - \frac{aq}{kq} \right)$$

$$\beta(q, p) = \frac{e^2 a^2}{8} \left(\frac{1}{kp} - \frac{1}{kq} \right)$$

The amplitude expressed via $\tilde{\Gamma}^\mu$

$$S_{fi}^{\text{dir}} = (ie)^2 \int d^4x_1 d^4x_2 \bar{\psi}_q(x_1) \gamma^\mu \psi_p(x_1) D_{\mu\nu}^c(x_1 - x_2) \bar{\psi}_f(x_2) \gamma^\nu \psi_g(x_2)$$

$$= (ie)^2 \int d^4x_1 d^4x_2 \frac{d^4l}{(2\pi)^4} \frac{-i}{l^2 + i0} g_{\mu\nu}$$

$$\left(\bar{u}_{q\lambda_2} \bar{E}_q(x_1) \gamma^\mu E_p(x_1) u_{p\lambda_1} e^{ilx_1} \right) \left(\bar{u}_{f\sigma_2} \bar{E}_f(x_2) \gamma^\mu E_g(x_2) v_{g\sigma_1} e^{-ilx_2} \right)$$

$$= ie^2 \int \frac{d^4l}{(2\pi)^4} \frac{g_{\mu\nu}}{l^2 + i0} \left(\bar{u}_{q\lambda_2} \int d^4s_1 (2\pi)^4 \delta(p + s_1 k - q - l) \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right)$$

$$\left(\bar{u}_{f\sigma_2} \int d^4s_2 (2\pi)^4 \delta(l + s_2 k - f - g) \tilde{\Gamma}^\mu(s_2, g, -f) v_{g\sigma_1} \right)$$

$$= ie^2 \int d^4s_1 d^4s_2 (2\pi)^4 \delta(p + (s_1 + s_2)k - q - f - g)$$

$$\frac{g_{\mu\nu}}{(p + s_1 k - q)^2 + i0} \left(\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right) \left(\bar{u}_{f\sigma_2} \tilde{\Gamma}^\mu(s_2, g, -f) v_{g\sigma_1} \right)$$

Let us introduce partial amplitude $m(s_1, s_2)$

(curly M in [arXiv 2605.12479], see Eq. (6.1), note that here $f \leftrightarrow g$)

$$S_{fi}^{\text{dir}} = \int ds_1 ds_2 i m(s_1, s_2) (2\pi)^4 \delta(g + f + q - p - (s_1 + s_2)k)$$

$$i m(s_1, s_2) = ie^2 \left[\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right] \frac{g_{\mu\nu}}{l^2 + i0} \left[\bar{u}_{f\sigma_2} \tilde{\Gamma}^\nu(s_2, f, -g) v_{g\sigma_1} \right]$$

$$m(s_1, s_2) = e^2 \frac{1}{l^2 + i0} \left[\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right] \left[\bar{u}_{g\sigma_2} \tilde{\Gamma}_\mu(s_2, f, -g) v_{g\sigma_1} \right]$$

$$m^+(t_1, t_2) =$$

$$e^2 \frac{1}{l'^2 - i0} \left[\bar{u}_{p\lambda_1} \left(\tilde{\Gamma}^\mu(t_1, q, p) \right)^+ u_{q\lambda_2} \right] \left[\bar{v}_{g\sigma_1} \left(\tilde{\Gamma}_\mu(t_2, f, -g) \right)^+ u_{f\sigma_2} \right]$$

$$l = p + s_1 k - q$$

$$l' = p + t_1 k - q$$

$$\sum u_{p\lambda}^{\alpha} \bar{u}_{p\lambda}^{\beta} = (p\gamma + m)^{\alpha\beta}$$

$$\sum v_{p\lambda}^{\alpha} \bar{v}_{p\lambda}^{\beta} = (p\gamma - m)^{\alpha\beta}$$

$$dW = \frac{dw}{VT 2 p^0};$$

$$\frac{1}{2} \sum_{\lambda\sigma} - \text{averaging over the initial electron spin states}$$

and summation over the final particle spin projections

$$\frac{1}{2} - \text{a symmetry factor to account for the two identical final electrons}$$

$$dw = \frac{1}{2} \times \frac{1}{2} \sum_{\lambda\sigma} \frac{d^3 f}{(2\pi)^3 2 f^0} \frac{d^3 g}{(2\pi)^3 2 g^0} \frac{d^3 q}{(2\pi)^3 2 q^0} (|S_{fi}^{dir}|)^2;$$

$$(|S_{fi}^{dir}|)^2 = \frac{VT}{L_{\phi}} \int ds_1 ds_2 dt_1 dt_2 M(s_1, s_2, t_1, t_2)$$

$$(2\pi)^4 \delta(g + f + q - p - (s_1 + s_2)k) (2\pi) \delta(s_1 + s_2 - t_1 - t_2)$$

$$M(s_1, s_2, t_1, t_2) = m(s_1, s_2) m^+(t_1, t_2)$$

Steps 1-2: Integrations using δ -functions

Step 1 : integrate over $d^3 f$ using δ - function

$$\int \frac{d^3 f}{(2\pi)^3 2 f^0} (2\pi)^4 \delta(g + f + q - p - (s_1 + s_2)k) =$$

$$\frac{2\pi}{2 f^0} \delta(g^0 + f^0 + q^0 - p^0 - (s_1 + s_2)k^0)$$

$\delta(g + f + q - p - (s_1 + s_2)k)$ also implies the conservation of :

$$kg + kf + kq - kp = 0, \quad g_{\perp} + f_{\perp} + q_{\perp} - p_{\perp} = 0$$

$$\text{Consider } g^0 + f^0 + q^0 - p^0 - (s_1 + s_2)k^0 = 0$$

$$\text{By substituting } f^0 = \sqrt{m^2 + \left(\vec{f}\right)^2} = \sqrt{m^2 + \left(\vec{p} + \tilde{s} \vec{k} - \vec{g} - \vec{q}\right)^2}, \text{ we get}$$

$$\sqrt{m^2 + \left(\vec{p} + (s_1 + s_2) \vec{k} - \vec{g} - \vec{q}\right)^2} = p^0 + (s_1 + s_2)k^0 - g^0 - q^0$$

If we square this expression and rearrange terms, we will arrive at

$$s_1 + s_2 = \tilde{s} = \frac{(pq + pg - qg - m^2)}{kp - kq - kg} = \frac{(pq + pg - qg - m^2)}{kf}$$

Hence,

$$\delta (g^0 + f^0 + q^0 - p^0 - (s_1 + s_2) k^0) = \delta (F(s_1 + s_2)) = \frac{\delta (s_1 + s_2 - \tilde{s})}{|F'(\tilde{s})|}$$

$$\text{where } F(s) = p^0 + s k^0 - g^0 - q^0 - \sqrt{m^2 + (\vec{p} + s \vec{k} - \vec{g} - \vec{q})^2}$$

By taking the derivative and substituting

$$\text{back } \sqrt{m^2 + (\vec{p} + \tilde{s} \vec{k} - \vec{g} - \vec{q})^2} = (p^0 + \tilde{s} k^0 - g^0 - q^0) = f^0,$$

it is straightforward to show that

$$F'(\tilde{s}) = \frac{(kf)}{f^0} = \frac{m^2 \chi_f}{\xi f^0}$$

hence

$$\delta (g^0 + f^0 + q^0 - p^0 - \tilde{s} k^0) = \frac{\xi f^0}{m^2 \chi_f} \delta (s_1 + s_2 - \tilde{s})$$

$$\int \frac{d^3 f}{(2\pi)^3 2 f^0} (2\pi)^4 \delta (g + f + q - p - (s_1 + s_2) k) = \frac{\pi \xi}{m^2 \chi_f} \delta (s_1 + s_2 - \tilde{s})$$

Step 2 : integrate over s_2, t_2

$$\begin{aligned} & \int ds_1 ds_2 dt_1 dt_2 (2\pi) \delta \\ & (s_1 + s_2 - t_1 - t_2) \frac{\pi \xi}{m^2 \chi_f} \delta (s_1 + s_2 - \tilde{s}) M(s_1, s_2, t_1, t_2) \\ &= \frac{2\pi^2 \xi}{m^2 \chi_f} \int ds_1 dt_1 dt_2 \delta (\tilde{s} - t_1 - t_2) M(s_1, \tilde{s} - s_1, t_1, t_2) \\ &= \frac{2\pi^2 \xi}{m^2 \chi_f} \int ds dt M(s_1, \tilde{s} - s, t, \tilde{s} - t), \text{ where } s = s_1, t = t_1 \end{aligned}$$

In the end, we are calculating :

$$dW = \frac{1}{2} \times \frac{1}{2} \frac{1}{2 p^0 L_\phi} \frac{2\pi^2 \xi}{(2\pi)^6 2 g^0 2 q^0 m^2 \chi_f} \sum_{\lambda\sigma} d^3 g d^3 q ds dt M(s, \tilde{s} - s, t, \tilde{s} - t)$$

$$M(s, \tilde{s} - s, t, \tilde{s} - t) = m(s, \tilde{s} - s) m^+(t, \tilde{s} - t)$$

$$m(s, \tilde{s} - s) = e^2 \frac{1}{l^2 + i0} [\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s, q, p) u_{p\lambda_1}] [\bar{u}_{g\sigma_2} \tilde{\Gamma}_\mu(\tilde{s} - s, f, -g) v_{g\sigma_1}]$$

$$m^+(t, \tilde{s} - t) =$$

$$e^2 \frac{1}{l'^2 - i0} [\bar{u}_{p\lambda_1} (\tilde{\Gamma}^{\mu 1}(t, q, p))^+ u_{q\lambda_2}] [\bar{v}_{g\sigma_1} (\tilde{\Gamma}_{\mu 1}(\tilde{s} - t, f, -g))^+ u_{f\sigma_2}]$$

$$l = p + s k - q$$

$$\mathbf{l}' = \mathbf{p} + \mathbf{t} \mathbf{k} - \mathbf{q}$$

Step 3: Spin sums

For technical convenience,

we first define the operation for a dummy matrix element mod squared

$$\left[\bar{u}_{q\lambda_2} \gamma^\mu u_{p\lambda_1} \right] \left[\bar{u}_{g\sigma_2} \gamma^\nu v_{g\sigma_1} \right] g_{\mu\nu} \left[\bar{u}_{p\lambda_1} \gamma^{\mu 1} u_{q\lambda_2} \right] \left[\bar{v}_{g\sigma_1} \gamma^{\nu 1} u_{f\sigma_2} \right] g_{\mu 1 \nu 1},$$

then perform the formal spin summation, and after this we substitute

$$\gamma^\mu \rightarrow \tilde{\Gamma}^\mu (s, q, p)$$

$$\gamma^\nu \rightarrow \tilde{\Gamma}^\nu (\tilde{s} - s, f, -g)$$

$$\gamma^{\mu 1} \rightarrow \left(\tilde{\Gamma}^{\mu 1} (t, q, p) \right)^+$$

$$\gamma^{\nu 1} \rightarrow \left(\tilde{\Gamma}^{\nu 1} (\tilde{s} - t, f, -g) \right)^+$$

```
In[20]:= DummyExpr = (SpinorUBarD[q, m].GAD[μ].SpinorUD[p, m])
              (SpinorUBarD[f, m].GAD[ν].SpinorVD[g, m])
DummyExprC =
  ComplexConjugate[DummyExpr, FCRenameDummyIndices → False] /. {μ → μ1, ν → ν1}
```

```
Out[20]=  $\bar{u}(q, m) \cdot \gamma^\mu \cdot u(p, m) \bar{u}(f, m) \cdot \gamma^\nu \cdot v(g, m)$ 
```

```
Out[21]=  $(\varphi(-g, m)) \cdot \gamma^{\nu 1} \cdot (\varphi(f, m)) (\varphi(p, m)) \cdot \gamma^{\mu 1} \cdot (\varphi(q, m))$ 
```

```
In[22]:= DummyExpr1 = FermionSpinSum[DummyExpr DummyExprC]
```

```
Out[22]=  $\text{tr}((\gamma \cdot f + m) \cdot \gamma^\nu \cdot (\gamma \cdot g - m) \cdot \gamma^{\nu 1}) \text{tr}((m + \gamma \cdot p) \cdot \gamma^{\mu 1} \cdot (m + \gamma \cdot q) \cdot \gamma^\mu)$ 
```

We define

$$\tilde{\Gamma}^\mu (s, q, p) = \gamma^\mu A_0 (s, q, p) - \frac{e}{2} \left(\frac{(\gamma a) (\gamma k) \gamma^\mu}{kq} + \frac{\gamma^\mu (\gamma k) (\gamma a)}{kp} \right) A_1 (s, q, p) - \frac{e^2 a^2 k^\mu (\gamma k)}{2 kq kp} A_2 (s, q, p)$$

$$\left(\tilde{\Gamma}^\mu (t, q, p) \right)^+ = \gamma^\mu B_0 (t, q, p) - \frac{e}{2} \left(\frac{\gamma^\mu (\gamma k) (\gamma a)}{kq} + \frac{(\gamma a) (\gamma k) \gamma^\mu}{kp} \right) B_1 (t, q, p) - \frac{e^2 a^2 k^\mu (\gamma k)}{2 kq kp} B_2 (t, q, p)$$

$$B_n = (A_n)^*$$

It what follows,

our goal is to integrate the expression for the trident rate;

This involves some variable changes

in the integrand and algebraic simplifications;

To track the calculation, we split the expression into variables

- Expr - the integrand
- Coeff -
the overall coefficient (also dependent on the integration variables)
- Diffs - the integration measure to track which integrations are left, namely

$$W = \int [\text{Diffs}] * \text{Coeff} * \text{Expr}$$

We will be modifying there three variables,
in each step adding a number to the name (DiffsN, CoeffN, ExprN)

Commands $\tilde{\Gamma}$ and $\tilde{\Gamma}^C$ are introduced
in definitions.nb in accordance with formulas for
 $\tilde{\Gamma}^\mu(s, q, p)$ and $(\tilde{\Gamma}^\mu(t, q, p))^+$, respectively;
chif[p] is also a handy command introduced in definitions.nb,
it returns symbol χp

```
In[23]:= Diffs = ds dt d3q d3g
Coeff = (1/2) (1/2) (1/Lphi/2/p0) 2 pi^2 xi / ((2 pi)^6 2 g0 2 q0 m^2 chif[f])
        e^4 / (l1v2 + i epsilon) / (l2v2 - i epsilon) /. {e^4 -> (4 pi alpha)^2}
Expr = DummyExpr1 /.
  {DiracGamma[LorentzIndex[mu, D], D] -> rtilde[mu, s, q, p],
   DiracGamma[LorentzIndex[nu, D], D] -> rtilde[mu, stilde - s, f, -g],
   DiracGamma[LorentzIndex[mu1, D], D] -> rtildeC[mu1, t, q, p],
   DiracGamma[LorentzIndex[nu1, D], D] -> rtildeC[mu1, stilde - t, f, -g]}
```

Out[23]= d3g d3q ds dt

$$\text{Out[24]= } \frac{\alpha^2 \xi}{64 \pi^2 g_0 L \phi m^2 p_0 q_0 \chi f (i\epsilon + l_1^2) (l_2^2 - i\epsilon)}$$

$$\begin{aligned} \text{Out[25]= } & \text{tr} \left((m + \gamma \cdot p) \left(-\frac{a^2 e^2 k^{\mu 1} \gamma \cdot k B_2(t, q, p)}{2 (k \cdot p) (k \cdot q)} - \frac{1}{2} e B_1(t, q, p) \left(\frac{(\gamma \cdot a) (\gamma \cdot k) \gamma^{\mu 1}}{k \cdot p} + \frac{\gamma^{\mu 1} (\gamma \cdot k) (\gamma \cdot a)}{k \cdot q} \right) + \gamma^{\mu 1} B_0(t, q, p) \right) (m + \gamma \cdot q) \right. \\ & \left. \left(-\frac{a^2 e^2 k^\mu \gamma \cdot k A_2(s, q, p)}{2 (k \cdot p) (k \cdot q)} - \frac{1}{2} e A_1(s, q, p) \left(\frac{\gamma^\mu (\gamma \cdot k) (\gamma \cdot a)}{k \cdot p} + \frac{(\gamma \cdot a) (\gamma \cdot k) \gamma^\mu}{k \cdot q} \right) + \gamma^\mu A_0(s, q, p) \right) \right) \\ & \text{tr} \left((\gamma \cdot f + m) \left(\frac{a^2 e^2 k^\mu \gamma \cdot k A_2(\text{stilde} - s, f, -g)}{2 (f \cdot k) (g \cdot k)} - \frac{1}{2} e A_1(\text{stilde} - s, f, -g) \left(\frac{(\gamma \cdot a) (\gamma \cdot k) \gamma^\mu}{f \cdot k} - \frac{\gamma^\mu (\gamma \cdot k) (\gamma \cdot a)}{g \cdot k} \right) + \right. \right. \\ & \left. \left. \gamma^\mu A_0(\text{stilde} - s, f, -g) \right) (\gamma \cdot g - m) \left(\frac{a^2 e^2 k^{\mu 1} \gamma \cdot k B_2(\text{stilde} - t, f, -g)}{2 (f \cdot k) (g \cdot k)} - \right. \right. \\ & \left. \left. \frac{1}{2} e B_1(\text{stilde} - t, f, -g) \left(\frac{\gamma^{\mu 1} (\gamma \cdot k) (\gamma \cdot a)}{f \cdot k} - \frac{(\gamma \cdot a) (\gamma \cdot k) \gamma^{\mu 1}}{g \cdot k} \right) + \gamma^{\mu 1} B_0(\text{stilde} - t, f, -g) \right) \right) \end{aligned}$$

```
In[26]:= Diffs1 = Diffs;
Coeff1 = Coeff;
```

The following step is saved as a pre - evaluated expression stored in TridentDiracSimplify.nb. To perform the actual computation, uncomment commented line. The calculation takes several minutes

```
In[28]:= (*Expr1=DiracSimplify[Expr];*)
NotebookEvaluate[
  NotebookOpen[NotebookDirectory[] <> "TridentDiracSimplify.nb"]];
```

Step 4: Rearranging terms in Expr and further simplifications

SortTerms and SortTermsSimplify are two handy commands that collect terms in Expr

```
In[29]:= SortTerms[X_] := Collect[X, {A0 B0, A0 B1, A0 B2, A1 B0, A1 B1, A1 B2, A2 B0,
  A2 B1, A2 B2, C0 D0, C0 D1, C0 C2, C1 D0, C1 D1, C1 C2, C2 D0, C2 D1, C2 C2}]
SortTermsSimplify[X_] :=
  Collect[X, {A0 B0, A0 B1, A0 B2, A1 B0, A1 B1, A1 B2, A2 B0, A2 B1, A2 B2,
    C0 D0, C0 D1, C0 C2, C1 D0, C1 D1, C1 C2, C2 D0, C2 D1, C2 C2}, Simplify]
```

For brevity, we substitute A_n and B_n functions :

$$A_n[s, q, p] \rightarrow A_n$$

$$B_n[t, q, p] \rightarrow B_n$$

$$A_n[\tilde{s} - s, f, -g] \rightarrow C_n$$

$$B_n[\tilde{s} - t, f, -g] \rightarrow D_n$$

Then terms in the expressions are collected according to the rules introduced in the previous input cell

```
In[31]:= Diffs2 = Diffs1;
Coeff2 = Coeff1;
Expr1 /. {A0[s, q, p] → A0, A1[s, q, p] → A1, A2[s, q, p] → A2,
  B0[t, q, p] → B0, B1[t, q, p] → B1, B2[t, q, p] → B2,
  A0[stilde - s, f, -g] → C0,
  A1[stilde - s, f, -g] → C1, A2[stilde - s, f, -g] → C2,
  B0[stilde - t, f, -g] → D0, B1[stilde - t, f, -g] → D1, B2[stilde - t, f, -g] → D2};
Expr2 = SortTerms[%];
Expr2 /. {D → 4}
```

$$\text{Out[35]} = \frac{16 A_2 B_2 C_0 D_0 \bar{a}^4 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^4}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} + A_1 B_2 \left(C_1 D_0 \left(\frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} + \frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} \right) + \right. \\ \left. C_0 D_0 \left(-\frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} - \right. \right. \\ \left. \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right) \right) +$$

$$\begin{aligned}
& \text{A2 B1} \left(\text{C0 D1} \left(\frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} + \frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} \right) + \right. \\
& \quad \text{C0 D0} \left(-\frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} - \right. \\
& \quad \quad \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right) \Bigg) + \\
& \quad \text{A0 B0} \left(\frac{16 \text{C2 D2 } \bar{a}^4 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^4}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{8 \text{C1 D2 } \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \text{C1 D2 } \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \right. \\
& \quad \frac{16 \text{C1 D2 } \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \frac{16 \text{C1 D2 } \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{8 \text{C1 D2 } \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \\
& \quad \frac{8 \text{C1 D2 } \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + 16 \text{C0 D2 } m^2 \bar{a}^2 e^2 + \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \\
& \quad \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} + \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} + \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} - \\
& \quad \frac{16 \text{C0 D2 } m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \frac{16 \text{C0 D2 } \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - 16 \text{C0 D2 } \bar{a}^2 (\bar{p} \cdot \bar{q}) e^2 + \\
& \quad \text{C1 D0} \left(16 e (\bar{a} \cdot \bar{f}) m^2 - 16 e (\bar{a} \cdot \bar{g}) m^2 + \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) m^2}{\bar{f} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) m^2}{\bar{g} \cdot \bar{k}} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) - \right. \\
& \quad 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{f} \cdot \bar{k}} + \\
& \quad \left. \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{g} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{f} \cdot \bar{k}} + \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{g} \cdot \bar{k}} \right) + \\
& \quad \text{C0 D1} \left(16 e (\bar{a} \cdot \bar{f}) m^2 - 16 e (\bar{a} \cdot \bar{g}) m^2 + \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) m^2}{\bar{f} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) m^2}{\bar{g} \cdot \bar{k}} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) - \right. \\
& \quad 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{f} \cdot \bar{k}} + \\
& \quad \left. \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{g} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{f} \cdot \bar{k}} + \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{g} \cdot \bar{k}} \right) + \\
& \quad \text{C2 D1} \left(\frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \right. \\
& \quad \left. \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} \right) + \\
& \quad \text{C0 D0} (-64 m^4 - 32 (\bar{f} \cdot \bar{g}) m^2 + 32 (\bar{p} \cdot \bar{q}) m^2 + 32 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 32 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{q})) + \\
& \quad \text{C2 D0} \left(16 m^2 \bar{a}^2 e^2 + \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} + \right.
\end{aligned}$$

$$\begin{aligned}
& \frac{8 \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} - \frac{16 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - 16 \bar{a}^2 (\bar{p} \cdot \bar{q}) e^2 \Bigg) + \\
\text{C1 D1} & \left(-64 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) e^2 + \frac{16 m^2 \bar{a}^2 (\bar{g} \cdot \bar{k}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} - \right. \\
& \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} + \frac{32 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \frac{32 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} - \\
& \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} + \frac{32 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \\
& \frac{64 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{32 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{32 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} + \\
& \left. \frac{32 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} + 32 \bar{a}^2 (\bar{p} \cdot \bar{q}) e^2 + \frac{16 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) e^2}{\bar{g} \cdot \bar{k}} \right) \Bigg) + \\
\text{A0 B1} & \left(\text{C2 D1} \left(\frac{8 \bar{a}^4 (\bar{k} \cdot \bar{p}) e^4}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^4 (\bar{k} \cdot \bar{p}) e^4}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^4 (\bar{k} \cdot \bar{q}) e^4}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^4 (\bar{k} \cdot \bar{q}) e^4}{\bar{g} \cdot \bar{k}} \right) + \right. \\
\text{C2 D0} & \left(-16 \bar{a}^2 (\bar{a} \cdot \bar{p}) e^3 - 16 \bar{a}^2 (\bar{a} \cdot \bar{q}) e^3 + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} + \right. \\
& \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} \right) + \\
\text{C0 D0} & \left(16 e (\bar{a} \cdot \bar{p}) m^2 + 16 e (\bar{a} \cdot \bar{q}) m^2 - \frac{16 e (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) m^2}{\bar{k} \cdot \bar{p}} - \frac{16 e (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) m^2}{\bar{k} \cdot \bar{q}} + 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) + \right. \\
& 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) - \frac{16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{k})}{\bar{k} \cdot \bar{p}} - \\
& \left. \frac{16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{q})}{\bar{k} \cdot \bar{p}} - \frac{16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{k})}{\bar{k} \cdot \bar{q}} - \frac{16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{p})}{\bar{k} \cdot \bar{q}} \right) + \\
\text{C1 D1} & \left(-16 \bar{a}^2 (\bar{a} \cdot \bar{p}) e^3 - 16 \bar{a}^2 (\bar{a} \cdot \bar{q}) e^3 + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \right. \\
& \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \\
& \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} + \\
& \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{p}) e^3}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} \right) + \text{C0 D1} \\
& \left(-8 \bar{a}^2 (\bar{f} \cdot \bar{p}) e^2 - 8 \bar{a}^2 (\bar{f} \cdot \bar{q}) e^2 + 8 \bar{a}^2 (\bar{g} \cdot \bar{p}) e^2 + 8 \bar{a}^2 (\bar{g} \cdot \bar{q}) e^2 - \frac{8 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \right. \\
& \left. \frac{8 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} - \frac{8 m^2 \bar{a}^2 (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} + \frac{8 m^2 \bar{a}^2 (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} + \right.
\end{aligned}$$

[illegible]

$$\begin{aligned}
& \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} + \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{p}) e^2}{\bar{k} \cdot \bar{q}} + \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{q}} - \\
& \left. \frac{16 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) e^2}{\bar{k} \cdot \bar{q}} + \frac{32 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^2}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} + \frac{64 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^2}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right) \Bigg) + \\
& A0 B2 \left(-16 C2 D0 \bar{a}^4 e^4 + C1 D0 \left(-16 \bar{a}^2 (\bar{a} \cdot \bar{f}) e^3 + 16 \bar{a}^2 (\bar{a} \cdot \bar{g}) e^3 + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \right. \right. \\
& \left. \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} \right) + \right. \\
& C0 D0 \left(16 m^2 \bar{a}^2 e^2 + 16 \bar{a}^2 (\bar{f} \cdot \bar{g}) e^2 + \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) (\bar{p} \cdot \bar{q}) e^2}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{p}} - \right. \\
& \left. \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{p}) e^2}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{q}} - \frac{16 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^2}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right) \Bigg) + \\
& A2 B0 \left(-16 C0 D2 \bar{a}^4 e^4 + C0 D1 \left(-16 \bar{a}^2 (\bar{a} \cdot \bar{f}) e^3 + 16 \bar{a}^2 (\bar{a} \cdot \bar{g}) e^3 + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \right. \right. \\
& \left. \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} \right) + \right. \\
& C0 D0 \left(16 m^2 \bar{a}^2 e^2 + 16 \bar{a}^2 (\bar{f} \cdot \bar{g}) e^2 + \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) (\bar{p} \cdot \bar{q}) e^2}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{p}} - \right. \\
& \left. \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{p}) e^2}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{q}} - \frac{16 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^2}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right) \Bigg)
\end{aligned}$$

To simplify further, contractions of momenta with k ,
such as (kp) , are replaced by χ - parameters

$$(kp) = \frac{m^2 \chi_p}{\xi}$$

ξ is defined from

$$\xi^2 = -\frac{e^2 a^2}{m^2}$$

Function `ktochiSubs` is defined in `Definitons`,
it generates substitutions of scalar products (kp) , (kq) , etc

In[36]:= `kp /. ktochiSubs[p, q, f, g]`

Out[36]= $\frac{m^2 \chi_p}{\xi}$

In[37]:= **Diff3 = Diff2;**

Coeff3 = Coeff2;

Expr3 = Expr2 /. ktociSubs[p, q, f, g] /. {av2 → -m² ξ² / e²};

% /. {D → 4}

$$\begin{aligned}
 \text{Out[40]} = & \frac{16 A2 B2 C0 D0 m^4 \chi f \chi g \xi^4}{\chi p \chi q} + A1 B2 \\
 & \left(C1 D0 \left(\frac{8 m^4 \chi f \xi^4}{\chi p} - \frac{8 m^4 \chi g \xi^4}{\chi p} + \frac{8 m^4 \chi f \xi^4}{\chi q} - \frac{8 m^4 \chi g \xi^4}{\chi q} \right) + C0 D0 \left(\frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi q} + \right. \right. \\
 & \quad \left. \left. \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi p} + \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p \chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi p \chi q} \right) \right) + A2 B1 \\
 & \left(C0 D1 \left(\frac{8 m^4 \chi f \xi^4}{\chi p} - \frac{8 m^4 \chi g \xi^4}{\chi p} + \frac{8 m^4 \chi f \xi^4}{\chi q} - \frac{8 m^4 \chi g \xi^4}{\chi q} \right) + C0 D0 \left(\frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi q} + \right. \right. \\
 & \quad \left. \left. \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi p} + \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p \chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi p \chi q} \right) \right) + \\
 & A0 B0 \left(-16 C0 D2 \xi^2 m^4 + \frac{16 C2 D2 \xi^4 \chi p \chi q m^4}{\chi f \chi g} + \frac{16 C0 D2 \xi^2 \chi p \chi q m^4}{\chi f \chi g} - \frac{16 C1 D2 e \xi^2 \chi p \chi q (\bar{a} \cdot \bar{f}) m^2}{\chi f \chi g} + \right. \\
 & \quad \frac{16 C1 D2 e \xi^2 \chi p \chi q (\bar{a} \cdot \bar{g}) m^2}{\chi f \chi g} - \frac{8 C1 D2 e \xi^2 \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi f} + \frac{8 C1 D2 e \xi^2 \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi g} - \\
 & \quad \frac{8 C1 D2 e \xi^2 \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi f} + \frac{8 C1 D2 e \xi^2 \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi g} + \frac{16 C0 D2 \xi^2 \chi p \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f \chi g} - \\
 & \quad \frac{8 C0 D2 \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f} - \frac{8 C0 D2 \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f} - \frac{8 C0 D2 \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g} - \frac{8 C0 D2 \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g} + \\
 & \quad 16 C0 D2 \xi^2 (\bar{p} \cdot \bar{q}) m^2 + C2 D1 \left(-\frac{16 e m^2 \chi p \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi g} + \frac{16 e m^2 \chi p \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi f \chi g} - \right. \\
 & \quad \left. \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi f} + \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi g} - \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi f} + \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi g} \right) + \\
 & C1 D0 \left(16 e (\bar{a} \cdot \bar{f}) m^2 + \frac{16 e \chi g (\bar{a} \cdot \bar{f}) m^2}{\chi f} - 16 e (\bar{a} \cdot \bar{g}) m^2 - \frac{16 e \chi f (\bar{a} \cdot \bar{g}) m^2}{\chi g} + \right. \\
 & \quad \frac{16 e \chi q (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p})}{\chi g} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) + \frac{16 e \chi p (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q})}{\chi g} - 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) - \\
 & \quad \left. \frac{16 e \chi q (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p})}{\chi f} + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi p (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q})}{\chi f} + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) \right) + \\
 & C0 D1 \left(16 e (\bar{a} \cdot \bar{f}) m^2 + \frac{16 e \chi g (\bar{a} \cdot \bar{f}) m^2}{\chi f} - 16 e (\bar{a} \cdot \bar{g}) m^2 - \frac{16 e \chi f (\bar{a} \cdot \bar{g}) m^2}{\chi g} + \right. \\
 & \quad \frac{16 e \chi q (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p})}{\chi g} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) + \frac{16 e \chi p (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q})}{\chi g} - 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) - \\
 & \quad \left. \frac{16 e \chi q (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p})}{\chi f} + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi p (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q})}{\chi f} + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) \right) +
 \end{aligned}$$

$$\begin{aligned}
& \text{C0 D0} (-64 m^4 - 32 (\bar{f} \cdot \bar{g}) m^2 + 32 (\bar{p} \cdot \bar{q}) m^2 + 32 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 32 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{q})) + \\
& \text{C1 D1} \left(-\frac{16 \xi^2 \chi g m^4}{\chi f} - \frac{32 \xi^2 \chi p \chi q m^4}{\chi f \chi g} - \frac{16 \xi^2 \chi f m^4}{\chi g} - \frac{32 \xi^2 \chi p \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f \chi g} + \right. \\
& \quad \frac{16 \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f} + \frac{16 \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi g} + \frac{16 \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f} + \frac{16 \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi g} + \\
& \quad \frac{16 \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi f} + \frac{16 \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g} + \frac{16 \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi f} + \frac{16 \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g} - \\
& \quad 32 \xi^2 (\bar{p} \cdot \bar{q}) m^2 - \frac{64 e^2 \chi p \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi f \chi g} + \frac{32 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p})}{\chi f} + \frac{32 e^2 \chi q (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p})}{\chi g} + \\
& \quad \left. \frac{32 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q})}{\chi f} + \frac{32 e^2 \chi p (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q})}{\chi g} - 64 e^2 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) \right) + \\
& \text{C2 D0} \left(-16 \xi^2 m^4 + \frac{16 \xi^2 \chi p \chi q m^4}{\chi f \chi g} + \frac{16 \xi^2 \chi p \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f \chi g} - \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f} - \right. \\
& \quad \left. \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f} - \frac{8 \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g} - \frac{8 \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g} + 16 \xi^2 (\bar{p} \cdot \bar{q}) m^2 \right) + \\
& \text{A1 B0} \left(\frac{8 \text{C1 D2 } m^4 \chi p \xi^4}{\chi f} - \frac{8 \text{C1 D2 } m^4 \chi p \xi^4}{\chi g} + \frac{8 \text{C1 D2 } m^4 \chi q \xi^4}{\chi f} - \frac{8 \text{C1 D2 } m^4 \chi q \xi^4}{\chi g} - \right. \\
& \quad \frac{8 \text{C0 D2 } e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} - \frac{8 \text{C0 D2 } e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} - \frac{8 \text{C0 D2 } e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} - \\
& \quad \frac{8 \text{C0 D2 } e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + 16 \text{C0 D2 } e m^2 (\bar{a} \cdot \bar{p}) \xi^2 + 16 \text{C0 D2 } e m^2 (\bar{a} \cdot \bar{q}) \xi^2 + \\
& \text{C1 D1} \left(-\frac{8 e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} + \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi g} - \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} + \right. \\
& \quad \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi g} + \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi f} - \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi f} - \\
& \quad \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + 16 e m^2 (\bar{a} \cdot \bar{p}) \xi^2 - \frac{8 e m^2 \chi g \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi f \chi p} - \frac{8 e m^2 \chi f \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi g \chi p} + \\
& \quad \left. 16 e m^2 (\bar{a} \cdot \bar{q}) \xi^2 - \frac{8 e m^2 \chi g \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi f \chi q} - \frac{8 e m^2 \chi f \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi g \chi q} \right) + \\
& \text{C0 D0} \left(16 e (\bar{a} \cdot \bar{p}) m^2 - \frac{16 e \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi p} + 16 e (\bar{a} \cdot \bar{q}) m^2 - \frac{16 e \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& \quad 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) - \frac{16 e \chi g (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) - \frac{16 e \chi g (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q})}{\chi p} + \\
& \quad \left. 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi f (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) - \frac{16 e \chi f (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q})}{\chi p} \right) + \\
& \text{C0 D1} \left(-\frac{8 \xi^2 \chi p m^4}{\chi f} + \frac{8 \xi^2 \chi p m^4}{\chi g} - \frac{8 \xi^2 \chi q m^4}{\chi f} + \frac{8 \xi^2 \chi q m^4}{\chi g} - \frac{8 \xi^2 \chi f m^4}{\chi p} + \frac{8 \xi^2 \chi g m^4}{\chi p} - \right.
\end{aligned}$$

$$\begin{aligned}
& \frac{8\xi^2\chi f m^4}{\chi q} + \frac{8\xi^2\chi g m^4}{\chi q} - \frac{8\xi^2\chi p(\bar{f}\cdot\bar{g})m^2}{\chi f} + \frac{8\xi^2\chi p(\bar{f}\cdot\bar{g})m^2}{\chi g} - \frac{8\xi^2\chi q(\bar{f}\cdot\bar{g})m^2}{\chi f} + \\
& \frac{8\xi^2\chi q(\bar{f}\cdot\bar{g})m^2}{\chi g} + 8\xi^2(\bar{f}\cdot\bar{p})m^2 + \frac{8\xi^2\chi g\chi q(\bar{f}\cdot\bar{p})m^2}{\chi f\chi p} + 8\xi^2(\bar{f}\cdot\bar{q})m^2 + \frac{8\xi^2\chi g\chi p(\bar{f}\cdot\bar{q})m^2}{\chi f\chi q} - \\
& 8\xi^2(\bar{g}\cdot\bar{p})m^2 - \frac{8\xi^2\chi f\chi q(\bar{g}\cdot\bar{p})m^2}{\chi g\chi p} - 8\xi^2(\bar{g}\cdot\bar{q})m^2 - \frac{8\xi^2\chi f\chi p(\bar{g}\cdot\bar{q})m^2}{\chi g\chi q} + \\
& \frac{8\xi^2\chi f(\bar{p}\cdot\bar{q})m^2}{\chi p} - \frac{8\xi^2\chi g(\bar{p}\cdot\bar{q})m^2}{\chi p} + \frac{8\xi^2\chi f(\bar{p}\cdot\bar{q})m^2}{\chi q} - \frac{8\xi^2\chi g(\bar{p}\cdot\bar{q})m^2}{\chi q} - \\
& \frac{16e^2\chi p(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{g})}{\chi f} + \frac{16e^2\chi p(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{g})}{\chi g} - \frac{16e^2\chi q(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{g})}{\chi f} + \frac{16e^2\chi q(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{g})}{\chi g} - \\
& 16e^2(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{p}) + \frac{16e^2\chi g\chi q(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{p})}{\chi f\chi p} + 16e^2(\bar{a}\cdot\bar{g})(\bar{a}\cdot\bar{p}) - \frac{16e^2\chi f\chi q(\bar{a}\cdot\bar{g})(\bar{a}\cdot\bar{p})}{\chi g\chi p} - \\
& 16e^2(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{q}) + \frac{16e^2\chi g\chi p(\bar{a}\cdot\bar{f})(\bar{a}\cdot\bar{q})}{\chi f\chi q} + 16e^2(\bar{a}\cdot\bar{g})(\bar{a}\cdot\bar{q}) - \frac{16e^2\chi f\chi p(\bar{a}\cdot\bar{g})(\bar{a}\cdot\bar{q})}{\chi g\chi q} + \\
& \frac{16e^2\chi f(\bar{a}\cdot\bar{p})(\bar{a}\cdot\bar{q})}{\chi p} - \frac{16e^2\chi g(\bar{a}\cdot\bar{p})(\bar{a}\cdot\bar{q})}{\chi p} + \frac{16e^2\chi f(\bar{a}\cdot\bar{p})(\bar{a}\cdot\bar{q})}{\chi q} - \frac{16e^2\chi g(\bar{a}\cdot\bar{p})(\bar{a}\cdot\bar{q})}{\chi q} \Big) + \\
\text{C1 D0} & \left(\frac{8\xi^2\chi p m^4}{\chi f} - \frac{8\xi^2\chi p m^4}{\chi g} + \frac{8\xi^2\chi q m^4}{\chi f} - \frac{8\xi^2\chi q m^4}{\chi g} + \frac{8\xi^2\chi f m^4}{\chi p} - \frac{8\xi^2\chi g m^4}{\chi p} + \right. \\
& \frac{8\xi^2\chi f m^4}{\chi q} - \frac{8\xi^2\chi g m^4}{\chi q} + \frac{8\xi^2\chi p(\bar{f}\cdot\bar{g})m^2}{\chi f} - \frac{8\xi^2\chi p(\bar{f}\cdot\bar{g})m^2}{\chi g} + \frac{8\xi^2\chi q(\bar{f}\cdot\bar{g})m^2}{\chi f} - \\
& \frac{8\xi^2\chi q(\bar{f}\cdot\bar{g})m^2}{\chi g} + 8\xi^2(\bar{f}\cdot\bar{p})m^2 - \frac{8\xi^2\chi g\chi q(\bar{f}\cdot\bar{p})m^2}{\chi f\chi p} + 8\xi^2(\bar{f}\cdot\bar{q})m^2 - \frac{8\xi^2\chi g\chi p(\bar{f}\cdot\bar{q})m^2}{\chi f\chi q} - \\
& 8\xi^2(\bar{g}\cdot\bar{p})m^2 + \frac{8\xi^2\chi f\chi q(\bar{g}\cdot\bar{p})m^2}{\chi g\chi p} - 8\xi^2(\bar{g}\cdot\bar{q})m^2 + \frac{8\xi^2\chi f\chi p(\bar{g}\cdot\bar{q})m^2}{\chi g\chi q} - \\
& \left. \frac{8\xi^2\chi f(\bar{p}\cdot\bar{q})m^2}{\chi p} + \frac{8\xi^2\chi g(\bar{p}\cdot\bar{q})m^2}{\chi p} - \frac{8\xi^2\chi f(\bar{p}\cdot\bar{q})m^2}{\chi q} + \frac{8\xi^2\chi g(\bar{p}\cdot\bar{q})m^2}{\chi q} \right) + \\
\text{A0 B1} & \left(\text{C2 D1} \left(\frac{8m^4\chi p\xi^4}{\chi f} - \frac{8m^4\chi p\xi^4}{\chi g} + \frac{8m^4\chi q\xi^4}{\chi f} - \frac{8m^4\chi q\xi^4}{\chi g} \right) + \text{C2 D0} \left(-\frac{8em^2\chi p(\bar{a}\cdot\bar{f})\xi^2}{\chi f} - \right. \right. \\
& \frac{8em^2\chi q(\bar{a}\cdot\bar{f})\xi^2}{\chi f} - \frac{8em^2\chi p(\bar{a}\cdot\bar{g})\xi^2}{\chi g} - \frac{8em^2\chi q(\bar{a}\cdot\bar{g})\xi^2}{\chi g} + 16em^2(\bar{a}\cdot\bar{p})\xi^2 + 16em^2(\bar{a}\cdot\bar{q})\xi^2 \Big) + \\
\text{C1 D1} & \left(-\frac{8em^2\chi p(\bar{a}\cdot\bar{f})\xi^2}{\chi f} + \frac{8em^2\chi p(\bar{a}\cdot\bar{f})\xi^2}{\chi g} - \frac{8em^2\chi q(\bar{a}\cdot\bar{f})\xi^2}{\chi f} + \frac{8em^2\chi q(\bar{a}\cdot\bar{f})\xi^2}{\chi g} + \right. \\
& \frac{8em^2\chi p(\bar{a}\cdot\bar{g})\xi^2}{\chi f} - \frac{8em^2\chi p(\bar{a}\cdot\bar{g})\xi^2}{\chi g} + \frac{8em^2\chi q(\bar{a}\cdot\bar{g})\xi^2}{\chi f} - \frac{8em^2\chi q(\bar{a}\cdot\bar{g})\xi^2}{\chi g} + \\
& 16em^2(\bar{a}\cdot\bar{p})\xi^2 - \frac{8em^2\chi g\chi q(\bar{a}\cdot\bar{p})\xi^2}{\chi f\chi p} - \frac{8em^2\chi f\chi q(\bar{a}\cdot\bar{p})\xi^2}{\chi g\chi p} + \\
& \left. 16em^2(\bar{a}\cdot\bar{q})\xi^2 - \frac{8em^2\chi g\chi p(\bar{a}\cdot\bar{q})\xi^2}{\chi f\chi q} - \frac{8em^2\chi f\chi p(\bar{a}\cdot\bar{q})\xi^2}{\chi g\chi q} \right) +
\end{aligned}$$

$$\begin{aligned}
& \text{C0 D0} \left(16 e (\bar{a} \cdot \bar{p}) m^2 - \frac{16 e \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi p} + 16 e (\bar{a} \cdot \bar{q}) m^2 - \frac{16 e \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) - \frac{16 e \chi g (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) - \frac{16 e \chi g (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q})}{\chi p} + \\
& \left. 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi f (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) - \frac{16 e \chi f (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q})}{\chi p} \right) + \\
& \text{C1 D0} \left(-\frac{8 \xi^2 \chi p m^4}{\chi f} + \frac{8 \xi^2 \chi p m^4}{\chi g} - \frac{8 \xi^2 \chi q m^4}{\chi f} + \frac{8 \xi^2 \chi q m^4}{\chi g} - \frac{8 \xi^2 \chi f m^4}{\chi p} + \frac{8 \xi^2 \chi g m^4}{\chi p} - \right. \\
& \frac{8 \xi^2 \chi f m^4}{\chi q} + \frac{8 \xi^2 \chi g m^4}{\chi q} - \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi f} + \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi g} - \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f} + \\
& \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi g} + 8 \xi^2 (\bar{f} \cdot \bar{p}) m^2 + \frac{8 \xi^2 \chi g \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f \chi p} + 8 \xi^2 (\bar{f} \cdot \bar{q}) m^2 + \frac{8 \xi^2 \chi g \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f \chi q} - \\
& 8 \xi^2 (\bar{g} \cdot \bar{p}) m^2 - \frac{8 \xi^2 \chi f \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g \chi p} - 8 \xi^2 (\bar{g} \cdot \bar{q}) m^2 - \frac{8 \xi^2 \chi f \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g \chi q} + \\
& \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi p} - \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p} + \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi q} - \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi q} - \\
& \frac{16 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi f} + \frac{16 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi g} - \frac{16 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi f} + \frac{16 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi g} - \\
& 16 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p}) + \frac{16 e^2 \chi g \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p})}{\chi f \chi p} + 16 e^2 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p}) - \frac{16 e^2 \chi f \chi q (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p})}{\chi g \chi p} - \\
& 16 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q}) + \frac{16 e^2 \chi g \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q})}{\chi f \chi q} + 16 e^2 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q}) - \frac{16 e^2 \chi f \chi p (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q})}{\chi g \chi q} + \\
& \left. \frac{16 e^2 \chi f (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p} - \frac{16 e^2 \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p} + \frac{16 e^2 \chi f (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi q} - \frac{16 e^2 \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi q} \right) + \\
& \text{C0 D1} \left(\frac{8 \xi^2 \chi p m^4}{\chi f} - \frac{8 \xi^2 \chi p m^4}{\chi g} + \frac{8 \xi^2 \chi q m^4}{\chi f} - \frac{8 \xi^2 \chi q m^4}{\chi g} + \frac{8 \xi^2 \chi f m^4}{\chi p} - \frac{8 \xi^2 \chi g m^4}{\chi p} + \right. \\
& \frac{8 \xi^2 \chi f m^4}{\chi q} - \frac{8 \xi^2 \chi g m^4}{\chi q} + \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi f} - \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi g} + \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f} - \\
& \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi g} + 8 \xi^2 (\bar{f} \cdot \bar{p}) m^2 - \frac{8 \xi^2 \chi g \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f \chi p} + 8 \xi^2 (\bar{f} \cdot \bar{q}) m^2 - \frac{8 \xi^2 \chi g \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f \chi q} - \\
& 8 \xi^2 (\bar{g} \cdot \bar{p}) m^2 + \frac{8 \xi^2 \chi f \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g \chi p} - 8 \xi^2 (\bar{g} \cdot \bar{q}) m^2 + \frac{8 \xi^2 \chi f \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g \chi q} - \\
& \left. \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi p} + \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p} - \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi q} + \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi q} \right) + \\
& \text{A0 B2} \left(-16 \text{C2 D0} m^4 \xi^4 + \text{C1 D0} \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \right. \right. \\
& \left. \left. \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \right.
\end{aligned}$$

$$\begin{aligned}
& \text{C0 D0} \left(-16 \xi^2 m^4 + \frac{16 \xi^2 \chi f \chi g m^4}{\chi p \chi q} - 16 \xi^2 (\bar{f} \cdot \bar{g}) m^2 + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& \quad \left. \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi q} - \frac{16 \xi^2 \chi f \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p \chi q} \right) + \\
& \text{A2 B0} \left(-16 \text{C0 D2} m^4 \xi^4 + \text{C0 D1} \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \right. \right. \\
& \quad \left. \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \\
& \text{C0 D0} \left(-16 \xi^2 m^4 + \frac{16 \xi^2 \chi f \chi g m^4}{\chi p \chi q} - 16 \xi^2 (\bar{f} \cdot \bar{g}) m^2 + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& \quad \left. \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi q} - \frac{16 \xi^2 \chi f \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p \chi q} \right) + \\
& \text{A1 B1} \left(\text{C1 D1} \left(-32 m^4 \xi^4 + \frac{8 m^4 \chi g \chi q \xi^4}{\chi f \chi p} + \frac{8 m^4 \chi f \chi q \xi^4}{\chi g \chi p} + \frac{8 m^4 \chi g \chi p \xi^4}{\chi f \chi q} + \frac{8 m^4 \chi f \chi p \xi^4}{\chi g \chi q} \right) + \right. \\
& \text{C1 D0} \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - \frac{8 e m^2 \chi g \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi p} - \frac{8 e m^2 \chi g \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi q} - \right. \\
& \quad 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 + \frac{8 e m^2 \chi f \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi p} + \frac{8 e m^2 \chi f \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi q} - \\
& \quad \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} - \\
& \quad \left. \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \\
& \text{C0 D1} \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - \frac{8 e m^2 \chi g \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi p} - \frac{8 e m^2 \chi g \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi q} - 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 + \right. \\
& \quad \frac{8 e m^2 \chi f \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi p} + \frac{8 e m^2 \chi f \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi q} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \\
& \quad \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi p} + \\
& \quad \left. \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \text{C0 D0} \\
& \left(\frac{16 \xi^2 \chi q m^4}{\chi p} + \frac{16 \xi^2 \chi p m^4}{\chi q} - \frac{32 \xi^2 \chi f \chi g m^4}{\chi p \chi q} + 32 \xi^2 (\bar{f} \cdot \bar{g}) m^2 - \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi p} + \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi q} + \right. \\
& \quad \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi p} - \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi q} - \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi p} + \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi q} + \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi p} - \\
& \quad \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi q} + \frac{32 \xi^2 \chi f \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p \chi q} + 64 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g}) - \frac{32 e^2 \chi g (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p})}{\chi p} -
\end{aligned}$$

$$\left. \begin{aligned} & \frac{32 e^2 \chi f (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p})}{\chi p} - \frac{32 e^2 \chi g (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q})}{\chi q} - \frac{32 e^2 \chi f (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q})}{\chi q} + \frac{64 e^2 \chi f \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p \chi q} \right) \end{aligned}$$

Step 5: Passing to invariant variables ρ_1 , τ_1 , ρ_2 , τ_2 in the integral

In the following step, we consider only the integrand, namely, $\text{Coeff} * \text{Expr}$. We will introduce the Jacobian of the transformation later

Recall the conservation laws

$$p + sk = q + l$$

$$\chi p = \chi q + \chi l$$

Variables ρ_1 , τ_1 read

$$\begin{aligned} \rho_1 &= \frac{e(qf p)}{\xi m^4 \chi l} = -\frac{e(lf p)}{\xi m^4 \chi l} = \frac{e(qf l)}{\xi m^4 \chi l} \\ \tau_1 &= -\frac{e(qF^* p)}{m^4 \chi l} = \frac{e(lF^* p)}{m^4 \chi l} = \frac{e(qF^* l)}{m^4 \chi l} \end{aligned}$$

Similarly, we have ρ_2 and τ_2

$$t k + l = f + g$$

$$\chi l = \chi f + \chi g$$

$$\begin{aligned} \rho_2 &= -\frac{e(gFf)}{\xi m^4 \chi l} = -\frac{e(gFl)}{\xi m^4 \chi l} = \frac{e(fFl)}{\xi m^4 \chi l} \\ \tau_2 &= -\frac{e(gF^* f)}{m^4 \chi l} = -\frac{e(gF^* l)}{m^4 \chi l} = \frac{e(fF^* l)}{m^4 \chi l} \end{aligned}$$

Also, notice that

$$(q F^2 p) = -a^2 k p k q$$

Expr contains contractions of momenta in pairs

$$(pq), (pf), (pg), (qf), (qg), (fg)$$

Let us express all such contractions through ρ_1 ,

τ_1 , ρ_2 , τ_2 and the χ parameters

The key idea is projecting a given 4 - momentum on a basis spanned by the field tensor and another 4 - momentum :

$$q^\mu = C_1 p^\mu + C_2 (Fp)^\mu + C_3 (F^* p)^\mu + C_4 (FFp)^\mu$$

$$\begin{aligned}
C_1 &= \frac{\chi_q}{\chi_p} \\
C_2 &= -\frac{e^2 (qFp)}{m^6 \chi_p^2} \\
C_3 &= -\frac{e^2 (qF^* p)}{m^6 \chi_p^2} \\
C_4 &= \frac{e^2}{2 m^4 \chi_p \chi_q} \left(1 - \frac{\chi_q^2}{\chi_p^2} + \frac{e^2 (qFp)^2}{m^8 \chi_p^2} + \frac{e^2 (qF^* p)^2}{m^8 \chi_p^2} \right)
\end{aligned}$$

This projection is specific to a crossed field,
as in such field $[F^{n>2}]^{\mu\nu} = [F^* F]^{\mu\nu} = 0$
(a non – crossed field may have higher – order terms in powers of F)

Then we substitute the projected vector to
the initial scalar product. Terms such as (qFp) are re –
expressed via invariant parameters

Note that expressing combinations such as (qg) via
the invariant parameters ρ_n, τ_n requires an intermediate
step. The issue is that q and g enter different vertices,
and their projection results in appearance of combinations
such as (pFg) , which do not enter the definitions of $\rho_n,$
 τ_n . We use that the vertices are connected by l .

We use a command `ExpandVector`, which is defined in `definitions.nb`,
`ExpandVector[p, q,  $\mu$ ]`,
which projects 4 – vector q on the basis spanned by q and F ,
and the resulting vector carries index μ ;
See the illustrating example below

Each projection is stored in a function of the Lorentz index μ ,
given by command
`qpexpansion[ $\mu$ ]`
where q is the vector being projected and p –
the momentum spanning the basis

When defining a projection, we also introduce related
substitutions of scalar combinations such as (qFp) ,
that reexpress them via invariant ρ_n, τ_n ;

These substitutions are stored in arrays, named, for example, as `subs $\rho_1\tau_1$ qp` ,
so that the last two letters are attributed
to the momenta in the same order as in '`qpexpansion`'

In[41]:= (***example***)

ExpandVector[p, q, μ]

$$\text{Out[41]} = \frac{e^2 \text{FFp}^\mu (e^2 (\text{FDp} \cdot q)^2 + e^2 (\text{Fp} \cdot q)^2 + m^8 \chi p^2 - m^8 \chi q^2)}{2 m^{12} \chi p^3 \chi q} - \frac{e^2 \text{FDp}^\mu (\text{FDp} \cdot q)}{m^6 \chi p^2} - \frac{e^2 \text{Fp}^\mu (\text{Fp} \cdot q)}{m^6 \chi p^2} + \frac{\chi q p^\mu}{\chi p}$$

(pq)

– project q on p

– contract with p, reexpress via ρ_n, τ_n

In[42]:= **subs $\rho_1\tau_1$ qp** = {Pair[Momentum[Fp, D], Momentum[q, D]] $\rightarrow$ $\rho_1 \xi m^4 \chi l / e$,

Pair[Momentum[FDp, D], Momentum[q, D]] $\rightarrow$ $-\tau_1 m^4 \chi l / e$ }

qpexpansion[μ] = Collect[ExpandVector[p, q, μ] /. **subs $\rho_1\tau_1$ qp**,

{pv[μ], Fpv[μ], FDpv[μ], FFpv[μ]}, Simplify]

pq = Contract[pv[μ] $\times$ qv[μ] /. {qv[μ] $\rightarrow$ qpexpansion[μ]}

$$\text{Out[42]} = \left\{ \text{Fp} \cdot q \rightarrow \frac{m^4 \xi \rho_1 \chi l}{e}, \text{FDp} \cdot q \rightarrow -\frac{m^4 \tau_1 \chi l}{e} \right\}$$

$$\text{Out[43]} = \frac{e^2 \text{FFp}^\mu (\xi^2 \rho_1^2 \chi l^2 + \tau_1^2 \chi l^2 + \chi p^2 - \chi q^2)}{2 m^4 \chi p^3 \chi q} + \frac{e \tau_1 \chi l \text{FDp}^\mu}{m^2 \chi p^2} - \frac{e \xi \rho_1 \chi l \text{Fp}^\mu}{m^2 \chi p^2} + \frac{\chi q p^\mu}{\chi p}$$

$$\text{Out[44]} = \frac{m^2 (\xi^2 \rho_1^2 \chi l^2 + \tau_1^2 \chi l^2 + \chi p^2 - \chi q^2)}{2 \chi p \chi q} + \frac{m^2 \chi q}{\chi p}$$

(lp), required later as an intermediate step

– project l on p

– contract with p, reexpress via ρ_n, τ_n

```
In[45]:= subs $\rho$ 1 $\tau$ 1lp = {Pair[Momentum[Fp, D], Momentum[l, D]]  $\rightarrow$  -  $\rho$ 1  $\xi$  m^4  $\chi$  l / e,
      Pair[Momentum[FDp, D], Momentum[l, D]]  $\rightarrow$   $\tau$ 1 m^4  $\chi$  l / e,
      Pair[Momentum[FFp, D], Momentum[l, D]]  $\rightarrow$  -av2 kl kp} /.
      ktochiSubs[p, q, f, g, l] /. {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2}
      lpexpansion[ $\mu$ _] = Collect[ExpandVector[p, l,  $\mu$ ] /. subs $\rho$ 1 $\tau$ 1lp,
      {lv[ $\mu$ ], Flv[ $\mu$ ], FDllv[ $\mu$ ], FFlv[ $\mu$ ]}, Simplify]
      lp = Contract[pv[ $\mu$ ]  $\times$  lv[ $\mu$ ] /. {lv[ $\mu$ ]  $\rightarrow$  lpexpansion[ $\mu$ ]}} /. subs $\rho$ 1 $\tau$ 1lp
```

$$\text{Out[45]} = \left\{ \text{Fp} \cdot l \rightarrow -\frac{m^4 \xi \rho l \chi l}{e}, \text{FDp} \cdot l \rightarrow \frac{m^4 \tau l \chi l}{e}, \text{FFp} \cdot l \rightarrow \frac{m^6 \chi l \chi p}{e^2} \right\}$$

$$\text{Out[46]} = \frac{e^2 \text{FFp}^\mu (\chi l^2 (\xi^2 \rho l^2 + \tau l^2 - 1) + \chi p^2) - 2 e m^2 \tau l \chi l^2 \chi p \text{FDp}^\mu + 2 m^2 \chi l^2 \chi p (e \xi \rho l \text{Fp}^\mu + m^2 \chi p p^\mu)}{2 m^4 \chi l \chi p^3}$$

$$\text{Out[47]} = \frac{m^6 \chi p^2 (\chi l^2 (\xi^2 \rho l^2 + \tau l^2 - 1) + \chi p^2) + 2 m^6 \chi l^2 \chi p^2}{2 m^4 \chi l \chi p^3}$$

Projection of p on l,
 required later as an intermediate step (particularly, subs ρ 1 τ 1pl)
 Note that subs ρ 1 τ 1pl is not exactly
 the same as subs ρ 1 τ 1lp! We use both further

```
In[48]:= subs $\rho$ 1 $\tau$ 1pl = {Pair[Momentum[Fl, D], Momentum[p, D]]  $\rightarrow$   $\rho$ 1  $\xi$  m^4  $\chi$  l / e,
      Pair[Momentum[F Dl, D], Momentum[p, D]]  $\rightarrow$  -  $\tau$ 1 m^4  $\chi$  l / e,
      Pair[Momentum[FFl, D], Momentum[p, D]]  $\rightarrow$  -av2 kl kp} /.
      ktochiSubs[p, q, f, g, l] /. {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2}
      plexpansion[ $\mu$ _] = Collect[ExpandVector[l, p,  $\mu$ ] /. subs $\rho$ 1 $\tau$ 1pl,
      {lv[ $\mu$ ], Flv[ $\mu$ ], FDllv[ $\mu$ ], FFlv[ $\mu$ ]}, Simplify]
```

$$\text{Out[48]} = \left\{ \text{Fl} \cdot p \rightarrow \frac{m^4 \xi \rho l \chi l}{e}, \text{FDl} \cdot p \rightarrow -\frac{m^4 \tau l \chi l}{e}, \text{FFl} \cdot p \rightarrow \frac{m^6 \chi l \chi p}{e^2} \right\}$$

$$\text{Out[49]} = \frac{e^2 \text{FFl}^\mu (\chi l^2 (\xi^2 \rho l^2 + \tau l^2 + 1) - \chi p^2)}{2 m^4 \chi l^3 \chi p} + \frac{e \tau l \text{FDl}^\mu}{m^2 \chi l} - \frac{e \xi \rho l \text{Fl}^\mu}{m^2 \chi l} + \frac{\chi p l^\mu}{\chi l}$$

(fg)
 -project g on f
 -contract with f, reexpress via ρ_n , τ_n

```
In[50]:= subsp2τ2gf = {Pair[Momentum[Ff, D], Momentum[g, D]] → -ρ2 ξ m^4 χl / e,
  Pair[Momentum[FDf, D], Momentum[g, D]] → -τ2 m^4 χl / e}
gfexpansion[μ_] = Collect[ExpandVector[f, g, μ] /. subsp2τ2gf,
  {fv[μ], Ffv[μ], FDfv[μ], FFfv[μ]}, Simplify]
fg = Contract[gfexpansion[μ] × fv[μ]]
```

$$\text{Out[50]} = \left\{ Ff \cdot g \rightarrow -\frac{m^4 \xi \rho 2 \chi l}{e}, Fd f \cdot g \rightarrow -\frac{m^4 \tau 2 \chi l}{e} \right\}$$

$$\text{Out[51]} = \frac{e^2 FFf^\mu (\chi l^2 (\xi^2 \rho 2^2 + \tau 2^2) + \chi f^2 - \chi g^2)}{2 m^4 \chi f^3 \chi g} + \frac{e \tau 2 \chi l FDf^\mu}{m^2 \chi f^2} + \frac{e \xi \rho 2 \chi l FFf^\mu}{m^2 \chi f^2} + \frac{\chi g f^\mu}{\chi f}$$

$$\text{Out[52]} = \frac{m^2 (\chi l^2 (\xi^2 \rho 2^2 + \tau 2^2) + \chi f^2 - \chi g^2)}{2 \chi f \chi g} + \frac{m^2 \chi g}{\chi f}$$

Projection of g on l, required later as an intermediate step

```
In[53]:= subsp2τ2gl = {Pair[Momentum[Fl, D], Momentum[g, D]] → -ρ2 ξ m^4 χl / e,
  Pair[Momentum[FDl, D], Momentum[g, D]] → -τ2 m^4 χl / e}
glexpansion[μ_] = Collect[ExpandVector[l, g, μ] /. subsp2τ2gl,
  {lv[μ], Flv[μ], FDlv[μ], FFlv[μ]}, Simplify]
```

$$\text{Out[53]} = \left\{ Fl \cdot g \rightarrow -\frac{m^4 \xi \rho 2 \chi l}{e}, Fd l \cdot g \rightarrow -\frac{m^4 \tau 2 \chi l}{e} \right\}$$

$$\text{Out[54]} = -\frac{e^2 FFf^\mu (\chi g^2 - \chi l^2 (\xi^2 \rho 2^2 + \tau 2^2 + 1))}{2 m^4 \chi g \chi l^3} + \frac{e \tau 2 FDl^\mu}{m^2 \chi l} + \frac{e \xi \rho 2 Fl^\mu}{m^2 \chi l} + \frac{\chi g l^\mu}{\chi l}$$

(pg)
 -project g on l
 -contract with p, reexpress via ρ_n ,
 τ_n using also substitutions $\text{subsp1}\tau\text{lp}l$
 and expression for (lp) calculated above

```
In[55]:= gpexpansion[μ_] =
  Collect[ExpandVector[p, g, μ], {pv[μ], Fpv[μ], FDPv[μ], FFpv[μ]}, Simplify]
pg = Simplify[Contract[pv[μ] × gv[μ] /. {gv[μ] → glexpansion[μ]}] /. subsp1τlp] /.
  {Contract[pv[μ] × lv[μ]] → lp}]
gp = pg;
```

$$\text{Out[55]} = \frac{FFp^\mu (e^4 (FDp \cdot g)^2 + e^4 (Fp \cdot g)^2 + e^2 m^8 (\chi p^2 - \chi g^2))}{2 m^{12} \chi g \chi p^3} - \frac{e^2 FDp^\mu (FDp \cdot g)}{m^6 \chi p^2} - \frac{e^2 Fp^\mu (Fp \cdot g)}{m^6 \chi p^2} + \frac{\chi g p^\mu}{\chi p}$$

$$\text{Out[56]} = \frac{m^2 (\chi g^2 (\xi^2 \rho 1^2 + \tau 1^2 + 1) + 2 \chi g \chi p (\xi^2 \rho 1 \rho 2 - \tau 1 \tau 2) + \chi p^2 (\xi^2 \rho 2^2 + \tau 2^2 + 1))}{2 \chi g \chi p}$$

(qg)
 -project q on p (calculated above)
 -project g on l (calculated above)

– contract, reexpress via ρ_n, τ_n
 – the result will contain (lp) ,
 which we calculated above and substitute here;
 – In the last step,
 we remove all terms containing $(F)^{n>2}$ and $(F^* F)$ as they are zero

```
In[58]:= qg = Simplify[
  Contract[qv[μ] × gv[μ] /. {qv[μ] → qpexpansion[μ], gv[μ] → glexpansion[μ]}] /.
    subsρ1τ1qp /. subsρ1τ1pl /. subsρ1τ1lp /. {Contract[pv[μ] × lv[μ]] →
      lp} /. {Pair[Momentum[FFp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[FFp, D], Momentum[FDl, D]] → 0,
      Pair[Momentum[FDp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[FFp, D], Momentum[Ffl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[FDl, D]] → 0,
      Pair[Momentum[FDp, D], Momentum[Ffl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[Ffl, D]] → av2 kp kl,
      Pair[Momentum[FDp, D], Momentum[FDl, D]] → av2 kp kl} /.
    ktouchSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}]
gq = qg;
```

$$\text{Out[58]} = \frac{m^2 (\xi^2 (\rho^1 \chi g (\chi^1 + \chi q) + \rho^2 \chi p \chi q)^2 + \tau^1 \chi g^2 (\chi^1 + \chi q)^2 - 2 \tau^1 \tau^2 \chi g \chi p \chi q (\chi^1 + \chi q) + \chi p^2 ((\tau^2 + 1) \chi q^2 + \chi g^2))}{2 \chi g \chi p^2 \chi q}$$

Projection of f on l, required later as an intermediate step

```
In[60]:= subsρ2τ2fl = {Pair[Momentum[Ffl, D], Momentum[f, D]] → ρ2 ξ m^4 χl / e,
  Pair[Momentum[FDl, D], Momentum[f, D]] → τ2 m^4 χl / e,
  Pair[Momentum[FFl, D], Momentum[f, D]] → -av2 kl kf} /.
  ktouchSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}
flexpansion[μ_] = Collect[ExpandVector[l, f, μ] /. subsρ2τ2fl,
  {lv[μ], Flv[μ], Fdlv[μ], FFlv[μ]}, Simplify]
```

$$\text{Out[60]} = \left\{ f \cdot Fl \rightarrow \frac{m^4 \xi \rho^2 \chi^1}{e}, f \cdot FDl \rightarrow \frac{m^4 \tau^2 \chi^1}{e}, f \cdot FFl \rightarrow \frac{m^6 \chi^f \chi^1}{e^2} \right\}$$

$$\text{Out[61]} = -\frac{e^2 FFf^\mu (\chi^f{}^2 - \chi^l{}^2 (\xi^2 \rho^2 + \tau^2 + 1))}{2 m^4 \chi^f \chi^l{}^3} - \frac{e \tau^2 FDl^\mu}{m^2 \chi^l} - \frac{e \xi \rho^2 Fl^\mu}{m^2 \chi^l} + \frac{\chi^f l^\mu}{\chi^l}$$

(pf)

– project f on l
 – contract, reexpress via ρ_n, τ_n ,
 also using $\text{subs}\rho_1\tau_1pl$ and scalar product (lp) calculated before

```
In[62]:= fpexpansion[μ_] =
  Collect[ExpandVector[p, f, μ], {pv[μ], Fpv[μ], FDpv[μ], FFpv[μ]}, Simplify]
pf = Simplify[Contract[pv[μ] × fv[μ] /. {fv[μ] → flexpansion[μ]}] /. subsρ1τ1pl /.
  {Contract[pv[μ] × lv[μ]] → lp}]
fp = pf;

Out[62]= 
$$\frac{FFp^\mu (e^4 (f \cdot FDP)^2 + e^4 (f \cdot Fp)^2 + e^2 m^8 (\chi p^2 - \chi f^2))}{2 m^{12} \chi f \chi p^3} - \frac{e^2 FDP^\mu (f \cdot FDP)}{m^6 \chi p^2} - \frac{e^2 Fp^\mu (f \cdot Fp)}{m^6 \chi p^2} + \frac{\chi f p^\mu}{\chi p}$$


Out[63]= 
$$\frac{m^2 (\chi f^2 (\xi^2 \rho 1^2 + \tau 1^2 + 1) + \chi f (2 \tau 1 \tau 2 \chi p - 2 \xi^2 \rho 1 \rho 2 \chi p) + \chi p^2 (\xi^2 \rho 2^2 + \tau 2^2 + 1))}{2 \chi f \chi p}$$

```

(qf)
 – project q on p (calculated above)
 – project f on l (calculated above)
 – contract, reexpress via ρ_n, τ_n ,
 – the result will contain (lp),
 which we calculated above and substitute here;
 – In the last step,
 we remove all terms containing $(F)^{n>2}$ and $(F^* F)$ as they are zero

```
In[65]:= qf = Simplify[
  Contract[qv[μ] × fv[μ] /. {qv[μ] → qpexpansion[μ], fv[μ] → flexpansion[μ]}] /.
    subsρ1τ1qp /. subsρ1τ1pl /. subsρ1τ1lp /. {Contract[pv[μ] × lv[μ]] →
      lp} /. {Pair[Momentum[FFp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[FFp, D], Momentum[FDl, D]] → 0,
      Pair[Momentum[FDp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[FFp, D], Momentum[Fl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[FDl, D]] → 0,
      Pair[Momentum[FDp, D], Momentum[Fl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[Fl, D]] → av2 kp kl,
      Pair[Momentum[FDp, D], Momentum[FDl, D]] → av2 kp kl} /.
    ktchiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}]
fq = qf;

Out[65]= 
$$\frac{m^2 (\xi^2 (\rho 2 \chi p \chi q - \rho 1 \chi f (\chi l + \chi q))^2 + \tau 1^2 \chi f^2 (\chi l + \chi q)^2 + 2 \tau 1 \tau 2 \chi f \chi p \chi q (\chi l + \chi q) + \chi p^2 ((\tau 2^2 + 1) \chi q^2 + \chi f^2))}{2 \chi f \chi p^2 \chi q}$$

```

All scalar products of the form

(aq), (al), ...

can be reduced to only two : (ap) or (af) plus invariant terms,

This is achieved again by projecting 4 –

momenta on the basis spanned by F and p or f

$$q^\mu = C_1 p^\mu + C_2 (Fp)^\mu + C_3 (F^* p)^\mu + C_4 (FFp)^\mu$$

If done correctly,
all terms containing (ap) and (af) in variable Expr will cancel ,
and the expression will depend only on gauge – invariant parameters

The result is stored in Expr4

```
In[67]:= aqsubs = {Pair[Momentum[a, D], Momentum[q, D]] → Contract[av[μ] × qpexpansion[μ]] /.  
          ktchiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}  
alsubs = {Pair[Momentum[a, D], Momentum[l, D]] → Contract[av[μ] × lpexpansion[μ]] /.  
          ktchiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}  
agsubs = {Pair[Momentum[a, D], Momentum[g, D]] → Contract[av[μ] × glexpansion[μ]] /.  
          alsubs /. ktchiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}  
afsubs = {Pair[Momentum[a, D], Momentum[f, D]] → Contract[av[μ] × flxpansion[μ]] /.  
          alsubs /. ktchiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}
```

$$\text{Out[67]} = \left\{ a \cdot q \rightarrow \frac{\chi q (a \cdot p)}{\chi p} - \frac{m^2 \xi^2 \rho 1 \chi l}{e \chi p} \right\}$$

$$\text{Out[68]} = \left\{ a \cdot l \rightarrow \frac{\chi l \left(m^2 \chi p (a \cdot p) + \frac{m^4 \xi^2 \rho 1 \chi p}{e} \right)}{m^2 \chi p^2} \right\}$$

$$\text{Out[69]} = \left\{ a \cdot g \rightarrow \frac{\chi g \left(m^2 \chi p (a \cdot p) + \frac{m^4 \xi^2 \rho 1 \chi p}{e} \right)}{m^2 \chi p^2} + \frac{m^2 \xi^2 \rho 2}{e} \right\}$$

$$\text{Out[70]} = \left\{ a \cdot f \rightarrow \frac{\chi f \left(m^2 \chi p (a \cdot p) + \frac{m^4 \xi^2 \rho 1 \chi p}{e} \right)}{m^2 \chi p^2} - \frac{m^2 \xi^2 \rho 2}{e} \right\}$$

```
In[71]:= Expr3 /. aqsubs /. agsubs /. afsubs /. {Pair[Momentum[p, D], Momentum[q, D]] → pq,  
          Pair[Momentum[p, D], Momentum[f, D]] → pf,  
          Pair[Momentum[p, D], Momentum[g, D]] → pg,  
          Pair[Momentum[q, D], Momentum[f, D]] → qf,  
          Pair[Momentum[q, D], Momentum[g, D]] → qg,  
          Pair[Momentum[f, D], Momentum[g, D]] → fg};  
Expand[%];  
Expr4 = SortTermsSimplify[Contract[%] /. ktchiSubs[p, q, f, g] /.  
          {av2 → -m^2 ξ^2 / e^2} (* /. {χq→χp-χl, χg→χl-χg} *)];  
% /. {D → 4}
```

$$\text{Out[74]} = \frac{16 A2 B2 C0 D0 m^4 \chi f \chi g \xi^4}{\chi p \chi q} +$$

$$A1 B2 \left(\frac{8 C1 D0 m^4 (\chi f - \chi g) (\chi p + \chi q) \xi^4}{\chi p \chi q} + \frac{8 C0 D0 m^4 (\rho 2 (\chi f - \chi g) \chi p (\chi p + \chi q) + 2 \rho 1 \chi f \chi g (\chi l + \chi p + \chi q)) \xi^4}{\chi p^2 \chi q} \right) +$$

$$A2 B1 \left(\frac{8 C0 D1 m^4 (\chi f - \chi g) (\chi p + \chi q) \xi^4}{\chi p \chi q} + \frac{8 C0 D0 m^4 (\rho 2 (\chi f - \chi g) \chi p (\chi p + \chi q) + 2 \rho 1 \chi f \chi g (\chi l + \chi p + \chi q)) \xi^4}{\chi p^2 \chi q} \right) +$$

$$\begin{aligned}
& \text{A0 B2} \left(-16 \text{C2 D0} \xi^4 m^4 + \frac{8 \text{C1 D0} \xi^4 (\rho 1 (\chi f - \chi g) (\chi l + 2 \chi q) - 4 \rho 2 \chi p \chi q) m^4}{\chi p \chi q} + \frac{1}{\chi f \chi g \chi p^2 \chi q} \right. \\
& \quad 8 \text{C0 D0} \xi^2 ((2 \rho 1^2 \chi f^2 (\chi l + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi p (\chi l + 2 \chi q) \chi g + \rho 2^2 (\chi f^2 + \chi g^2 - \chi l^2) \chi p^2 \chi q) \xi^2 + \\
& \quad 2 \tau 1^2 \chi f^2 \chi g^2 (\chi l + \chi q) - \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l + 2 \chi q) + \\
& \quad \left. \chi p ((\chi g^2 - \chi l^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2))) m^4 \right) + \\
& \text{A2 B0} \left(-16 \text{C0 D2} \xi^4 m^4 + \frac{8 \text{C0 D1} \xi^4 (\rho 1 (\chi f - \chi g) (\chi l + 2 \chi q) - 4 \rho 2 \chi p \chi q) m^4}{\chi p \chi q} + \frac{1}{\chi f \chi g \chi p^2 \chi q} \right. \\
& \quad 8 \text{C0 D0} \xi^2 ((2 \rho 1^2 \chi f^2 (\chi l + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi p (\chi l + 2 \chi q) \chi g + \rho 2^2 (\chi f^2 + \chi g^2 - \chi l^2) \chi p^2 \chi q) \xi^2 + \\
& \quad 2 \tau 1^2 \chi f^2 \chi g^2 (\chi l + \chi q) - \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l + 2 \chi q) + \\
& \quad \left. \chi p ((\chi g^2 - \chi l^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2))) m^4 \right) + \\
& \text{A1 B1} \left(\frac{4 \text{C1 D1} \xi^4 ((2 \chi p^2 + 2 \chi q^2) \chi f^2 - 8 \chi g \chi p \chi q \chi f + \chi g^2 (2 \chi p^2 + 2 \chi q^2)) m^4}{\chi f \chi g \chi p \chi q} + \frac{1}{\chi f \chi g \chi p^2 \chi q} \right. \\
& \quad 4 \text{C1 D0} \xi^4 (2 \rho 1 \chi f (\chi f - \chi g) \chi g (\chi p + \chi q) (\chi l + \chi p + \chi q) + \\
& \quad \rho 2 \chi p ((2 \chi p^2 + 2 \chi q^2) \chi f^2 - 8 \chi g \chi p \chi q \chi f + \chi g^2 (2 \chi p^2 + 2 \chi q^2))) m^4 + \\
& \quad \frac{1}{\chi f \chi g \chi p^2 \chi q} 4 \text{C0 D1} \xi^4 (2 \rho 1 \chi f (\chi f - \chi g) \chi g (\chi p + \chi q) (\chi l + \chi p + \chi q) + \\
& \quad \rho 2 \chi p ((2 \chi p^2 + 2 \chi q^2) \chi f^2 - 8 \chi g \chi p \chi q \chi f + \chi g^2 (2 \chi p^2 + 2 \chi q^2))) m^4 + \frac{1}{\chi f \chi g \chi p^3 \chi q} \\
& \quad 4 \text{C0 D0} \xi^2 ((\rho 2^2 (4 \chi p \chi q \chi l^2 + 2 \chi f^2 (\chi p - \chi q)^2 + 2 \chi g^2 (\chi p - \chi q)^2 - 16 \chi f \chi g \chi p \chi q) \chi p^2 + \\
& \quad 4 \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi g (\chi p + \chi q) (\chi l + \chi p + \chi q) \chi p + 4 \rho 1^2 \chi f^2 \chi g^2 (\chi l + \chi p + \chi q)^2) \xi^2 + \\
& \quad 4 \tau 1^2 \chi f^2 \chi g^2 (\chi l - \chi p + \chi q)^2 + 4 \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi p - \chi q) (\chi l - \chi p + \chi q) + \\
& \quad \chi p^2 ((-4 \chi p \chi q \tau 2^2 + (2 \tau 2^2 + 2) \chi p^2 + (2 \tau 2^2 + 2) \chi q^2) \chi f^2 + 2 \chi g (2 \chi p^2 + 2 \chi q^2) \chi f + \\
& \quad \left. 4 \tau 2^2 \chi l^2 \chi p \chi q + \chi g^2 (-4 \chi p \chi q \tau 2^2 + (2 \tau 2^2 + 2) \chi p^2 + (2 \tau 2^2 + 2) \chi q^2))) m^4 \right) + \\
& \text{A1 B0} \left(-\frac{8 \text{D2} \xi^4 (\text{C1} (\chi f - \chi g) \chi p (\chi p + \chi q) + \text{C0} \rho 2 (\chi f - \chi g) \chi p (\chi p + \chi q) + 2 \text{C0} \rho 1 \chi f \chi g (\chi l + \chi p + \chi q)) m^4}{\chi f \chi g \chi p} + \right. \\
& \quad \frac{1}{\chi f \chi g \chi p \chi q} 4 \text{C1 D1} \xi^4 \\
& \quad (\rho 1 ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \chi f^2 + 2 \chi g (-2 \chi l \chi q - 2 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) - \\
& \quad 4 \rho 2 (\chi f - \chi g) \chi p \chi q (\chi p + \chi q)) m^4 + \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} \\
& \quad 4 \text{C0 D1} \xi^2 ((2 \rho 1^2 \chi f^2 (\chi f - \chi g) (\chi l + \chi p + \chi q) (\chi l + 2 \chi q) \chi g^2 + \rho 1 \rho 2 \chi f \chi p ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \chi f^2 + \\
& \quad 2 \chi g (-6 \chi l \chi q - 6 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) \chi g - \\
& \quad \rho 2^2 (\chi f - \chi g) (\chi f^2 + 6 \chi g \chi f + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q)) \xi^2 + \\
& \quad 2 \tau 1^2 \chi f^2 (\chi f - \chi g) \chi g^2 \chi l (\chi l - \chi p + \chi q) - \tau 2^2 (\chi f - \chi g) (\chi f^2 + 2 \chi g \chi f + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q) + \\
& \quad 2 \tau 1 \tau 2 \chi f \chi g \chi p ((\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 + 2 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q))) m^4 + \\
& \quad \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} 4 \text{C1 D0} \xi^2 ((2 \rho 1^2 \chi f^2 (\chi f - \chi g) (\chi l + \chi q) (\chi p + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f \chi p ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \\
& \quad \chi f^2 + 2 \chi g (-2 \chi l \chi q - 2 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) \chi g + \\
& \quad \rho 2^2 (\chi f - \chi g) (\chi f^2 + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q)) \xi^2 + 2 \tau 1^2 \chi f^2 (\chi f - \chi g) \chi g^2 (\chi l + \chi q) (\chi p + \chi q) +
\end{aligned}$$

$$\begin{aligned}
& (\chi f - \chi g) \chi p (\chi p + \chi q) ((\chi g^2 - \chi l^2) \chi p \chi q \tau^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau^2 + 2 \chi g^2)) - \\
& 2 \tau 1 \tau 2 \chi f \chi g \chi p ((\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 - 2 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q))) \\
& m^4 + \frac{1}{\chi f \chi g \chi p^3 \chi q} 4 \text{C0 D0 } \xi^2 \\
& ((4 \chi f^2 \chi g^2 (\chi l + \chi q) (\chi l + \chi p + \chi q) \rho 1^3 + 2 \rho 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l^2 + 2 (\chi p + 2 \chi q) \chi l + 3 \chi q (\chi p + \chi q)) \rho 1^2 + \\
& \rho 2^2 \chi p^2 (2 (\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 - 8 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + 2 \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q))) \rho 1 - \\
& 2 \rho 2^3 (\chi f - \chi g) \chi p^3 \chi q (\chi p + \chi q)) \xi^2 + 2 \rho 2 \chi p (\chi f (\chi f - \chi g) \chi g (\chi l^2 + 2 \chi q \chi l + \chi q (\chi p + \chi q)) \tau 1^2 + \\
& 4 \tau 2 \chi f \chi g \chi p \chi q (\chi l + \chi p + \chi q) \tau 1 + (\chi f - \chi g) \chi p (\chi p + \chi q) (\chi f \chi g - (\tau^2 + 1) \chi p \chi q)) + \\
& \rho 1 (4 \tau 1^2 \chi f^2 (\chi l + \chi q) (\chi l + \chi p + \chi q) \chi g^2 - 4 \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi p (\chi l + \chi q) (\chi p + \chi q) \chi g + \\
& \chi p ((4 (\chi l + \chi p + \chi q) \chi g^2 + \chi p ((2 \tau^2 + 2) \chi l \chi p + 2 (\tau^2 + 1) \chi q (\chi p + \chi q))) \chi f^2 + \\
& 4 \chi g \chi l \chi p (\chi p - \chi q) \chi f + \chi g^2 \chi p ((2 \tau^2 + 2) \chi l \chi p + 2 (\tau^2 + 1) \chi q (\chi p + \chi q)))) m^4) + \\
& \text{A0 B1} \left(- \frac{8 \text{C2 D1 } \xi^4 (\chi f - \chi g) (\chi p + \chi q) m^4}{\chi f \chi g} - \frac{8 \text{C2 D0 } \xi^4 (\rho 2 (\chi f - \chi g) \chi p (\chi p + \chi q) + 2 \rho 1 \chi f \chi g (\chi l + \chi p + \chi q)) m^4}{\chi f \chi g \chi p} + \right. \\
& \frac{1}{\chi f \chi g \chi p \chi q} 4 \text{C1 D1 } \xi^4 \\
& (\rho 1 ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \chi f^2 + 2 \chi g (-2 \chi l \chi q - 2 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) - \\
& 4 \rho 2 (\chi f - \chi g) \chi p \chi q (\chi p + \chi q)) m^4 + \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} \\
& 4 \text{C1 D0 } \xi^2 ((2 \rho 1^2 \chi f^2 (\chi f - \chi g) (\chi l + \chi p + \chi q) (\chi l + 2 \chi q) \chi g^2 + \rho 1 \rho 2 \chi f \chi p ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \chi f^2 + \\
& 2 \chi g (-6 \chi l \chi q - 6 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) \chi g - \\
& \rho 2^2 (\chi f - \chi g) (\chi f^2 + 6 \chi g \chi f + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q)) \xi^2 + \\
& 2 \tau 1^2 \chi f^2 (\chi f - \chi g) \chi g^2 \chi l (\chi l - \chi p + \chi q) - \tau 2^2 (\chi f - \chi g) (\chi f^2 + 2 \chi g \chi f + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q) + \\
& 2 \tau 1 \tau 2 \chi f \chi g \chi p ((\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 + 2 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q))) m^4 + \\
& \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} 4 \text{C0 D1 } \xi^2 ((2 \rho 1^2 \chi f^2 (\chi f - \chi g) (\chi l + \chi q) (\chi p + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f \chi p ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \\
& \chi f^2 + 2 \chi g (-2 \chi l \chi q - 2 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) \chi g + \\
& \rho 2^2 (\chi f - \chi g) (\chi f^2 + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q)) \xi^2 + 2 \tau 1^2 \chi f^2 (\chi f - \chi g) \chi g^2 (\chi l + \chi q) (\chi p + \chi q) + \\
& (\chi f - \chi g) \chi p (\chi p + \chi q) ((\chi g^2 - \chi l^2) \chi p \chi q \tau^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau^2 + 2 \chi g^2)) - \\
& 2 \tau 1 \tau 2 \chi f \chi g \chi p ((\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 - 2 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q))) \\
& m^4 + \frac{1}{\chi f \chi g \chi p^3 \chi q} 4 \text{C0 D0 } \xi^2 \\
& ((4 \chi f^2 \chi g^2 (\chi l + \chi q) (\chi l + \chi p + \chi q) \rho 1^3 + 2 \rho 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l^2 + 2 (\chi p + 2 \chi q) \chi l + 3 \chi q (\chi p + \chi q)) \rho 1^2 + \\
& \rho 2^2 \chi p^2 (2 (\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 - 8 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + 2 \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q))) \rho 1 - \\
& 2 \rho 2^3 (\chi f - \chi g) \chi p^3 \chi q (\chi p + \chi q)) \xi^2 + 2 \rho 2 \chi p (\chi f (\chi f - \chi g) \chi g (\chi l^2 + 2 \chi q \chi l + \chi q (\chi p + \chi q)) \tau 1^2 + \\
& 4 \tau 2 \chi f \chi g \chi p \chi q (\chi l + \chi p + \chi q) \tau 1 + (\chi f - \chi g) \chi p (\chi p + \chi q) (\chi f \chi g - (\tau^2 + 1) \chi p \chi q)) + \\
& \rho 1 (4 \tau 1^2 \chi f^2 (\chi l + \chi q) (\chi l + \chi p + \chi q) \chi g^2 - 4 \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi p (\chi l + \chi q) (\chi p + \chi q) \chi g + \\
& \chi p ((4 (\chi l + \chi p + \chi q) \chi g^2 + \chi p ((2 \tau^2 + 2) \chi l \chi p + 2 (\tau^2 + 1) \chi q (\chi p + \chi q))) \chi f^2 + \\
& 4 \chi g \chi l \chi p (\chi p - \chi q) \chi f + \chi g^2 \chi p ((2 \tau^2 + 2) \chi l \chi p + 2 (\tau^2 + 1) \chi q (\chi p + \chi q)))) m^4) + \\
& \text{A0 B0} \left(\frac{16 \text{C2 D2 } \xi^4 \chi p \chi q m^4}{\chi f \chi g} - \frac{8 \text{C2 D1 } \xi^4 (\rho 1 (\chi f - \chi g) (\chi l + 2 \chi q) - 4 \rho 2 \chi p \chi q) m^4}{\chi f \chi g} - \right.
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{\chi f^2 \chi g^2 \chi p} \\
& 8 \text{ C2 D0 } \xi^2 ((2 \rho 1^2 \chi f^2 (\chi 1 + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi p (\chi 1 + 2 \chi q) \chi g + \rho 2^2 (\chi f^2 + \chi g^2 - \chi 1^2) \chi p^2 \chi q) \xi^2 + \\
& \quad 2 \tau 1^2 \chi f^2 \chi g^2 (\chi 1 + \chi q) - \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi 1 + 2 \chi q) + \\
& \quad \chi p ((\chi g^2 - \chi 1^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2))) m^4 - \\
& \frac{1}{\chi f^2 \chi g^2 \chi p} 8 \text{ D2 } \xi^2 (C1 \chi f \chi g \chi p (\rho 1 (\chi f - \chi g) (\chi 1 + 2 \chi q) - 4 \rho 2 \chi p \chi q) \xi^2 + \\
& \quad C0 ((2 \rho 1^2 \chi f^2 (\chi 1 + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi p (\chi 1 + 2 \chi q) \chi g + \rho 2^2 (\chi f^2 + \chi g^2 - \chi 1^2) \chi p^2 \chi q) \xi^2 + \\
& \quad 2 \tau 1^2 \chi f^2 \chi g^2 (\chi 1 + \chi q) - \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi 1 + 2 \chi q) + \\
& \quad \chi p ((\chi g^2 - \chi 1^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2)))) m^4 + \\
& \frac{1}{\chi f^2 \chi g^2 \chi p \chi q} 4 \text{ C1 D1 } \xi^2 (2 \chi g \chi p^2 \chi f^3 + 2 \chi g \chi q^2 \chi f^3 - 4 \chi g \chi p \chi q \chi f^3 + 4 \tau 2^2 \chi p^2 \chi q^2 \chi f^2 + \\
& \quad 2 \chi g^3 \chi p^2 \chi f + 2 \chi g^3 \chi q^2 \chi f + 8 \tau 2^2 \chi g \chi p^2 \chi q^2 \chi f - 4 \chi g^3 \chi p \chi q \chi f + \\
& \quad \tau 1^2 \chi g ((2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2) \chi f^2 + 2 \chi g (4 \chi q^2 + 4 \chi 1 \chi q) \chi f + \chi g^2 (2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2)) \chi f + \\
& \quad 4 \tau 2^2 \chi g^2 \chi p^2 \chi q^2 - 4 \tau 2^2 \chi 1^2 \chi p^2 \chi q^2 + \\
& \quad \xi^2 (\chi f \chi g ((2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2) \chi f^2 + 2 \chi g (-4 \chi q^2 - 4 \chi 1 \chi q) \chi f + \chi g^2 (2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2)) \rho 1^2 - \\
& \quad 8 \rho 2 \chi f (\chi f - \chi g) \chi g \chi p \chi q (\chi 1 + 2 \chi q) \rho 1 + 4 \rho 2^2 (\chi f^2 + 6 \chi g \chi f + \chi g^2 - \chi 1^2) \chi p^2 \chi q^2)) m^4 + \\
& \frac{1}{\chi f \chi g \chi p^2 \chi q} 4 \text{ C1 D0 } \xi^2 ((2 \chi f (\chi f - \chi g) \chi g (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2) \rho 1^3 + \\
& \quad \rho 2 \chi p ((2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2) \chi f^2 + 2 \chi g (-8 \chi q^2 - 8 \chi 1 \chi q) \chi f + \chi g^2 (2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2)) \rho 1^2 - \\
& \quad 6 \rho 2^2 (\chi f - \chi g) \chi p^2 \chi q (\chi 1 + 2 \chi q) \rho 1 + 8 \rho 2^3 \chi p^3 \chi q^2) \xi^2 + \\
& \quad \rho 2 \chi p (((2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2) \chi f^2 + \chi g^2 (2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2)) \tau 1^2 + \\
& \quad 4 \tau 2 (\chi f - \chi g) \chi p \chi q (\chi 1 + 2 \chi q) \tau 1 + 2 \chi g^2 \chi p^2 + 2 \chi f^2 (\chi p - \chi q)^2 + \\
& \quad 2 \chi g^2 \chi q^2 + 8 \tau 2^2 \chi p^2 \chi q^2 + 8 \chi p^2 \chi q^2 - 4 \chi g^2 \chi p \chi q - 8 \chi f \chi g \chi p \chi q) + \\
& \quad 2 \rho 1 (\chi f (\chi f - \chi g) \chi g (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2) \tau 1^2 + 8 \tau 2 \chi f \chi g \chi p \chi q (\chi 1 + \chi q) \tau 1 + \\
& \quad (\chi f - \chi g) (\chi f \chi g (\chi p^2 + \chi q (\chi 1 + \chi q)) - (\tau 2^2 + 1) \chi p^2 \chi q (\chi 1 + 2 \chi q)))) m^4 + \\
& \frac{1}{\chi f \chi g \chi p^2 \chi q} 4 \text{ C0 D1 } \xi^2 ((2 \chi f (\chi f - \chi g) \chi g (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2) \rho 1^3 + \\
& \quad \rho 2 \chi p ((2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2) \chi f^2 + 2 \chi g (-8 \chi q^2 - 8 \chi 1 \chi q) \chi f + \chi g^2 (2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2)) \rho 1^2 - \\
& \quad 6 \rho 2^2 (\chi f - \chi g) \chi p^2 \chi q (\chi 1 + 2 \chi q) \rho 1 + 8 \rho 2^3 \chi p^3 \chi q^2) \xi^2 + \\
& \quad \rho 2 \chi p (((2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2) \chi f^2 + \chi g^2 (2 \chi 1^2 + 4 \chi q \chi 1 + 4 \chi q^2)) \tau 1^2 + \\
& \quad 4 \tau 2 (\chi f - \chi g) \chi p \chi q (\chi 1 + 2 \chi q) \tau 1 + 2 \chi g^2 \chi p^2 + 2 \chi f^2 (\chi p - \chi q)^2 + \\
& \quad 2 \chi g^2 \chi q^2 + 8 \tau 2^2 \chi p^2 \chi q^2 + 8 \chi p^2 \chi q^2 - 4 \chi g^2 \chi p \chi q - 8 \chi f \chi g \chi p \chi q) + \\
& \quad 2 \rho 1 (\chi f (\chi f - \chi g) \chi g (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2) \tau 1^2 + 8 \tau 2 \chi f \chi g \chi p \chi q (\chi 1 + \chi q) \tau 1 + \\
& \quad (\chi f - \chi g) (\chi f \chi g (\chi p^2 + \chi q (\chi 1 + \chi q)) - (\tau 2^2 + 1) \chi p^2 \chi q (\chi 1 + 2 \chi q)))) m^4 + \\
& \frac{1}{\chi f \chi g \chi p^3 \chi q} 4 \text{ C0 D0 } ((4 \chi f^2 \chi g^2 (\chi 1 + \chi q)^2 \rho 1^4 + 4 \rho 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2) \rho 1^3 + \\
& \quad \rho 2^2 \chi p^2 (2 (\chi 1^2 + 2 \chi q \chi 1 + 2 \chi q^2) \chi f^2 - 16 \chi g \chi q (\chi 1 + \chi q) \chi f + 2 \chi g^2 (\chi 1^2 + 2 \chi q \chi 1 + 2 \chi q^2)) \rho 1^2 - \\
& \quad 4 \rho 2^3 (\chi f - \chi g) \chi p^3 \chi q (\chi 1 + 2 \chi q) \rho 1 + 4 \rho 2^4 \chi p^4 \chi q^2) \xi^4 + \\
& \quad (((4 ((2 \tau 1^2 + 1) \chi 1^2 + 2 (2 \chi q \tau 1^2 + \chi q) \chi 1 + \chi p^2 + 2 \tau 1^2 \chi q^2 + \chi q^2) \chi g^2 - 4 \tau 1 \tau 2 \chi p \\
& \quad (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2) \chi g + \chi p^2 ((2 \tau 2^2 + 2) \chi 1^2 + 4 (\tau 2^2 + 1) \chi q \chi 1 + 4 (\tau 2^2 + 1) \chi q^2)) \\
& \quad \chi f^2 + 2 \chi g \chi p (2 \chi p \chi 1^2 + 2 \tau 1 \tau 2 \chi g (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2)) \chi f + \\
& \quad \chi g^2 \chi p^2 ((2 \tau 2^2 + 2) \chi 1^2 + 4 (\tau 2^2 + 1) \chi q \chi 1 + 4 (\tau 2^2 + 1) \chi q^2)) \rho 1^2 + \\
& \quad 4 \rho 2 \chi p (\chi f (\chi f - \chi g) \chi g (\chi 1^2 + 3 \chi q \chi 1 + 2 \chi q^2) \tau 1^2 + 8 \tau 2 \chi f \chi g \chi p \chi q (\chi 1 + \chi q) \tau 1 +
\end{aligned}$$

$$\begin{aligned}
& (\chi_f - \chi_g) (\chi_f \chi_g (\chi p^2 + \chi q (\chi l + \chi q)) - (\tau^2 + 1) \chi p^2 \chi q (\chi l + 2 \chi q)) \rho l + \\
& \rho^2 \chi p^2 ((2 (\chi l^2 + 2 \chi q \chi l + 2 \chi q^2) \chi f^2 + 2 \chi g^2 (\chi l^2 + 2 \chi q \chi l + 2 \chi q^2)) \tau l^2 + \\
& 4 \tau^2 (\chi_f - \chi_g) \chi p \chi q (\chi l + 2 \chi q) \tau l + 2 \chi f^2 \chi p^2 + 2 \chi g^2 \chi p^2 + \\
& 2 \chi f^2 \chi q^2 + 2 \chi g^2 \chi q^2 + 8 \tau^2 \chi p^2 \chi q^2 + 8 \chi p^2 \chi q^2 - 4 \chi l^2 \chi p \chi q) \xi^2 + \\
& 4 \tau l^4 \chi f^2 \chi g^2 (\chi l + \chi q)^2 - 4 \tau l^3 \tau^2 \chi f (\chi_f - \chi_g) \chi g \chi p (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) + \\
& \chi p^2 (4 \chi p^2 \chi q^2 \tau^2 + 8 \chi p^2 \chi q^2 \tau^2 - 4 \chi l^2 \chi p \chi q \tau^2 + 4 \chi p^2 \chi q^2 + \\
& 2 \chi f \chi g (2 \chi p^2 - 8 \chi q \chi p + 2 \chi q^2) + \chi g^2 ((2 \tau^2 + 2) \chi p^2 - 4 \chi q \chi p + (2 \tau^2 + 2) \chi q^2) + \\
& \chi f^2 (4 \chi g^2 + (2 \tau^2 + 2) \chi p^2 + (2 \tau^2 + 2) \chi q^2 - 4 \chi p \chi q)) + \\
& 4 \tau l \tau^2 (\chi_f - \chi_g) \chi p ((\tau^2 + 1) \chi p^2 \chi q (\chi l + 2 \chi q) - \chi f \chi g (\chi p^2 + \chi q (\chi l + \chi q))) + \\
& \tau l^2 ((4 (\chi l^2 + 2 \chi q \chi l + \chi p^2 + \chi q^2) \chi g^2 + \chi p^2 ((2 \tau^2 + 2) \chi l^2 + 4 (\tau^2 + 1) \chi q \chi l + 4 (\tau^2 + 1) \chi q^2)) \chi f^2 + \\
& 2 \chi g \chi p^2 (-8 \chi q^2 \tau^2 - 8 \chi l \chi q \tau^2 + 2 \chi l^2) \chi f + \\
& \chi g^2 \chi p^2 ((2 \tau^2 + 2) \chi l^2 + 4 (\tau^2 + 1) \chi q \chi l + 4 (\tau^2 + 1) \chi q^2))) m^4 \Big)
\end{aligned}$$

Check

No terms linear in a, as expected

```
In[75]:= Collect[Coefficient[Expr4, Pair[Momentum[a, D], Momentum[p, D]], e]
```

```
Out[75]= 0
```

Finally, in this step, recall that

$$l = p + s k - q$$

$$l' = p + t k - q$$

Namely,

$$l^2 = 2 m^2 + 2 s [(kp) - (kq)] - 2 (pq)$$

$$(l')^2 = 2 m^2 + 2 t [(kp) - (kq)] - 2 (pq)$$

so we need to substitute (pq) here as well

$$\frac{1}{(l^2 + i\epsilon)(l'^2 + i\epsilon)} \text{ enters Coeff, therefore we rewrite it}$$

Note that we still did not pass to

new variables in the integral measure ,

we will introduce the Jacobian determinant in the following steps

```
In[76]:= Diffs4 = Diffs3;
```

```
Coeff4 = Simplify[
```

```
  Coeff3 /. {l1v2 -> 2 m^2 + 2 s (kp - kq) - 2 pq, l2v2 -> 2 m^2 + 2 t (kp - kq) - 2 pq} /.
  ktochiSubs[p, q, f, g, l]]
```

```
Out[77]= (a^2 \xi^3 \chi p^2 \chi q^2)/(64 \pi^2 g0 L \phi m^2 p0 q0 \chi f (m^2 (\xi^3 \rho l^2 \chi l^2 + \xi (\tau l^2 \chi l^2 + (\chi p - \chi q)^2) + 2 s \chi p \chi q (\chi q - \chi p)) - i \epsilon \xi \chi p \chi q)
  (i \epsilon \xi \chi p \chi q + m^2 (\xi^3 \rho l^2 \chi l^2 + \xi (\tau l^2 \chi l^2 + (\chi p - \chi q)^2) + 2 t \chi p \chi q (\chi q - \chi p))))
```

Step 6: Recovering Airy functions

Let us now substitute the explicit form of A_n , B_n , C_n , D_n

$$A_0(s, q, p) = \int \frac{d\phi}{2\pi} e^{is\phi} e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} = \frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \text{Ai}(y)$$

$$y = y(s, q, p) = \frac{1}{[4\beta(q, p)]^{1/3}} \left[s - \frac{\alpha^2(q, p)}{16\beta(q, p)} \right]$$

Recall that

$$M(s, \tilde{s} - s, t, \tilde{s} - t) = m(s, \tilde{s} - s) m^+(t, \tilde{s} - t)$$

$$m(s, \tilde{s} - s) = e^2 \frac{1}{l^2 + i0} [\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s, q, p) u_{p\lambda_1}] [\bar{u}_{g\sigma_2} \tilde{\Gamma}_\mu(\tilde{s} - s, f, -g) v_{g\sigma_1}]$$

$$m^+(t, \tilde{s} - t) =$$

$$e^2 \frac{1}{l'^2 - i0} [\bar{u}_{p\lambda_1} (\tilde{\Gamma}^{\mu 1}(t, q, p))^+ u_{q\lambda_2}] [\bar{v}_{g\sigma_1} (\tilde{\Gamma}_{\mu 1}(\tilde{s} - t, f, -g))^+ u_{f\sigma_2}]$$

Steps are the following

1. Introduce the Airy function arguments explicitly and rewrite them via invariant variables
2. Reexpress previously introduced A_n , B_n , C_n , D_n explicitly via A_n , B_n functions
3. Substitute the definition of A_n , B_n
4. Simplify the overall expression and collect terms

1. Airy function arguments

First, we rewrite

$$\tilde{s} = \frac{\xi}{m^2 \chi_f} (pq + pg - qg - m^2)$$

in terms of invariants. For this,

we use the results for scalar products obtained above

Recall the definitions

$$p + sk = q + l$$

$$\chi_p = \chi_q + \chi_l$$

$$\rho_1 = \frac{e(qFp)}{\xi m^4 \chi_l} = -\frac{e(lFp)}{\xi m^4 \chi_l} = \frac{e(qFl)}{\xi m^4 \chi_l}$$

$$\tau_1 = -\frac{e(qF^* p)}{m^4 \chi_l} = \frac{e(lF^* p)}{m^4 \chi_l} = -\frac{e(qF^* l)}{m^4 \chi_l}$$

$$t_k + l = f + g$$

$$\chi_l = \chi_f + \chi_g$$

$$\rho_2 = -\frac{e(gFf)}{\xi m^4 \chi_l} = -\frac{e(gFl)}{\xi m^4 \chi_l}$$

$$\tau_2 = -\frac{e(gF^* f)}{m^4 \chi_l} = -\frac{e(gF^* l)}{m^4 \chi_l}$$

Commands `yArg`, `αdef`, and `βdef` are introduced in `definitions.nb`. They are used to write the Airy function arguments explicitly

```
In[78]:= yArg[s, q, p]
Expand[%]
```

$$\text{Out[78]} = 2 \sqrt[3]{2} \beta(q, p)^{2/3} \left(\frac{s}{4 \beta(q, p)} - \frac{\alpha(q, p)^2}{64 \beta(q, p)^2} \right)$$

$$\text{Out[79]} = \frac{s}{2^{2/3} \sqrt[3]{\beta(q, p)}} - \frac{\alpha(q, p)^2}{16 \times 2^{2/3} \beta(q, p)^{4/3}}$$

```
In[80]:= Simplify[(ξ / m^2 / χf) (pq + pg - qg - m^2)];
Expand[% /. {χl → χp - χq}];
Collect[%, {1 / χf / χg, 1 / χf / χp / χq, ρ1, τ1, ρ2, τ2}, Simplify]
stildeDef = Simplify[% /. {χp - χq → χl, χg - χp + χq → -χf}]
```

$$\text{Out[82]} = -\frac{\xi^3 \rho_1^2 (\chi p - \chi q) (\chi g - \chi p + \chi q)}{2 \chi f \chi p \chi q} + \frac{\frac{1}{2} \xi^3 \rho_2^2 (\chi p - \chi q) + \frac{1}{2} \xi \tau_2^2 (\chi p - \chi q) + \frac{1}{2} \xi (\chi p - \chi q)}{\chi f \chi g} -$$

$$\frac{\xi \tau_1^2 (\chi p - \chi q) (\chi g - \chi p + \chi q)}{2 \chi f \chi p \chi q} - \frac{\xi (\chi p - \chi q) (\chi g - \chi p + \chi q)}{2 \chi f \chi p \chi q}$$

$$\text{Out[83]} = \frac{\xi \chi l (\chi f \chi g (\xi^2 \rho_1^2 + \tau_1^2 + 1) + \chi p \chi q (\xi^2 \rho_2^2 + \tau_2^2 + 1))}{2 \chi f \chi g \chi p \chi q}$$

Now, we rewrite α_n and β_n in terms of invariants
by replacing contractions with $a^\mu [(ap), (af), \text{ etc}]$

```

In[84]:=  $\alpha$ def[q, p] /. {Pair[Momentum[a, D], Momentum[q, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  qv[ $\mu$ ]}
 $\alpha$ qpInv =
  Simplify[Contract[% /. {qv[ $\mu$ ]  $\rightarrow$  qpexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2} /. {xp - xq  $\rightarrow$  x $l$ , xq - xp  $\rightarrow$  -x $l$ }
 $\beta$ def[q, p] /. {Pair[Momentum[a, D], Momentum[q, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  qv[ $\mu$ ]}
 $\beta$ qpInv =
  Simplify[Contract[% /. {qv[ $\mu$ ]  $\rightarrow$  qpexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2} /. {xp - xq  $\rightarrow$  x $l$ , xq - xp  $\rightarrow$  -x $l$ }

```

$$\text{Out[84]} = -e \left(\frac{a \cdot p}{k \cdot p} - \frac{a^\mu q^\mu}{k \cdot q} \right)$$

$$\text{Out[85]} = -\frac{\xi^3 \rho_1 \chi^1}{\chi p \chi q}$$

$$\text{Out[86]} = \frac{1}{8} a^2 e^2 \left(\frac{1}{k \cdot p} - \frac{1}{k \cdot q} \right)$$

$$\text{Out[87]} = \frac{\xi^3 \chi^1}{8 \chi p \chi q}$$

```

In[88]:=  $\alpha$ def[f, -g] /. {Pair[Momentum[a, D], Momentum[g, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  gv[ $\mu$ ]}
 $\alpha$ fgInv =
  Simplify[Contract[% /. {gv[ $\mu$ ]  $\rightarrow$  gfexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2} /. {xf + xg  $\rightarrow$  x $l$ }
 $\beta$ def[f, -g] /. {Pair[Momentum[a, D], Momentum[g, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  gv[ $\mu$ ]}
 $\beta$ fgInv =
  Simplify[Contract[% /. {gv[ $\mu$ ]  $\rightarrow$  gfexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2} /. {xf + xg  $\rightarrow$  x $l$ }

```

$$\text{Out[88]} = -e \left(\frac{a^\mu g^\mu}{g \cdot k} - \frac{a \cdot f}{f \cdot k} \right)$$

$$\text{Out[89]} = -\frac{\xi^3 \rho_2 \chi^1}{\chi f \chi g}$$

$$\text{Out[90]} = \frac{1}{8} a^2 e^2 \left(-\frac{1}{f \cdot k} - \frac{1}{g \cdot k} \right)$$

$$\text{Out[91]} = \frac{\xi^3 \chi^1}{8 \chi f \chi g}$$

With this, we can write the Airy function arguments explicitly

Notations

ysqp == y (s, q, p)

ytqp == y (t, q, p)

In[92]:= `yArg[s, q, p] /. {α[q, p] → αqpInv, β[q, p] → βqpInv};`

$$ysqp = \frac{1}{\xi} \left(\frac{\chi l}{2 \chi p \chi q} \right)^{-1/3} \text{Simplify} \left[\text{Expand}[\%] \left(\frac{\xi^3 \chi l}{2 \chi p \chi q} \right)^{1/3} \right]$$

`yArg[t, q, p] /. {α[q, p] → αqpInv, β[q, p] → βqpInv};`

$$ytqp = \frac{1}{\xi} \left(\frac{\chi l}{2 \chi p \chi q} \right)^{-1/3} \text{Simplify} \left[\text{Expand}[\%] \left(\frac{\xi^3 \chi l}{2 \chi p \chi q} \right)^{1/3} \right]$$

Out[93]=

$$\frac{\sqrt[3]{2} \left(s - \frac{\xi^3 \rho l^2 \chi l}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi l}{\chi p \chi q}}}$$

Out[95]=

$$\frac{\sqrt[3]{2} \left(t - \frac{\xi^3 \rho l^2 \chi l}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi l}{\chi p \chi q}}}$$

Notation

`ysfg == y (s, f, -g)`

`ystsfg == y (s̃ - s, f, -g)`

`ytfg == y (t, f, -g)`

`ysttfg == y (s̃ - t, f, -g)`

In[96]:= $\text{yArg}[s, f, -g] /. \{\alpha[f, -g] \rightarrow \alpha f g \text{Inv}, \beta[f, -g] \rightarrow \beta f g \text{Inv}\};$

$$\text{ysfg} = \frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Expand} \left[\% \left(\frac{\xi^3 \chi^l}{2 \chi^f \chi^g} \right)^{1/3} \right]$$

$\text{ystsfg} =$

$$\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Collect} \left[\text{Expand} \left[\left(\text{ysfg} / \left(\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \right) \right) /. \{s \rightarrow \text{stildeDef} - s\} \right], \{\chi^p, \chi^q, \chi^f, \chi^g\}, \text{Simplify} \right]$$

$\text{yArg}[t, f, -g] /. \{\alpha[f, -g] \rightarrow \alpha f g \text{Inv}, \beta[f, -g] \rightarrow \beta f g \text{Inv}\};$

$$\text{ytf} = \frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Expand} \left[\% \left(\frac{\xi^3 \chi^l}{2 \chi^f \chi^g} \right)^{1/3} \right]$$

$\text{ysttf} =$

$$\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Collect} \left[\text{Expand} \left[\left(\text{ytf} / \left(\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \right) \right) /. \{t \rightarrow \text{stildeDef} - t\} \right], \{\chi^p, \chi^q, \chi^f, \chi^g\}, \text{Simplify} \right]$$

Out[97]=
$$\frac{\sqrt[3]{2} \left(s - \frac{\xi^3 \rho^2 \chi^l}{2 \chi^f \chi^g} \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

Out[98]=
$$\frac{\sqrt[3]{2} \left(\frac{\xi \chi^l (\xi^2 \rho^2 l^2 + \tau l^2 + 1)}{2 \chi^p \chi^q} + \frac{\xi (\tau^2 + 1) \chi^l}{2 \chi^f \chi^g} - s \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

Out[100]=
$$\frac{\sqrt[3]{2} \left(t - \frac{\xi^3 \rho^2 \chi^l}{2 \chi^f \chi^g} \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

Out[101]=
$$\frac{\sqrt[3]{2} \left(\frac{\xi \chi^l (\xi^2 \rho^2 l^2 + \tau l^2 + 1)}{2 \chi^p \chi^q} + \frac{\xi (\tau^2 + 1) \chi^l}{2 \chi^f \chi^g} - t \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

2. Express A_n , B_n , C_n , D_n explicitly

Let us define the substitutions to revert pre

$$A_n \rightarrow A_n[s, q, p]$$

$$B_n \rightarrow B_n[t, q, p]$$

$$C_n \rightarrow A_n[\tilde{s} - s, f, -g]$$

$$D_n \rightarrow B_n[\tilde{s} - t, f, -g]$$

Note that `AnsubsNoFactor` does not contain the factors

$$1 / (4 \beta)^{1/3} \text{ and the exponentials } \text{Exp} \left[i \left(\frac{\alpha s}{8 \beta} - \frac{\alpha^3}{192 \beta^2} \right) \right]$$

We will store the former in Coeff5 and the phase of the latter in Phase5

```
In[102]:= SubsAnBack = {A0 → A0[s, q, p], A1 → A1[s, q, p], A2 → A2[s, q, p],
  B0 → B0[t, q, p], B1 → B1[t, q, p], B2 → B2[t, q, p],
  C0 → A0[-s + stilde, f, -g],
  C1 → A1[-s + stilde, f, -g], C2 → A2[-s + stilde, f, -g],
  D0 → B0[stilde - t, f, -g], D1 → B1[stilde - t, f, -g], D2 → B2[stilde - t, f, -g]}
SubsyArgsInAi =
  {AiryAi[y[s, q, p]] → AiryAi[ys1], AiryAi[y[t, q, p]] → AiryAi[yt1],
  AiryAi[y[-s + stilde, f, -g]] → AiryAi[ys2],
  AiryAi[y[stilde - t, f, -g]] → AiryAi[yt2],
  AiryAiPrime[y[s, q, p]] → AiryAiPrime[ys1],
  AiryAiPrime[y[t, q, p]] → AiryAiPrime[yt1],
  AiryAiPrime[y[-s + stilde, f, -g]] → AiryAiPrime[ys2],
  AiryAiPrime[y[stilde - t, f, -g]] → AiryAiPrime[yt2]}
Subsaβ = {α[q, p] → αqpInv, β[q, p] → βqpInv, α[f, -g] → αfgInv, β[f, -g] → βfgInv}
SubsyArgExplicit = {y[s, q, p] → ysqp, y[t, q, p] → ytcp,
  y[-s + stilde, f, -g] → ystsfg, y[stilde - t, f, -g] → ysttfg}

Out[102]:= {A0 → A0(s, q, p), A1 → A1(s, q, p), A2 → A2(s, q, p), B0 → B0(t, q, p), B1 → B1(t, q, p),
  B2 → B2(t, q, p), C0 → A0(stilde - s, f, -g), C1 → A1(stilde - s, f, -g), C2 → A2(stilde - s, f, -g),
  D0 → B0(stilde - t, f, -g), D1 → B1(stilde - t, f, -g), D2 → B2(stilde - t, f, -g)}

Out[103]:= {Ai(y(s, q, p)) → Ai(ys1), Ai(y(t, q, p)) → Ai(yt1), Ai(y(stilde - s, f, -g)) → Ai(ys2),
  Ai(y(stilde - t, f, -g)) → Ai(yt2), Ai'(y(s, q, p)) → Ai'(ys1), Ai'(y(t, q, p)) → Ai'(yt1),
  Ai'(y(stilde - s, f, -g)) → Ai'(ys2), Ai'(y(stilde - t, f, -g)) → Ai'(yt2)}

Out[104]:= {α(q, p) → - $\frac{\xi^3 \rho^1 \chi^1}{\chi p \chi q}$ , β(q, p) →  $\frac{\xi^3 \chi^1}{8 \chi p \chi q}$ , α(f, -g) → - $\frac{\xi^3 \rho^2 \chi^1}{\chi^f \chi g}$ , β(f, -g) →  $\frac{\xi^3 \chi^1}{8 \chi^f \chi g}$ }

Out[105]:= {y(s, q, p) →  $\frac{\sqrt[3]{2} \left( s - \frac{\xi^3 \rho^1 \chi^1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi p \chi q}}}$ , y(t, q, p) →  $\frac{\sqrt[3]{2} \left( t - \frac{\xi^3 \rho^1 \chi^1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi p \chi q}}}$ ,
  y(stilde - s, f, -g) →  $\frac{\sqrt[3]{2} \left( \frac{\xi \chi^1 (\xi^2 \rho^1 \chi^1 + \tau^1 \chi^1 + 1)}{2 \chi p \chi q} + \frac{\xi (\tau^2 + 1) \chi^1}{2 \chi^f \chi g} - s \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi^f \chi g}}}$ , y(stilde - t, f, -g) →  $\frac{\sqrt[3]{2} \left( \frac{\xi \chi^1 (\xi^2 \rho^1 \chi^1 + \tau^1 \chi^1 + 1)}{2 \chi p \chi q} + \frac{\xi (\tau^2 + 1) \chi^1}{2 \chi^f \chi g} - t \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi^f \chi g}}}$ }
```

Symbol AnsubsNoFactor is introduced in definitions.nb

It contains the substitutions to re -

introduce A_n and B_n explicitly (see the input cell below),
expressed via Airy functions

Note that Airy functions are expressed

using Mathematica's AiryAi and AiryAiPrime,
which obey usual Airy function properties

In[106]:= **AnsSubsNoFactor**

AnFactor = 1 / (4 β [q, p] 4 β [f, -g] * 4 β [q, p] 4 β [f, -g]) ^ (1 / 3)

Out[106]= $\{A_0(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)), B_0(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)),$

$$A_1(s_-, q_-, p_-) \rightarrow \frac{\alpha(q, p) \text{Ai}(y(s, q, p))}{8 \beta(q, p)} - \frac{i \text{Ai}'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}}, B_1(s_-, q_-, p_-) \rightarrow \frac{\alpha(q, p) \text{Ai}(y(s, q, p))}{8 \beta(q, p)} + \frac{i \text{Ai}'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}},$$

$$A_2(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) - \frac{i \alpha(q, p) \text{Ai}'(y(s, q, p))}{4 \times 2^{2/3} \beta(q, p)^{4/3}},$$

$$B_2(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) + \frac{i \alpha(q, p) \text{Ai}'(y(s, q, p))}{4 \times 2^{2/3} \beta(q, p)^{4/3}} \}$$

Out[107]=
$$\frac{1}{4 \times 2^{2/3} \sqrt[3]{\beta(f, -g)^2 \beta(q, p)^2}}$$

Some default assumptions used in command Simplify below

In[108]:= **assump ξ χ = { $\xi > 0$, $\chi p > 0$, $\chi q > 0$, $\chi f > 0$, $\chi g > 0$ }**

assumpAll = { $u > 0$, $v > 4$, $\chi p > 0$, $\chi q > 0$, $\chi g > 0$, $\chi f > 0$, $\chi f > 0$, $\xi > 0$ }

\$Assumptions = assumpAll;

Out[108]= { $\xi > 0$, $\chi p > 0$, $\chi q > 0$, $\chi f > 0$, $\chi g > 0$ }

Out[109]= { $u > 0$, $v > 4$, $\chi p > 0$, $\chi q > 0$, $\chi g > 0$, $\chi f > 0$, $\chi f > 0$, $\xi > 0$ }

```

In[111]:= (*a small test*)
A2 B2 C2 D2 /. SubsAnBack /. AnsubsNoFactor /. SubsyArgsInAi
% /. Subsaβ /. SubsyArgExplicit

```

$$\begin{aligned}
\text{Out[111]} = & \left(\text{Ai}(\text{ys1}) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) - \frac{i \text{Ai}'(\text{ys1}) \alpha(q, p)}{4 \times 2^{2/3} \beta(q, p)^{4/3}} \right) \\
& \left(\text{Ai}(\text{yt1}) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(t, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) + \frac{i \text{Ai}'(\text{yt1}) \alpha(q, p)}{4 \times 2^{2/3} \beta(q, p)^{4/3}} \right) \\
& \left(\text{Ai}(\text{ys2}) \left(\frac{\alpha(f, -g)^2}{64 \beta(f, -g)^2} - \frac{y(\text{stilde} - s, f, -g)}{2 \sqrt[3]{2} \beta(f, -g)^{2/3}} \right) - \frac{i \text{Ai}'(\text{ys2}) \alpha(f, -g)}{4 \times 2^{2/3} \beta(f, -g)^{4/3}} \right) \\
& \left(\text{Ai}(\text{yt2}) \left(\frac{\alpha(f, -g)^2}{64 \beta(f, -g)^2} - \frac{y(\text{stilde} - t, f, -g)}{2 \sqrt[3]{2} \beta(f, -g)^{2/3}} \right) + \frac{i \text{Ai}'(\text{yt2}) \alpha(f, -g)}{4 \times 2^{2/3} \beta(f, -g)^{4/3}} \right) \\
\text{Out[112]} = & \left(\text{Ai}(\text{ys1}) \left(\rho 1^2 - \frac{2 \left(s - \frac{\xi^3 \rho 1^2 \chi 1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi p \chi q} \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{2/3}}} \right) + \frac{2 i \sqrt[3]{2} \xi^3 \rho 1 \chi 1 \text{Ai}'(\text{ys1})}{\chi p \chi q \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{4/3}} \right) \\
& \left(\text{Ai}(\text{yt1}) \left(\rho 1^2 - \frac{2 \left(t - \frac{\xi^3 \rho 1^2 \chi 1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi p \chi q} \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{2/3}}} \right) - \frac{2 i \sqrt[3]{2} \xi^3 \rho 1 \chi 1 \text{Ai}'(\text{yt1})}{\chi p \chi q \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{4/3}} \right) \\
& \left(\text{Ai}(\text{ys2}) \left(\rho 2^2 - \frac{2 \left(\frac{\xi \chi 1 (\xi^2 \rho 1^2 + \tau 1^2 + 1)}{2 \chi p \chi q} + \frac{\xi (\tau 2^2 + 1) \chi 1}{2 \chi f \chi g} - s \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi f \chi g} \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{2/3}}} \right) + \frac{2 i \sqrt[3]{2} \xi^3 \rho 2 \chi 1 \text{Ai}'(\text{ys2})}{\chi f \chi g \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{4/3}} \right) \\
& \left(\text{Ai}(\text{yt2}) \left(\rho 2^2 - \frac{2 \left(\frac{\xi \chi 1 (\xi^2 \rho 1^2 + \tau 1^2 + 1)}{2 \chi p \chi q} + \frac{\xi (\tau 2^2 + 1) \chi 1}{2 \chi f \chi g} - t \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi f \chi g} \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{2/3}}} \right) - \frac{2 i \sqrt[3]{2} \xi^3 \rho 2 \chi 1 \text{Ai}'(\text{yt2})}{\chi f \chi g \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{4/3}} \right)
\end{aligned}$$

```

In[113]:= Simplify[(AnPhase[s, q, p] +
  AnPhase[st - s, f, -g] - AnPhase[t, q, p] - AnPhase[st - t, f, -g])]
Phase5 = Simplify[% /. {st -> stildeDef} /. Subsaβ, Assumptions -> assumptξχ]

```

$$\text{Out[113]} = \frac{(s - t) (\beta(f, -g) \alpha(q, p) - \alpha(f, -g) \beta(q, p))}{8 \beta(f, -g) \beta(q, p)}$$

$$\text{Out[114]} = -((\rho 1 - \rho 2) (s - t))$$

```

In[115]:= Diffs5 = Diffs4;
Coeff5 = Coeff4;
Expr4 * AnFactor;
Expand[%] /. SubsAnBack /. AnsubsNoFactor /. SubsyArgsInAi /. Subsaβ /.
  SubsyArgExplicit;
Expr5 = Expand[%];

```

```
In[120]:= test = Coefficient[Expr5, AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2]];
Coefficient[test, ρ1];
Simplify[%, Assumptions → {u1 > 0, u2 > 1}]
Out[122]= 0
```

Let us now pass to new variables in the integral measure

$$u = \frac{\chi_l}{\chi_q}$$

$$v^2 = \frac{\chi_l^2}{\chi_g \chi_f}$$

$$\chi_g = \frac{\chi_l}{2} \left(1 + \sigma \sqrt{1 - \frac{4}{v}} \right), \quad \chi_f = \frac{\chi_l}{2} \left(1 - \sigma \sqrt{1 - \frac{4}{v}} \right),$$

$\sigma = \pm 1$ – two branches, we sum over them

We also should take into account the integral measure also transforms as

$$d^3g d^3q \rightarrow |J_1^{-1}| |J_2^{-1}| d\rho_1 d\tau_1 du d\rho_2 d\tau_2 dv$$

$$|J_1^{-1}| = q_0 m^2 \xi \frac{u^2}{(1+u)^3}$$

$$|J_2^{-1}| = g_0 m^2 \xi \frac{\chi_l}{\chi_g} \frac{1}{v \sqrt{v(v-4)}}$$

```
In[123]:= jac1 = q0 m^2 ξ  $\frac{u^2}{(1+u)^3}$ 
```

```
jac2 = g0 m^2 ξ  $\frac{\chi_l}{\chi_g} \frac{1}{v \text{Sqrt}[v(v-4)]}$ 
```

```
Out[123]=  $\frac{m^2 \xi q_0 u^2}{(u+1)^3}$ 
```

```
Out[124]=  $\frac{g_0 m^2 \xi \chi_l}{v \sqrt{(v-4)v} \chi_g}$ 
```

```
In[125]:= subsu1 = {χq → χp / (1 + u), χl → χp u / (1 + u)}
subsu2σ = {χg → (1 / 2) χl (1 + σ Sqrt[1 - 4 / v]), χf → (1 / 2) χl (1 - σ Sqrt[1 - 4 / v])}
```

```
Out[125]=  $\left\{ \chi_q \rightarrow \frac{\chi_p}{u+1}, \chi_l \rightarrow \frac{u \chi_p}{u+1} \right\}$ 
```

```
Out[126]=  $\left\{ \chi_g \rightarrow \frac{1}{2} \chi_l \left( \sigma \sqrt{1 - \frac{4}{v}} + 1 \right), \chi_f \rightarrow \frac{1}{2} \chi_l \left( 1 - \sigma \sqrt{1 - \frac{4}{v}} \right) \right\}$ 
```

```

In[127]:= Coeff5 * jac1 * jac2
% /. subsu2σ /. subsu1;
Simplify[% /. {σ → 1}, Assumptions → assumpAll];
Simplify[% /. {σ → -1}, Assumptions → assumpAll];
% - %% (*must be 0*)
Coeff6 = %%
Diffs6 = Diffs5 /. {d3q → dρ1 dτ1 du, d3g → dρ2 dτ2 dv}
Phase6 = Phase5

```

$$\text{Out[127]} = \frac{(\alpha^2 m^2 \xi^5 u^2 \chi l \chi p^2 \chi q^2) / (64 \pi^2 L \phi p0 (u+1)^3 v \sqrt{(v-4)v} \chi f \chi g (m^2 (\xi^3 \rho l^2 \chi l^2 + \xi (\tau l^2 \chi l^2 + (\chi p - \chi q)^2) + 2 s \chi p \chi q (\chi q - \chi p)) - i \epsilon \xi \chi p \chi q) (i \epsilon \xi \chi p \chi q + m^2 (\xi^3 \rho l^2 \chi l^2 + \xi (\tau l^2 \chi l^2 + (\chi p - \chi q)^2) + 2 t \chi p \chi q (\chi q - \chi p))))}{0}$$

Out[131]= 0

$$\text{Out[132]} = (\alpha^2 m^2 \xi^5 u) \left/ \left(16 \pi^2 L \phi p0 \left(\sqrt{\frac{v-4}{v}} - 1 \right) \left(\sqrt{\frac{v-4}{v}} + 1 \right) v \sqrt{(v-4)v} \chi p \right. \right. \\ \left. \left. (i \epsilon \xi (u+1) - m^2 u (\xi u (\xi^2 \rho l^2 + \tau l^2 + 1) - 2 s \chi p)) (i \epsilon \xi (u+1) + m^2 u (\xi u (\xi^2 \rho l^2 + \tau l^2 + 1) - 2 t \chi p)) \right) \right)$$

Out[133]= dρ1 dρ2 ds dt dτ1 dτ2 du dv

Out[134]= -((ρ1 - ρ2)(s - t))

Summation over two σ branches

```

In[135]:= Expr6σ = Expr5 /. subsu2σ /. subsu1;
Expr6 = (Expr6σ /. {σ → 1}) + (Expr6σ /. {σ → -1});

```

Summation over two σ branches

```

In[137]:= AAAA = Coefficient[Expr6, AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2]]
          AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2];
ApAAA = Coefficient[Expr6, AiryAi'[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2]]
        AiryAi'[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2];
AApAA = Coefficient[Expr6, AiryAi[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi[yt2]]
        AiryAi[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi[yt2];
AAApA = Coefficient[Expr6, AiryAi[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi[yt2]]
        AiryAi[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi[yt2];
AAAAp = Coefficient[Expr6, AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi'[yt2]]
        AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi'[yt2];
ApApAA = Coefficient[Expr6, AiryAi'[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi[yt2]]
        AiryAi'[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi[yt2];
ApAApA = Coefficient[Expr6, AiryAi'[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi[yt2]]
        AiryAi'[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi[yt2];
ApAAAp = Coefficient[Expr6, AiryAi'[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi'[yt2]]
        AiryAi'[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi'[yt2];
AApApA = Coefficient[Expr6, AiryAi[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi[yt2]]
        AiryAi[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi[yt2];
AApAAp = Coefficient[Expr6, AiryAi[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi'[yt2]]
        AiryAi[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi'[yt2];
AAApAp = Coefficient[Expr6, AiryAi[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi'[yt2]]
        AiryAi[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi'[yt2];
ApApApA = Coefficient[Expr6, AiryAi'[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi[yt2]]
        AiryAi'[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi[yt2];
ApApAAp = Coefficient[Expr6, AiryAi'[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi'[yt2]]
        AiryAi'[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi'[yt2];
ApAApAp = Coefficient[Expr6, AiryAi'[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi'[yt2]]
        AiryAi'[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi'[yt2];
AApApAp = Coefficient[Expr6, AiryAi[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi'[yt2]]
        AiryAi[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi'[yt2];
ApApApAp =
  Coefficient[Expr6, AiryAi'[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi'[yt2]]
    AiryAi'[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi'[yt2];
Expr6List = {AAAA, ApAAA, AApAA, AAApA, AAAAp,
             ApApAA, ApAApA, ApAAAp, AApApA, AApAAp, AAAPAp,
             ApApApA, ApApAAp, ApAApAp, AApApAp, ApApApAp};

In[154]:= (*checking, must be 0*)
          Expand[Expr6 - Total[Expr6List]]

Out[154]= 0

In[155]= Expr7
Out[155]= Expr7

```

```
In[156]:= Expr7List = Simplify[Expr6List /. {D -> 4},
  Assumptions -> {u > 0, v > 4, xp > 0, xq > 0, xg > 0, xf > 0, xi > 0}];
Expr7 = Total[Expr7List];
Diffs7 = Diffs6
Coeff7 = Coeff6
Phase7 = Phase6
Expr7
```

```
Out[158]= dρ1 dρ2 ds dt dτ1 dτ2 du dv
```

$$\text{Out[159]} = (a^2 m^2 \xi^5 u) \left/ \left(16 \pi^2 L \phi p_0 \left(\sqrt{\frac{v-4}{v}} - 1 \right) \left(\sqrt{\frac{v-4}{v}} + 1 \right) v \sqrt{(v-4)v} \chi p \right. \right. \right. \\ \left. \left. \left(i \epsilon \xi (u+1) - m^2 u (\xi^2 \rho_1^2 + \tau_1^2 + 1) - 2 s \chi p \right) (i \epsilon \xi (u+1) + m^2 u (\xi^2 \rho_1^2 + \tau_1^2 + 1) - 2 t \chi p) \right) \right)$$

$$\text{Out[160]} = -((\rho_1 - \rho_2)(s - t))$$

$$\text{Out[161]} = -\frac{1}{\xi^6 (u+1)^{5/3} v^{2/3}} 8 \sqrt[3]{2} m^4 \left(\sqrt{\frac{v-4}{v}} - 1 \right) \left(\sqrt{\frac{v-4}{v}} + 1 \right) \chi p^{4/3} \text{Ai}(y_{s1}) \text{Ai}(y_{s2}) \text{Ai}(y_{t1}) \\ \text{Ai}(y_{t2}) \left(-2 \xi u (8 \chi p (s+t) (\xi^2 \rho_1^2 + \tau_1^2 + 1) - \xi \tau_1^2 (\tau_2^2 + 1) v^2 + 4 \xi \tau_1^2 \tau_2^2 v) + 32 s t \chi p^2 + \right. \\ \left. \xi^2 u^2 (8 (\xi^2 \rho_1^2 + \tau_1^2 + 1)^2 + (\tau_1^2 + 1)(\tau_2^2 + 1) v^2 - 2 (\tau_1^2 + 1) \tau_2^2 v) + 2 \xi^2 \tau_1^2 (\tau_2^2 + 1) v^2 - 8 \xi^2 \tau_1^2 \tau_2^2 v \right) + \\ \frac{64 \times 2^{2/3} m^4 \chi p^{8/3} (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}'(y_{s1}) \text{Ai}'(y_{s2}) \text{Ai}'(y_{t1}) \text{Ai}'(y_{t2})}{\xi^4 (u+1)^{7/3} v^{4/3}} + \\ \frac{64 m^4 \tau_1 \tau_2 \chi p^2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}'(y_{s1}) \text{Ai}'(y_{s2}) \text{Ai}(y_{t1}) \text{Ai}(y_{t2})}{\xi^4 (u+1)^2 v} + \\ \frac{64 m^4 \tau_1 \tau_2 \chi p^2 (u^2 (v-2) + 2 u v + 2 v) \text{Ai}(y_{s1}) \text{Ai}'(y_{s2}) \text{Ai}'(y_{t1}) \text{Ai}(y_{t2})}{\xi^4 (u+1)^2 v} + \\ \frac{64 m^4 \tau_1 \tau_2 \chi p^2 (u^2 (v-2) + 2 u v + 2 v) \text{Ai}'(y_{s1}) \text{Ai}(y_{s2}) \text{Ai}(y_{t1}) \text{Ai}'(y_{t2})}{\xi^4 (u+1)^2 v} + \\ \frac{64 m^4 \tau_1 \tau_2 \chi p^2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}(y_{s1}) \text{Ai}(y_{s2}) \text{Ai}'(y_{t1}) \text{Ai}'(y_{t2})}{\xi^4 (u+1)^2 v} + \\ \frac{64 m^4 u^{2/3} \chi p^2 \text{Ai}(y_{s1}) \text{Ai}'(y_{s2}) \text{Ai}(y_{t1}) \text{Ai}'(y_{t2}) ((\tau_1^2 + 1) u^2 (v-2) + 2 \tau_1^2 u v + 2 \tau_1^2 v)}{\xi^4 (u+1)^{7/3} v^{4/3}} - \\ \frac{16 m^4 \left(\sqrt{\frac{v-4}{v}} - 1 \right) \left(\sqrt{\frac{v-4}{v}} + 1 \right) \sqrt[3]{v} \chi p^2 \text{Ai}'(y_{s1}) \text{Ai}(y_{s2}) \text{Ai}'(y_{t1}) \text{Ai}(y_{t2}) ((\tau_2^2 + 1) (u^2 + 2 u + 2) v - 2 \tau_2^2 u^2)}{\xi^4 u^{2/3} (u+1)^{5/3}}$$

```
In[162]:= Coefficient[Expr7, {ρ1, ρ1^2, ρ1^3, ρ1^4}];
Coefficient[%, AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2]]
```

$$\text{Out[163]} = \left\{ 0, -\frac{512 \sqrt[3]{2} m^4 s u \chi p^{7/3}}{\xi^3 (u+1)^{5/3} v^{5/3}} - \frac{512 \sqrt[3]{2} m^4 t u \chi p^{7/3}}{\xi^3 (u+1)^{5/3} v^{5/3}} + \frac{512 \sqrt[3]{2} m^4 \tau_1^2 u^2 \chi p^{4/3}}{\xi^2 (u+1)^{5/3} v^{5/3}} + \frac{512 \sqrt[3]{2} m^4 u^2 \chi p^{4/3}}{\xi^2 (u+1)^{5/3} v^{5/3}}, \right. \\ \left. 0, \frac{256 \sqrt[3]{2} m^4 u^2 \chi p^{4/3}}{(u+1)^{5/3} v^{5/3}} \right\}$$

```
In[164]:= Coefficient[Expr7, {ρ2, ρ2^2, ρ2^3, ρ2^4}]
Out[164]= {0, 0, 0, 0}
```

Step 7: Shift of s and t and intermediate results

We shift s and t variables to simplify Coeff7 and remove the explicit dependence on τi and ρi in the denominator

Note that we use the property

$$i \in * X = i \in \text{for } X > 0$$

```
In[165]:= shift = {s → s +  $\frac{u \xi}{2 \chi p} (1 + \tau 1^2 + \xi^2 \rho 1^2)$ , t → t +  $\frac{u \xi}{2 \chi p} (1 + \tau 1^2 + \xi^2 \rho 1^2)$ };
factor8 = (8 m^2 u χ p / ξ^3 / (1 + u))^2;
Coeff8 = Simplify[Coeff7 * factor8 /. {iε → 0} /. shift] /.
  {1 / s → 1 / (s + iε), 1 / t → 1 / (t - iε)}
Diffs8 = Diffs7;
Phase8 = Phase7;
```

```
Out[167]= 
$$\frac{\alpha^2 m^2 u}{4 \pi^2 L \phi \xi p_0 (u + 1)^2 \sqrt{(v - 4) v} \chi p (i\epsilon + s) (t - i\epsilon)}$$

```

In[170]:= Expr81 =

```

Expand[Simplify[Expr7 / factor8 /. shift] /. {  $\left(\sqrt{\frac{v-4}{v}} - 1\right) \left(\sqrt{\frac{v-4}{v}} + 1\right) \rightarrow -\frac{4}{v}$  }];
(Simplify[Coefficient[Expr81, AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2]],
  Assumptions → assumpAll] AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2] +
  (*AAApAp*)
Simplify[Coefficient[Expr81, AiryAi[ys1] AiryAi[ys2]
  AiryAiPrime[yt1] AiryAiPrime[yt2]], Assumptions → assumpAll]
  AiryAi[ys1] AiryAi[ys2] AiryAiPrime[yt1] AiryAiPrime[yt2] +
  (*ApApAA*)
Simplify[Coefficient[Expr81, AiryAiPrime[ys1]
  AiryAiPrime[ys2] AiryAi[yt1] AiryAi[yt2]], Assumptions → assumpAll]
  AiryAiPrime[ys1] AiryAiPrime[ys2] AiryAi[yt1] AiryAi[yt2] +
  (*ApAApA*)
Simplify[Coefficient[Expr81, AiryAiPrime[ys1] AiryAi[ys2]
  AiryAiPrime[yt1] AiryAi[yt2]], Assumptions → assumpAll]
  AiryAiPrime[ys1] AiryAi[ys2] AiryAiPrime[yt1] AiryAi[yt2] +
  (*AAPAAp*)
Simplify[Coefficient[Expr81, AiryAi[ys1] AiryAiPrime[ys2]
  AiryAi[yt1] AiryAiPrime[yt2]], Assumptions → assumpAll]
  AiryAi[ys1] AiryAiPrime[ys2] AiryAi[yt1] AiryAiPrime[yt2] +
  (*ApAAAp*)
Simplify[Coefficient[Expr81, AiryAi[ys1] AiryAiPrime[ys2]
  AiryAiPrime[yt1] AiryAi[yt2]], Assumptions → assumpAll]
  AiryAi[ys1] AiryAiPrime[ys2] AiryAiPrime[yt1] AiryAi[yt2] +
  (*AApApA*)
Simplify[Coefficient[Expr81, AiryAiPrime[ys1] AiryAi[ys2]
  AiryAi[yt1] AiryAiPrime[yt2]], Assumptions → assumpAll]
  AiryAiPrime[ys1] AiryAi[ys2] AiryAi[yt1] AiryAiPrime[yt2] +
  (*ApApApAp*)
Simplify[Coefficient[Expr81, AiryAiPrime[ys1] AiryAiPrime[ys2]
  AiryAiPrime[yt1] AiryAiPrime[yt2]], Assumptions → assumpAll]
  AiryAiPrime[ys1] AiryAiPrime[ys2] AiryAiPrime[yt1] AiryAiPrime[yt2]));
Expr8 = PowerExpand[%]

Simplify[Expr81 - Expr8] (*must be 0*)

```


$$\begin{aligned}
\text{Out[172]} = & \frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}(yt1) \text{Ai}(yt2) (32 s t \chi p^2 + \\
& \xi^2 (\tau_1^2 + 1) u^2 v (-2 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 u v (-4 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 (\tau_2^2 + 1) v^2 - 8 \xi^2 \tau_1^2 \tau_2^2 v) + \\
& \frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}'(ys1) \text{Ai}'(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2)}{u^2 \sqrt[3]{u+1} v^{4/3}} + \\
& \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}'(ys1) \text{Ai}'(ys2) \text{Ai}(yt1) \text{Ai}(yt2)}{u^2 v} + \\
& \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u v + 2 v) \text{Ai}(ys1) \text{Ai}'(ys2) \text{Ai}'(yt1) \text{Ai}(yt2)}{u^2 v} + \\
& \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u v + 2 v) \text{Ai}'(ys1) \text{Ai}(ys2) \text{Ai}(yt1) \text{Ai}'(yt2)}{u^2 v} + \\
& \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2)}{u^2 v} + \\
& \frac{\xi^2 \text{Ai}(ys1) \text{Ai}'(ys2) \text{Ai}(yt1) \text{Ai}'(yt2) ((\tau_1^2 + 1) u^2 (v-2) + 2 \tau_1^2 u v + 2 \tau_1^2 v)}{u^{4/3} \sqrt[3]{u+1} v^{4/3}} + \\
& \frac{\xi^2 \sqrt[3]{u+1} \text{Ai}'(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}(yt2) ((\tau_2^2 + 1) (u^2 + 2 u + 2) v - 2 \tau_2^2 u^2)}{u^{8/3} v^{2/3}}
\end{aligned}$$

Out[173]= 0

In[174]:= (*must be all zeros*)
Coefficient[Expr8, {ρ1, ρ1^2, ρ1^3, ρ1^4}]

Out[174]= {0, 0, 0, 0}

Let us collect the intermediate result for Expr, expressed term by term (collected for different Airy function combinations)

In[175]:= Coefficient[Expr8, AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2]] /.
{τ1 → τ1, τ2 → τ2}

$$\text{Out[175]} = \frac{\sqrt[3]{u+1} (32 s t \chi p^2 + \xi^2 (\tau_1^2 + 1) u^2 v (-2 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 u v (-4 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 (\tau_2^2 + 1) v^2 - 8 \xi^2 \tau_1^2 \tau_2^2 v)}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}}$$

In[176]:= (Coefficient[Expr8, AiryAi[ys1] AiryAi[ys2] AiryAiPrime[yt1] AiryAiPrime[yt2]] /.
{τ1 → τ1, τ2 → τ2}) * AiryAi[ys1] AiryAi[ys2] AiryAiPrime[yt1] AiryAiPrime[yt2]

$$\text{Out[176]} = \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2)}{u^2 v}$$

In[177]:= Coefficient[Expr8, AiryAiPrime[ys1] AiryAiPrime[ys2] AiryAi[yt1] AiryAi[yt2]] /.
{τ1 → τ1, τ2 → τ2}

$$\text{Out[177]} = \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4))}{u^2 v}$$

In[178]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAi**[ys2] **AiryAiPrime**[yt1] **AiryAi**[yt2]] /.
 { $\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2$ }

$$\text{Out[178]} = \frac{\xi^2 \sqrt[3]{u+1} ((\tau_2^2 + 1)(u^2 + 2u + 2)v - 2\tau_2^2 u^2)}{u^{8/3} v^{2/3}}$$

In[179]:= **Coefficient**[Expr8, **AiryAi**[ys1] **AiryAiPrime**[ys2] **AiryAi**[yt1] **AiryAiPrime**[yt2]] /.
 { $\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2$ }

$$\text{Out[179]} = \frac{\xi^2 ((\tau_1^2 + 1)u^2(v-2) + 2\tau_1^2 uv + 2\tau_1^2 v)}{u^{4/3} \sqrt[3]{u+1} v^{4/3}}$$

In[180]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAi**[ys2] **AiryAi**[yt1] **AiryAiPrime**[yt2]] /.
 { $\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2$ }

$$\text{Out[180]} = \frac{\xi^2 \tau_1 \tau_2 (u^2(v-2) + 2uv + 2v)}{u^2 v}$$

In[181]:= **Coefficient**[Expr8, **AiryAi**[ys1] **AiryAiPrime**[ys2] **AiryAiPrime**[yt1] **AiryAi**[yt2]] /.
 { $\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2$ }

$$\text{Out[181]} = \frac{\xi^2 \tau_1 \tau_2 (u^2(v-2) + 2uv + 2v)}{u^2 v}$$

In[182]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAiPrime**[ys2]
AiryAiPrime[yt1] **AiryAiPrime**[yt2]] /. { $\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2$ }

$$\text{Out[182]} = \frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2(v-2) + 2u(v-4) + 2(v-4))}{u^2 \sqrt[3]{u+1} v^{4/3}}$$

In[183]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAiPrime**[ys2]
AiryAiPrime[yt1] **AiryAiPrime**[yt2]] /. { $\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2$ }

$$\text{Out[183]} = \frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2(v-2) + 2u(v-4) + 2(v-4))}{u^2 \sqrt[3]{u+1} v^{4/3}}$$

Explicit expressions for the y -
 arguments of the Airy functions after the shift
 They depend on τ_1^2 , but are independent of ρ_1

```
In[184]:= ys1new = Simplify[ysqp /. subsu2σ /. subsu1 /. shift] /. {σ^2 → 1}
ys2new = Simplify[ystsfg /. subsu2σ /. subsu1 /. shift] /. {σ^2 → 1}
yt1new = Simplify[ytqp /. subsu2σ /. subsu1 /. shift] /. {σ^2 → 1}
yt2new = Simplify[ysttfg /. subsu2σ /. subsu1 /. shift] /. {σ^2 → 1}
```

$$\text{Out[184]} = \frac{2 s \chi p + \xi (\tau^2 + 1) u}{2^{2/3} \xi \sqrt[3]{u \chi p^2}}$$

$$\text{Out[185]} = -\frac{4 s u \chi p - 2 \xi (\tau^2 + 1) (u + 1) v}{2 \times 2^{2/3} \xi u \chi p \sqrt[3]{\frac{u v + v}{u \chi p}}}$$

$$\text{Out[186]} = \frac{2 t \chi p + \xi (\tau^2 + 1) u}{2^{2/3} \xi \sqrt[3]{u \chi p^2}}$$

$$\text{Out[187]} = -\frac{4 t u \chi p - 2 \xi (\tau^2 + 1) (u + 1) v}{2 \times 2^{2/3} \xi u \chi p \sqrt[3]{\frac{u v + v}{u \chi p}}}$$

The integral over τ_1 is symmetric from $-\infty$ to $+\infty$. The arguments of the Airy functions are even functions of τ_1 , hence, all terms proportional to τ_1 in odd power are zero upon integration. We remove them in this step

```
In[188]:= Expr9 = Simplify[(Expr8 + (Expr8 /. {τ1 → -τ1})) / 2]
Diffs9 = Diffs8
Coeff9 = Coeff8
Phase9 = Phase8
```

$$\text{Out[188]} = \frac{1}{2 u^{8/3} \sqrt[3]{u+1} v^{5/3}} \left(\sqrt[3]{2} \left(\frac{u(u+1)}{\chi p} \right)^{2/3} \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}(yt1) \text{Ai}(yt2) (32 s t \chi p^2 + \xi^2 (\tau^2 + 1) u^2 v (-2 \tau^2 + \tau^2 v + v) + \right. \\ \left. 2 \xi^2 \tau^2 u v (-4 \tau^2 + \tau^2 v + v) + 2 \xi^2 \tau^2 (\tau^2 + 1) v^2 - 8 \xi^2 \tau^2 \tau^2 v) + \right. \\ \left. 2 \xi^2 (u+1)^{2/3} v \text{Ai}'(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}(yt2) ((\tau^2 + 1) (u^2 + 2 u + 2) v - 2 \tau^2 u^2) + \right. \\ \left. 2 \times 2^{2/3} \xi^2 \sqrt[3]{v} (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) (u \chi p)^{2/3} \text{Ai}'(ys1) \text{Ai}'(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2) + \right. \\ \left. 2 \xi^2 u^{4/3} \sqrt[3]{v} \text{Ai}(ys1) \text{Ai}'(ys2) \text{Ai}(yt1) \text{Ai}'(yt2) ((\tau^2 + 1) u^2 (v-2) + 2 \tau^2 u v + 2 \tau^2 v) \right)$$

```
Out[189]= dρ1 dρ2 ds dt dτ1 dτ2 du dv
```

$$\text{Out[190]} = \frac{\alpha^2 m^2 u}{4 \pi^2 L \phi \xi p_0 (u+1)^2 \sqrt{(v-4) v} \chi p (i\epsilon + s) (t - i\epsilon)}$$

```
Out[191]= -((ρ1 - ρ2) (s - t))
```

Step 8: Integration over ρ_1, ρ_2, s and t ; one- and two-step contributions

Integrals over $d\rho_1 d\rho_2 ds dt$ are considered together and are regularised as described in section

6.1 in [arXiv 2605.12479] (eqs. (6.36) , (6.37))

The procedure amounts to

$$\int d\rho_1 d\rho_2 ds dt \text{Coeff}_9 * \text{Expr}_9 * \text{Exp}[-i(\rho_1 - \rho_2)(s - t)] \rightarrow I_{1st} + I_{2st}$$

where

$$I_{1st} = 2\pi L_\phi \int \frac{ds}{s^2} \text{Coeff}_{1st} * (G[s] - G[-s] - 2G[0])$$

$$I_{2st} = 2\pi^2 L_\phi^2 \text{Coeff}_{2st} * G[0]$$

correspond to one - and two - step contributions, respectively

factor10 is an overall factor which we transfer from the Expr to Coeff

We introduce Gs and split Expr into two contributions,
Expr_{1st} and Expr_{2st}, which we track separately

Phase is zero in this step, so we drop the corresponding variable

```
In[192]:= factor10 = 8 m^4;
Gs = Expr9 / factor10 /. {t -> s} /. {yt1 -> ys1, yt2 -> ys2}
Phase10 = Phase9 /. {t -> s}
```

$$\text{Out[193]} = \frac{1}{16 m^4 u^{8/3} \sqrt[3]{u+1} v^{5/3}} \left(\sqrt[3]{2} \left(\frac{u(u+1)}{\chi p} \right)^{2/3} \text{Ai}(ys1)^2 \text{Ai}(ys2)^2 (32 s^2 \chi p^2 + \xi^2 (\tau 1^2 + 1) u^2 v (-2 \tau 2^2 + \tau 2^2 v + v) + 2 \xi^2 \tau 1^2 \right. \right. \\ \left. \left. u v (-4 \tau 2^2 + \tau 2^2 v + v) + 2 \xi^2 \tau 1^2 (\tau 2^2 + 1) v^2 - 8 \xi^2 \tau 1^2 \tau 2^2 v) + \right. \right. \\ \left. \left. 2 \xi^2 (u+1)^{2/3} v \text{Ai}'(ys1)^2 \text{Ai}(ys2)^2 ((\tau 2^2 + 1) (u^2 + 2 u + 2) v - 2 \tau 2^2 u^2) + \right. \right. \\ \left. \left. 2 \times 2^{2/3} \xi^2 \sqrt[3]{v} (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) (u \chi p)^{2/3} \text{Ai}'(ys1)^2 \text{Ai}'(ys2)^2 + \right. \right. \\ \left. \left. 2 \xi^2 u^{4/3} \sqrt[3]{v} \text{Ai}(ys1)^2 \text{Ai}'(ys2)^2 ((\tau 1^2 + 1) u^2 (v-2) + 2 \tau 1^2 u v + 2 \tau 1^2 v) \right) \right)$$

Out[194]= 0

One - step contribution

ys1 and ys2 are $y_{1,2}(s)$ arguments, introduced above

yms1 and yms1 are the same arguments with $s \rightarrow -s$

y01 and y02 are given by $y_{1,2}(0)$

```

In[195]:= Diff101st = Diff9 /. {dt → 1, dρ1 → 1, dρ2 → 1}
Coeff101st = Simplify[Coeff9 * 2 π Lφ * factor10 /. {t → s} /. {ie → 0}]
Expr101st =
  Gs + (Gs /. {s → -s, ys1 → yms1, ys2 → yms2}) - 2 (Gs /. {s → 0, ys1 → y01, ys2 → y02})

```

```
Out[195]= ds dτ1 dτ2 du dv
```

$$\text{Out[196]} = \frac{4 \alpha^2 m^6 u}{\pi \xi p_0 s^2 (u+1)^2 \sqrt{(v-4)v} \chi p}$$

$$\begin{aligned} \text{Out[197]} = & -\frac{1}{8 m^4 u^{8/3} \sqrt[3]{u+1} v^{5/3}} \\ & \left(\sqrt[3]{2} (-8 v \tau_1^2 \tau_2^2 \xi^2 + 2 v^2 \tau_1^2 (\tau_2^2 + 1) \xi^2 + 2 u v \tau_1^2 (v \tau_2^2 - 4 \tau_2^2 + v) \xi^2 + u^2 v (\tau_1^2 + 1) (v \tau_2^2 - 2 \tau_2^2 + v) \xi^2) \right. \\ & \left(\frac{u(u+1)}{\chi p} \right)^{2/3} \text{Ai}(y_{01})^2 \text{Ai}(y_{02})^2 + 2 (u+1)^{2/3} v \xi^2 ((u^2 + 2 u + 2) v (\tau_2^2 + 1) - 2 u^2 \tau_2^2) \text{Ai}'(y_{01})^2 \text{Ai}(y_{02})^2 + \\ & 2 u^{4/3} \sqrt[3]{v} \xi^2 ((v-2)(\tau_1^2 + 1) u^2 + 2 v \tau_1^2 u + 2 v \tau_1^2) \text{Ai}(y_{01})^2 \text{Ai}'(y_{02})^2 + \\ & \left. 2 \times 2^{2/3} ((v-2) u^2 + 2 (v-4) u + 2 (v-4)) \sqrt[3]{v} \xi^2 (u \chi p)^{2/3} \text{Ai}'(y_{01})^2 \text{Ai}'(y_{02})^2 \right) + \\ & \frac{1}{16 m^4 u^{8/3} \sqrt[3]{u+1} v^{5/3}} \left(\sqrt[3]{2} \left(\frac{u(u+1)}{\chi p} \right)^{2/3} (-8 v \tau_1^2 \tau_2^2 \xi^2 + 2 v^2 \tau_1^2 (\tau_2^2 + 1) \xi^2 + 2 u v \tau_1^2 (v \tau_2^2 - 4 \tau_2^2 + v) \xi^2 + \right. \\ & u^2 v (\tau_1^2 + 1) (v \tau_2^2 - 2 \tau_2^2 + v) \xi^2 + 32 s^2 \chi p^2) \text{Ai}(y_{ms1})^2 \text{Ai}(y_{ms2})^2 + \\ & 2 (u+1)^{2/3} v \xi^2 ((u^2 + 2 u + 2) v (\tau_2^2 + 1) - 2 u^2 \tau_2^2) \text{Ai}'(y_{ms1})^2 \text{Ai}(y_{ms2})^2 + \\ & 2 u^{4/3} \sqrt[3]{v} \xi^2 ((v-2)(\tau_1^2 + 1) u^2 + 2 v \tau_1^2 u + 2 v \tau_1^2) \text{Ai}(y_{ms1})^2 \text{Ai}'(y_{ms2})^2 + \\ & \left. 2 \times 2^{2/3} ((v-2) u^2 + 2 (v-4) u + 2 (v-4)) \sqrt[3]{v} \xi^2 (u \chi p)^{2/3} \text{Ai}'(y_{ms1})^2 \text{Ai}'(y_{ms2})^2 \right) + \\ & \frac{1}{16 m^4 u^{8/3} \sqrt[3]{u+1} v^{5/3}} \left(\sqrt[3]{2} \left(\frac{u(u+1)}{\chi p} \right)^{2/3} (-8 v \tau_1^2 \tau_2^2 \xi^2 + 2 v^2 \tau_1^2 (\tau_2^2 + 1) \xi^2 + \right. \\ & 2 u v \tau_1^2 (v \tau_2^2 - 4 \tau_2^2 + v) \xi^2 + u^2 v (\tau_1^2 + 1) (v \tau_2^2 - 2 \tau_2^2 + v) \xi^2 + 32 s^2 \chi p^2) \text{Ai}(y_{s1})^2 \text{Ai}(y_{s2})^2 + \\ & 2 (u+1)^{2/3} v \xi^2 ((u^2 + 2 u + 2) v (\tau_2^2 + 1) - 2 u^2 \tau_2^2) \text{Ai}'(y_{s1})^2 \text{Ai}(y_{s2})^2 + \\ & 2 u^{4/3} \sqrt[3]{v} \xi^2 ((v-2)(\tau_1^2 + 1) u^2 + 2 v \tau_1^2 u + 2 v \tau_1^2) \text{Ai}(y_{s1})^2 \text{Ai}'(y_{s2})^2 + \\ & \left. 2 \times 2^{2/3} ((v-2) u^2 + 2 (v-4) u + 2 (v-4)) \sqrt[3]{v} \xi^2 (u \chi p)^{2/3} \text{Ai}'(y_{s1})^2 \text{Ai}'(y_{s2})^2 \right) \end{aligned}$$

Two – step contribution

Note that the integral over s is carried out in this term

```
In[198]:= DiffS102 st = DiffS9 /. {ds → 1, dt → 1, dρ1 → 1, dρ2 → 1}
Coeff102 st = Simplify[Coeff9 * 2 π^2 Lφ^2 * factor10 /. {t → 1, s → 1} /. {ie → 0}]
Expr102 st = Gs /. {s → 0, ys1 → y01, ys2 → y02}
```

```
Out[198]= dτ1 dτ2 du dv
```

$$\text{Out[199]} = \frac{4 \alpha^2 L \phi m^6 u}{\xi p_0 (u+1)^2 \sqrt{(v-4)v} \chi p}$$

$$\begin{aligned} \text{Out[200]} = & \frac{1}{16 m^4 u^{8/3} \sqrt[3]{u+1} v^{5/3}} \left(\sqrt[3]{2} \left(\frac{u(u+1)}{\chi p} \right)^{2/3} \text{Ai}(y_{01})^2 \text{Ai}(y_{02})^2 \right. \\ & (\xi^2 (\tau^2 + 1) u^2 v (-2 \tau^2 + \tau^2 v + v) + 2 \xi^2 \tau^2 u v (-4 \tau^2 + \tau^2 v + v) + 2 \xi^2 \tau^2 (\tau^2 + 1) v^2 - 8 \xi^2 \tau^2 \tau^2 v) + \\ & 2 \xi^2 (u+1)^{2/3} v \text{Ai}'(y_{01})^2 \text{Ai}(y_{02})^2 ((\tau^2 + 1)(u^2 + 2u + 2)v - 2 \tau^2 u^2) + \\ & 2 \times 2^{2/3} \xi^2 \sqrt[3]{v} (u^2 (v-2) + 2u(v-4) + 2(v-4)) (u \chi p)^{2/3} \text{Ai}'(y_{01})^2 \text{Ai}'(y_{02})^2 + \\ & \left. 2 \xi^2 u^{4/3} \sqrt[3]{v} \text{Ai}(y_{01})^2 \text{Ai}'(y_{02})^2 ((\tau^2 + 1)u^2 (v-2) + 2 \tau^2 u v + 2 \tau^2 v) \right) \end{aligned}$$

Step 9: Integration over τ

Integrate over τ

$$\int d\tau \tau^{2k} \text{Ai}^2(a + x \tau^2) = \frac{\Gamma(k + 1/2)}{2^{2k/3+1} \pi^{1/2} x^{k+1/2}} \text{Ai}_{k+1}(2^{2/3} a)$$

$$\int d\tau \text{Ai}^2(a + x \tau^2) = \frac{1}{2 x^{1/2}} \text{Ai}_1(2^{2/3} a)$$

$$\int d\tau \tau^2 \text{Ai}^2(a + x \tau^2) =$$

$$\frac{1}{4 \times 2^{2/3} x^{3/2}} \text{Ai}_2(2^{2/3} a) = \frac{1}{4 \times 2^{2/3} x^{3/2}} [-\text{Ai}'(2^{2/3} a) - 2^{2/3} a \text{Ai}_1(2^{2/3} a)]$$

$$\int d\tau \text{Ai}'^2(a + x \tau^2) = \frac{1}{4 \times 2^{2/3} x^{1/2}} [-3 \text{Ai}'(2^{2/3} a) - 2^{2/3} a \text{Ai}_1(2^{2/3} a)]$$

$$\text{Ai}_2(y) = -\text{Ai}'(y) - y \text{Ai}_1(y)$$

$ys_i = a_i + x_i \tau_i$, see a_i and x_i below

We will perform the integration for G_s

and express the overall result via integrated G_s

Explicit representation for the Airy function arguments

```

In[201]:= Expand[PowerExpand[ys1new]]
a1expl = (u / 2 / xp) ^ (2 / 3) Expand[Simplify[(% / (u / 2 / xp) ^ (2 / 3) /. {τ1 → 0})]]
x1expl = Coefficient[%%, τ1 ^ 2]
Out[201]= 
$$\frac{\sqrt[3]{2} s \sqrt[3]{xp}}{\xi \sqrt[3]{u}} + \frac{\tau^2 u^{2/3}}{2^{2/3} xp^{2/3}} + \frac{u^{2/3}}{2^{2/3} xp^{2/3}}$$

Out[202]= 
$$\frac{\left(\frac{u}{xp}\right)^{2/3} \left(\frac{2 s xp}{\xi u} + 1\right)}{2^{2/3}}$$

Out[203]= 
$$\frac{u^{2/3}}{2^{2/3} xp^{2/3}}$$

In[204]:= 2 ^ (2 / 3)  $\left(\frac{(u + 1) v}{4 u xp}\right)^{2/3} *$ 
Collect[Expand[PowerExpand[ys2new / 2 ^ (2 / 3) /  $\left(\frac{(u + 1) v}{4 u xp}\right)^{2/3}$ ]], τ2, Simplify]
a2expl = % /. {τ2 → 0}
x2expl = Coefficient[%%, τ2 ^ 2]
Out[204]= 
$$\frac{\left(\frac{(u+1)v}{u xp}\right)^{2/3} \left(-\frac{2 s u xp}{\xi u v + \xi v} + \tau^2 + 1\right)}{2^{2/3}}$$

Out[205]= 
$$\frac{\left(\frac{(u+1)v}{u xp}\right)^{2/3} \left(1 - \frac{2 s u xp}{\xi u v + \xi v}\right)}{2^{2/3}}$$

Out[206]= 
$$\frac{\left(\frac{(u+1)v}{u xp}\right)^{2/3}}{2^{2/3}}$$

In[207]:= 2 ^ (2 / 3) Simplify[a1expl, Assumptions → assumpAll]
Simplify[% /. {s → 0}, Assumptions → assumpAll]
2 ^ (2 / 3)  $\left(\frac{(u + 1) v}{4 u xp}\right)^{2/3}$  Simplify[a2expl /  $\left(\frac{(u + 1) v}{4 u xp}\right)^{2/3}$ , Assumptions → assumpAll]
Simplify[% /. {s → 0}, Assumptions → assumpAll]
Out[207]= 
$$\left(\frac{u}{xp}\right)^{2/3} \left(\frac{2 s xp}{\xi u} + 1\right)$$

Out[208]= 
$$\left(\frac{u}{xp}\right)^{2/3}$$

Out[209]= 
$$\left(\frac{(u + 1) v}{u xp}\right)^{2/3} \left(1 - \frac{2 s u xp}{\xi u v + \xi v}\right)$$

Out[210]= 
$$\left(\frac{(u + 1) v}{u xp}\right)^{2/3}$$


```

Technical step to facilitate power

expansion (probably replaceable via PowerExpand)

```
In[211]:= subsFactoruvxp =
{
 $\sqrt[3]{u^a - (u+1)^b - v^c - xp^d} \rightarrow u^{a/3} (u+1)^{b/3} v^{c/3} xp^{d/3}, \sqrt[3]{u^a - (u+1)^b - v^c} \rightarrow u^{a/3} (u+1)^{b/3} v^{c/3},$ 
 $\sqrt[3]{u^a - v^c - xp^d} \rightarrow u^{a/3} v^{c/3} xp^{d/3}, \sqrt[3]{(u+1)^b - v^c - xp^d} \rightarrow (u+1)^{b/3} v^{c/3} xp^{d/3},$ 
 $\left(\frac{(u+1)v}{u xp}\right)^{2/3} \rightarrow \frac{(u+1)^{2/3} v^{2/3}}{u^{2/3} xp^{2/3}}, \sqrt[3]{(u+1)u^a} \rightarrow \sqrt[3]{u+1} u^{a/3}, \sqrt[3]{\frac{(u+1)v}{u xp}} \rightarrow \frac{\sqrt[3]{u+1} \sqrt[3]{v}}{\sqrt[3]{u} \sqrt[3]{xp}},$ 
 $(u xp^2)^{2/3} \rightarrow u^{2/3} xp^{4/3}, \sqrt[3]{u xp^2} \rightarrow \sqrt[3]{u} xp^{2/3}, \sqrt[3]{u^5 v xp^2} \rightarrow u^{5/3} \sqrt[3]{v} xp^{2/3},$ 
 $\sqrt[3]{u^2 v xp^2} \rightarrow u^{2/3} \sqrt[3]{v} xp^{2/3}, \left(\left(1 + \frac{1}{u}\right)v\right)^x \rightarrow (1+u)^x v^x / u^x,$ 
 $\sqrt[3]{\frac{u}{u+1}} \rightarrow u^{1/3} / (1+u)^{1/3}, \sqrt[3]{\frac{u+1}{xp}} \rightarrow (1+u)^{1/3} / xp^{1/3},$ 
 $\left(\frac{u+1}{xp}\right)^x \rightarrow (1+u)^x / xp^x, \left(\frac{v}{xp}\right)^x \rightarrow v^x / xp^x\}$ 
```

```
Out[211]= {
 $\sqrt[3]{u^a - (u+1)^b - v^c - xp^d} \rightarrow u^{a/3} (u+1)^{b/3} v^{c/3} xp^{d/3}, \sqrt[3]{u^a - (u+1)^b - v^c} \rightarrow u^{a/3} (u+1)^{b/3} v^{c/3},$ 
 $\sqrt[3]{u^a - v^c - xp^d} \rightarrow u^{a/3} v^{c/3} xp^{d/3}, \sqrt[3]{(u+1)^b - v^c - xp^d} \rightarrow (u+1)^{b/3} v^{c/3} xp^{d/3}, \left(\frac{(u+1)v}{u xp}\right)^{2/3} \rightarrow \frac{(u+1)^{2/3} v^{2/3}}{u^{2/3} xp^{2/3}},$ 
 $\sqrt[3]{(u+1)u^a} \rightarrow \sqrt[3]{u+1} u^{a/3}, \sqrt[3]{\frac{(u+1)v}{u xp}} \rightarrow \frac{\sqrt[3]{u+1} \sqrt[3]{v}}{\sqrt[3]{u} \sqrt[3]{xp}}, (u xp^2)^{2/3} \rightarrow u^{2/3} xp^{4/3}, \sqrt[3]{u xp^2} \rightarrow \sqrt[3]{u} xp^{2/3},$ 
 $\sqrt[3]{u^5 v xp^2} \rightarrow u^{5/3} \sqrt[3]{v} xp^{2/3}, \sqrt[3]{u^2 v xp^2} \rightarrow u^{2/3} \sqrt[3]{v} xp^{2/3}, \left(\left(\frac{1}{u} + 1\right)v\right)^x \rightarrow u^{-x} (u+1)^x v^x,$ 
 $\sqrt[3]{\frac{u}{u+1}} \rightarrow \frac{\sqrt[3]{u}}{\sqrt[3]{u+1}}, \sqrt[3]{\frac{u+1}{xp}} \rightarrow \frac{\sqrt[3]{u+1}}{\sqrt[3]{xp}}, \left(\frac{u+1}{xp}\right)^x \rightarrow (u+1)^x xp^{-x}, \left(\frac{v}{xp}\right)^x \rightarrow v^x xp^{-x}\}$ 
```

The following expression is used later for collecting terms

```
In[212]:= collectAirya = {AiryAiPrime[2^{2/3} a1] AiryAiPrime[2^{2/3} a2],
  Ai1[2^{2/3} a1] Ai1[2^{2/3} a2],
  AiryAiPrime[2^{2/3} a1] Ai1[2^{2/3} a2],
  Ai1[2^{2/3} a1] AiryAiPrime[2^{2/3} a2]};

```

Integration of Gs over $\tau_{1,2}$ is

implemented in this step via substitutions in Gs

Result is stored in GsIntegrated


```
In[213]:= Collect[Gs, {τ1, AiryAi[ys1]^2, AiryAiPrime[ys1]^2}, Simplify] /.
  {τ1^2 AiryAi[ys1]^2 →
    
$$\frac{1}{4 \times 2^{2/3} x1 \exp^{3/2}} (-\text{AiryAiPrime}[2^{2/3} a1] - 2^{2/3} a1 \exp^{1/2} \text{Ai}_1[2^{2/3} a1]) \} /.
  \{ \text{AiryAi}[ys1]^2 \rightarrow \frac{1}{2 x1 \exp^{1/2}} \text{Ai}_1[2^{2/3} a1], \text{AiryAiPrime}[ys1]^2 \rightarrow
    \frac{1}{4 \times 2^{2/3} x1 \exp^{1/2}} (-3 \text{AiryAiPrime}[2^{2/3} a1] - 2^{2/3} a1 \exp^{1/2} \text{Ai}_1[2^{2/3} a1]) \};$$

```

```
Expand[%];
```

```
Collect[%, {τ2, AiryAi[ys2]^2, AiryAiPrime[ys2]^2}, Simplify] /.
  {τ2^2 AiryAi[ys2]^2 →
    
$$\frac{1}{4 \times 2^{2/3} x2 \exp^{3/2}} (-\text{AiryAiPrime}[2^{2/3} a2] - 2^{2/3} a2 \exp^{1/2} \text{Ai}_1[2^{2/3} a2]) \} /.
  \{ \text{AiryAi}[ys2]^2 \rightarrow \frac{1}{2 x2 \exp^{1/2}} \text{Ai}_1[2^{2/3} a2], \text{AiryAiPrime}[ys2]^2 \rightarrow
    \frac{1}{4 \times 2^{2/3} x2 \exp^{1/2}} (-3 \text{AiryAiPrime}[2^{2/3} a2] - 2^{2/3} a2 \exp^{1/2} \text{Ai}_1[2^{2/3} a2]) \};$$

```

```
Simplify[% /. subsFactoruvxp];
```

```
Collect[Expand[%],
```

```
  collectAirya, Simplify] /. subsFactoruvxp;
```

```
GsIntegrated = Collect[%,
```

```
  collectAirya, Simplify] /. subsFactoruvxp
```

$$\begin{aligned} \text{Out[218]} = & -\frac{1}{8 m^4 u^3 (u+1) v^2} \text{Ai}_1(2^{2/3} a1) \text{Ai}_1(2^{2/3} a2) (u (v (\xi^2 + 2 s^2 \chi p^2 + 4 \xi s \chi p) - 12 s^2 \chi p^2) + \\ & u^3 (\xi + s \chi p) (v (\xi + s \chi p) - 2 s \chi p) + 2 u^2 (v (\xi + s \chi p)^2 - s \chi p (\xi + 6 s \chi p)) + 2 \xi s v \chi p) - \\ & \frac{\xi (u^2 + 2 u + 2) \chi p^{2/3} \text{Ai}'(2^{2/3} a1) \text{Ai}_1(2^{2/3} a2) (v (\xi + s u \chi p + \xi u) - 2 s u \chi p)}{8 m^4 u^{8/3} (u+1) v^2} + \\ & \frac{\xi (v-2) \chi p^{2/3} \text{Ai}_1(2^{2/3} a1) \text{Ai}'(2^{2/3} a2) (u^2 (\xi + s \chi p) + u (\xi + 2 s \chi p) + 2 s \chi p)}{8 m^4 \sqrt[3]{u^7} (u+1)^2 v^5} + \\ & \frac{\xi^2 \chi p^{4/3} \text{Ai}'(2^{2/3} a1) \text{Ai}'(2^{2/3} a2) (u^2 (v-2) + u (2 v-5) + 2 v-5)}{8 m^4 u^2 (u+1)^{2/3} v^{5/3}} \end{aligned}$$

One step term ;

Expressed via an integral over s

(equivalent of variable T in [arXiv 2605.12479] in Eq. (6.36))

```
In[219]:= Diffs111st = Diffs101st /. {dτ1 → 1, dτ2 → 1}
```

```
Coeff111st = Coeff101st
```

```
Expr111st = (GsIntegrated /. {a1 → a1expl, a2 → a2expl}) +
  (GsIntegrated /. {a1 → a1expl, a2 → a2expl} /. {s → -s}) -
  2 (GsIntegrated /. {a1 → a1expl, a2 → a2expl} /. {s → 0})
```

```
Out[219]= ds du dv
```

$$\begin{aligned}
& \text{Out[220]} = \frac{4 \alpha^2 m^6 u}{\pi \xi p_0 s^2 (u+1)^2 \sqrt{(v-4)v} \chi p} \\
& \text{Out[221]} = \frac{((v-2)u^2 + (2v-5)u + 2v-5) \chi p^{4/3} \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2s u \chi p}{u v \xi + v \xi}\right)\right) \xi^2}{8 m^4 u^2 (u+1)^{2/3} v^{5/3}} + \\
& \frac{((v-2)u^2 + (2v-5)u + 2v-5) \chi p^{4/3} \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2s u \chi p}{u v \xi + v \xi} + 1\right)\right) \xi^2}{8 m^4 u^2 (u+1)^{2/3} v^{5/3}} + \\
& \frac{(v-2) \chi p^{2/3} ((\xi - s\chi p)u^2 + (\xi - 2s\chi p)u - 2s\chi p) \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2s u \chi p}{u v \xi + v \xi} + 1\right)\right) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \xi}{8 m^4 \sqrt[3]{u^7 (u+1)^2 v^5}} + \\
& \frac{(v-2) \chi p^{2/3} ((\xi + s\chi p)u^2 + (\xi + 2s\chi p)u + 2s\chi p) \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2s u \chi p}{u v \xi + v \xi}\right)\right) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \xi}{8 m^4 \sqrt[3]{u^7 (u+1)^2 v^5}} - \\
& \frac{(u^2 + 2u + 2) \chi p^{2/3} (v(u\xi + \xi + s u \chi p) - 2s u \chi p) \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2s u \chi p}{u v \xi + v \xi}\right)\right) \xi}{8 m^4 u^{8/3} (u+1) v^2} - \\
& \frac{(u^2 + 2u + 2) \chi p^{2/3} (2s u \chi p + v(u\xi + \xi - s u \chi p)) \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2s u \chi p}{u v \xi + v \xi} + 1\right)\right) \xi}{8 m^4 u^{8/3} (u+1) v^2} - \\
& 2 \left(\frac{((v-2)u^2 + (2v-5)u + 2v-5) \chi p^{4/3} \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \xi^2}{8 m^4 u^2 (u+1)^{2/3} v^{5/3}} + \right. \\
& \frac{(v-2)(\xi u^2 + \xi u) \chi p^{2/3} \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \xi}{8 m^4 \sqrt[3]{u^7 (u+1)^2 v^5}} - \\
& \frac{(u^2 + 2u + 2)(u\xi + \xi) \chi p^{2/3} \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \xi}{8 m^4 u^{8/3} (u+1) v} - \\
& \left. \frac{(v\xi^2 u^3 + 2v\xi^2 u^2 + v\xi^2 u) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right)}{8 m^4 u^3 (u+1) v^2} \right) - \frac{1}{8 m^4 u^3 (u+1) v^2} \\
& ((\xi + s\chi p)(v(\xi + s\chi p) - 2s\chi p)u^3 + 2(v(\xi + s\chi p)^2 - s\chi p(\xi + 6s\chi p))u^2 + (v(\xi^2 + 4s\chi p\xi + 2s^2\chi p^2) - 12s^2\chi p^2)u + \\
& 2sv\xi\chi p) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2s u \chi p}{u v \xi + v \xi}\right)\right) - \frac{1}{8 m^4 u^3 (u+1) v^2} \\
& ((\xi - s\chi p)(2s\chi p + v(\xi - s\chi p))u^3 + 2(v(\xi - s\chi p)^2 + s\chi p(\xi - 6s\chi p))u^2 + (v(\xi^2 - 4s\chi p\xi + 2s^2\chi p^2) - 12s^2\chi p^2)u - \\
& 2sv\xi\chi p) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2s u \chi p}{u v \xi + v \xi} + 1\right)\right)
\end{aligned}$$

Two – step term

```
In[222]:= Diffss112 st = Diffss102 st /. {dτ1 → 1, dτ2 → 1}
Coeff112 st = Coeff102 st
Expr112 st = GsIntegrated /. {a1 → a1expl, a2 → a2expl} /. {s → 0}
Out[222]= du dv
```

$$\text{Out[223]} = \frac{4 \alpha^2 L \phi m^6 u}{\xi p_0 (u+1)^2 \sqrt{(v-4)v} \chi p}$$

$$\begin{aligned} \text{Out[224]} = & - \frac{\xi (u^2 + 2u + 2) \chi p^{2/3} (\xi + \xi u) \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 u^{8/3} (u+1)v} + \\ & \frac{\xi (v-2) \chi p^{2/3} (\xi u^2 + \xi u) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right) - \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) (\xi^2 u^3 v + 2 \xi^2 u^2 v + \xi^2 u v) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 \sqrt[3]{u^7} (u+1)^2 v^5} - \frac{\text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) (\xi^2 u^3 v + 2 \xi^2 u^2 v + \xi^2 u v) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 u^3 (u+1) v^2} + \\ & \frac{\xi^2 \chi p^{4/3} (u^2 (v-2) + u (2v-5) + 2v-5) \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 u^2 (u+1)^{2/3} v^{5/3}} \end{aligned}$$

Step 10: Rewriting the expressions to shape them into final result

We pass to variable $\lambda = s \frac{2 m^2 u \chi p}{\xi (1+u)}$,

which is the virtuality of the intermediate photon

```
In[225]:= a1expl
targ = u ^ (2 / 3) / χp ^ (2 / 3) (1 + (1 + u) / u ^ 2 * λ / m ^ 2)
Simplify[2 ^ (2 / 3) * %% == targ]
stoλ = Solve[%, s][[1]]
ξarg = Simplify[2 ^ (2 / 3) a2expl /. stoλ]
jacstoλ = D[s /. stoλ, λ]
```

$$\text{Out[225]} = \frac{\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2 s \chi p}{\xi u} + 1\right)}{2^{2/3}}$$

$$\text{Out[226]} = \frac{u^{2/3} \left(\frac{\lambda (u+1)}{m^2 u^2} + 1\right)}{\chi p^{2/3}}$$

$$\text{Out[227]} = 2 m s u \chi p = \frac{\lambda \xi (u+1)}{m}$$

$$\text{Out[228]} = \left\{ s \rightarrow \frac{\lambda \xi + \lambda \xi u}{2 m^2 u \chi p} \right\}$$

$$\text{Out[229]} = \frac{(m^2 v - \lambda) \left(\frac{(u+1)v}{u \chi p}\right)^{2/3}}{m^2 v}$$

$$\text{Out[230]} = \frac{\xi + \xi u}{2 m^2 u \chi p}$$

factor12 is an overall factor, which we transfer from Expr to Coeff

$\phi\lambda$ is the new function resulting from
passing from s to λ in Gs and diving it by factor12

The Airy function arguments are now

$$t = 2^{2/3} a1$$

$$\xi = 2^{2/3} a2$$

```
In[231]:= factor12 = (u + 1) ξ^2 / (8 u^2 v m^4);
φλ = Collect[GsIntegrated / factor12 /. stoλ, collectAirya, Simplify] /.
{a1 → t / 2^(2/3), a2 → ξ / 2^(2/3)}
```

$$\begin{aligned} \text{Out[232]} = & -\frac{(u^2 + 2u + 2) \text{Ai}_1(\xi) \text{Ai}'(t) \left(\frac{\chi p}{u}\right)^{2/3} (2m^2 v + \lambda(v-2))}{2m^2 (u+1)v} + \frac{(v-2) \text{Ai}_1(t) \text{Ai}'(\xi) (2m^2 u^2 + \lambda(u^2 + 2u + 2)) \left(\frac{\chi p}{u v + v}\right)^{2/3}}{2m^2 u^{4/3}} + \\ & \frac{\text{Ai}_1(\xi) \text{Ai}_1(t) (-4m^4 u^2 v - 4\lambda m^2 (u^2 (v-1) + u v + v) + \lambda^2 (-u^2 (v-2) - 2u(v-6) - 2(v-6)))}{4m^4 u^2 v} + \\ & \frac{\chi p^{4/3} \text{Ai}'(\xi) \text{Ai}'(t) (u^2 (v-2) + u(2v-5) + 2v-5)}{(u+1)^{5/3} v^{2/3}} \end{aligned}$$

```
In[233]:= Diffs12_1st = Diffs11_1st /. {ds → dλ}
Coeff12_1st = Simplify[(Coeff11_1st * jacstoλ * factor12) /. stoλ]
Expr12_1st = (φλ /. {a1 → targ / 2^(2/3), a2 → ξarg / 2^(2/3)}) +
(φλ /. {a1 → targ / 2^(2/3), a2 → ξarg / 2^(2/3)} /. {λ → -λ}) -
2 (φλ /. {a1 → targ / 2^(2/3), a2 → ξarg / 2^(2/3)} /. {λ → 0})
```

Out[233]= dλ du dv

$$\text{Out[234]} = \frac{\alpha^2 m^4}{\pi \lambda^2 p0 (u+1)^2 v \sqrt{(v-4)v}}$$

$$\begin{aligned} \text{Out[235]} = & -\frac{(u^2 + 2u + 2) \text{Ai}_1(\xi) \text{Ai}'(t) \left(\frac{\chi p}{u}\right)^{2/3} (2m^2 v - \lambda(v-2))}{2m^2 (u+1)v} - \frac{(u^2 + 2u + 2) \text{Ai}_1(\xi) \text{Ai}'(t) \left(\frac{\chi p}{u}\right)^{2/3} (2m^2 v + \lambda(v-2))}{2m^2 (u+1)v} + \\ & \frac{(v-2) \text{Ai}_1(t) \text{Ai}'(\xi) (2m^2 u^2 - \lambda(u^2 + 2u + 2)) \left(\frac{\chi p}{u v + v}\right)^{2/3}}{2m^2 u^{4/3}} + \frac{(v-2) \text{Ai}_1(t) \text{Ai}'(\xi) (2m^2 u^2 + \lambda(u^2 + 2u + 2)) \left(\frac{\chi p}{u v + v}\right)^{2/3}}{2m^2 u^{4/3}} + \\ & \frac{\text{Ai}_1(\xi) \text{Ai}_1(t) (-4m^4 u^2 v - 4\lambda m^2 (u^2 (v-1) + u v + v) + \lambda^2 (-u^2 (v-2) - 2u(v-6) - 2(v-6)))}{4m^4 u^2 v} + \\ & \frac{\text{Ai}_1(\xi) \text{Ai}_1(t) (-4m^4 u^2 v + 4\lambda m^2 (u^2 (v-1) + u v + v) + \lambda^2 (-u^2 (v-2) - 2u(v-6) - 2(v-6)))}{4m^4 u^2 v} - \\ & 2 \left(-\text{Ai}_1(\xi) \text{Ai}_1(t) + u^{2/3} (v-2) \text{Ai}_1(t) \text{Ai}'(\xi) \left(\frac{\chi p}{u v + v}\right)^{2/3} - \frac{(u^2 + 2u + 2) \text{Ai}_1(\xi) \text{Ai}'(t) \left(\frac{\chi p}{u}\right)^{2/3}}{u+1} + \right. \\ & \left. \frac{\chi p^{4/3} \text{Ai}'(\xi) \text{Ai}'(t) (u^2 (v-2) + u(2v-5) + 2v-5)}{(u+1)^{5/3} v^{2/3}} \right) + \frac{2 \chi p^{4/3} \text{Ai}'(\xi) \text{Ai}'(t) (u^2 (v-2) + u(2v-5) + 2v-5)}{(u+1)^{5/3} v^{2/3}} \end{aligned}$$

```
In[236]:= Diff12st = Diff11st
          Coeff12st = Factor[Simplify[(Coeff11st * factor12) /. stoλ]]
          Expr12st = (φλ /. {a1 → targ / 2 ^ (2 / 3), a2 → ζarg / 2 ^ (2 / 3)} /. {λ → 0})
```

```
Out[236]= du dv
```

$$\text{Out[237]} = \frac{\alpha^2 L \phi m^2 \xi}{2 p_0 u (u+1) \sqrt{v-4} v^{3/2} \chi p}$$

$$\begin{aligned} \text{Out[238]} = & -\text{Ai}_1(\zeta) \text{Ai}_1(t) + u^{2/3} (v-2) \text{Ai}_1(t) \text{Ai}'(\zeta) \left(\frac{\chi p}{u v + v} \right)^{2/3} - \\ & \frac{(u^2 + 2 u + 2) \text{Ai}_1(\zeta) \text{Ai}'(t) \left(\frac{\chi p}{u} \right)^{2/3}}{u+1} + \frac{\chi p^{4/3} \text{Ai}'(\zeta) \text{Ai}'(t) (u^2 (v-2) + u (2 v-5) + 2 v-5)}{(u+1)^{5/3} v^{2/3}} \end{aligned}$$

The last two cells are our final result,
which can be found in [arXiv 2605.12479]

For the one – step contribution

($\phi_{\text{tree}}(\lambda)$ is equivalent of $\phi\lambda$ used in this notebook), and $\chi = \chi p$

$$W_{\text{dir}}^{1\text{st}} = \frac{\alpha^2 m^4}{\pi p^0} \int_0^\infty du \int_4^\infty dv \int_0^\infty \frac{d\lambda}{\lambda^2} \frac{\phi_{\text{tree}}(\lambda) + \phi_{\text{tree}}(-\lambda) - 2\phi_{\text{tree}}(0)}{(1+u)^2 v^{3/2} \sqrt{v-4}}, \quad (6.54)$$

with

$$\begin{aligned} \phi_{\text{tree}}(\lambda) = & \chi^{4/3} \frac{u^2(v-2) + (1+u)(2v-5)}{(1+u)^{5/3} v^{2/3}} \text{Ai}'(t) \text{Ai}'(\zeta) \\ & + \left[\frac{\chi u}{(1+u)v} \right]^{2/3} \left(1 + \frac{\lambda}{m^2} \frac{u^2 + 2u + 2}{2u^2} \right) (v-2) \text{Ai}_1(t) \text{Ai}'(\zeta) \\ & - \left(\frac{\chi}{u} \right)^{2/3} \frac{u^2 + 2u + 2}{1+u} \left(1 + \frac{\lambda}{m^2} \frac{v-2}{2v} \right) \text{Ai}'(t) \text{Ai}_1(\zeta) \\ & - \left[1 + \frac{\lambda}{m^2} \left(\frac{u^2 + u + 1}{u^2} - \frac{1}{v} \right) \right. \\ & \quad \left. + \frac{\lambda^2}{m^4} \frac{u^2(v-2) + 2(1+u)(v-6)}{4u^2 v} \right] \text{Ai}_1(t) \text{Ai}_1(\zeta). \end{aligned} \quad (6.55)$$

$$W_{\text{dir}}^{1\text{st}} = \frac{\alpha^2 m^4}{\pi p^0} \int_0^\infty du \int_4^\infty dv \int_0^\infty \frac{d\lambda}{\lambda^2} \frac{\phi_{\text{tree}}(\lambda) + \phi_{\text{tree}}(-\lambda) - 2\phi_{\text{tree}}(0)}{(1+u)^2 v^{3/2} \sqrt{v-4}}, \quad (6.54)$$

with

$$\begin{aligned} \phi_{\text{tree}}(\lambda) = & \chi^{4/3} \frac{u^2(v-2) + (1+u)(2v-5)}{(1+u)^{5/3} v^{2/3}} \text{Ai}'(t) \text{Ai}'(\zeta) \\ & + \left[\frac{\chi u}{(1+u)v} \right]^{2/3} \left(1 + \frac{\lambda}{m^2} \frac{u^2 + 2u + 2}{2u^2} \right) (v-2) \text{Ai}_1(t) \text{Ai}'(\zeta) \\ & - \left(\frac{\chi}{u} \right)^{2/3} \frac{u^2 + 2u + 2}{1+u} \left(1 + \frac{\lambda}{m^2} \frac{v-2}{2v} \right) \text{Ai}'(t) \text{Ai}_1(\zeta) \\ & - \left[1 + \frac{\lambda}{m^2} \left(\frac{u^2 + u + 1}{u^2} - \frac{1}{v} \right) \right. \\ & \quad \left. + \frac{\lambda^2}{m^4} \frac{u^2(v-2) + 2(1+u)(v-6)}{4u^2 v} \right] \text{Ai}_1(t) \text{Ai}_1(\zeta). \end{aligned} \quad (6.55)$$

The two –

step result obtained here is equivalent to the following expression

$$W_{\text{dir},2\text{-loop}}^{2\text{st}}(\chi) = \frac{\alpha^2 m^2 \theta(\chi)}{2p^0 \chi} \xi L_\varphi \int_0^\infty du \int_4^\infty dv \frac{\phi(0)}{u(1+u)v^{3/2} \sqrt{v-4}}, \quad (6.53)$$

$$W_{\text{dir},2\text{-loop}}^{2\text{st}}(\chi) = \frac{\alpha^2 m^2 \theta(\chi)}{2p^0 \chi} \xi L_\varphi \int_0^\infty du \int_4^\infty dv \frac{\phi(0)}{u(1+u)v^{3/2} \sqrt{v-4}}, \quad (6.53)$$